

Heaven's Light Is Our Guide



CRISISCLIP-X: EXPLAINABLE CLIP-BASED MULTIMODAL DISASTER CLASSIFICATION

Thesis submitted in partial fulfillment for the requirement of the degree of

Bachelor of Science in

Electrical & Computer Engineering

by

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July, 2026

Acknowledgement

This thesis has been submitted to the Department of Electrical and Computer Engineering of Rajshahi University of Engineering & Technology (RUET), Rajshahi-6204, Bangladesh, for the partial fulfillment of the requirements for the degree of B.Sc. in Electrical & Computer Engineering. The thesis is entitled “**CRISISCLIP-X: EXPLAINABLE CLIP-BASED MULTIMODAL DISASTER CLASSIFICATION**”.

First and foremost, I offer my sincere gratitude and indebtedness to my thesis supervisor, Moloy Kumar Ghosh, Assistant Professor, Department of Electrical & Computer Engineering, who has supported me throughout my thesis with patience and knowledge. I shall ever remain grateful to him for his valuable guidance, advice, encouragement, and cordial contribution to my thesis.

I wish to thank once again Dr. G.K.M. Hasanuzzaman as the head of the Department of Electrical & Computer Engineering for his support and encouragement and also for providing all kinds of laboratory facilities.

I would like to express my deep gratitude to Hafsa Binte Kibria Assistant Professor, Department of Electrical & Computer Engineering. She has always encouraged me during the research. I am also grateful to the administration of Rajshahi University of Engineering & Technology for providing a self-sufficient Under Graduate (UG) lab.

Finally, I want to thank the most important and closest persons of my life and my parents for giving a big support to me.

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CERTIFICATE

This is to certify that the thesis entitled “*CRISISCLIP-X: EXPLAINABLE CLIP-BASED MULTIMODAL DISASTER CLASSIFICATION*” by Nabila Ferdous (Roll No. 2010035), has been carried out under my direct supervision. To the best of my knowledge, this thesis is an original one and has not been submitted anywhere for any degree or diploma.

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Abstract

This thesis proposes a multimodal disaster-information classification framework for analysing social-media posts containing images and text. Using the *CrisisMMD* and *MEDIC* datasets, the framework performs three tasks: humanitarian-category classification, damage-severity assessment, and informativeness classification. Humanitarian information is classified into *Infrastructure and Utility Damage*, *Other Relevant Information*, *Not Humanitarian*, *Rescue/Volunteering/Donation Effort*, and *Affected Individuals*. Damage severity is classified as *Severe*, *Mild*, or *None*, while informativeness is classified as *Informative* or *Non-informative*. The numerical findings are based on the implemented *CrisisMMD* experiments. The methodology uses pretrained CLIP ViT-B/32 to learn aligned representations from disaster images and social-media text. Zero-shot prompt engineering is evaluated, followed by bottleneck adapters, Low-Rank Adaptation (LoRA), and partial CLIP fine-tuning. Image and text features are normalised and combined using concatenation or task-specific cross-modal fusion. CLIP was selected because it represents images and text in a shared semantic space, while adapters and LoRA preserve pretrained knowledge with fewer trainable parameters, lower memory use, and reduced computational cost. Label smoothing, focal loss, weighted sampling, dropout, learning-rate scheduling, and early stopping address class imbalance and overfitting. Performance is evaluated using accuracy, precision, recall, F1-score, confusion matrices, latency, GPU memory, and trainable parameters. Explainable AI methods, including Grad-CAM-style heatmaps, attention inspection, Integrated Gradients, and token attribution, are used to interpret visual and textual evidence. The results show that task-specific adaptation performs better than zero-shot prediction. Humanitarian-classification accuracy increased from 66.49% to 90.58% with adapters, with a macro-F1 score of 0.8324. For informativeness classification, LoRA achieved 87.17% accuracy and a weighted F1-score of 0.8715 while training 1.03% of the model parameters. For severity assessment, adapters achieved 71.27% accuracy, while LoRA obtained a macro-F1 score of approximately 0.58. The framework can help emergency agencies filter social-media data, identify affected individuals and damaged infrastructure, and prioritise rescue and relief activities. However, inaccurate or unverified predictions may negatively affect humanitarian decisions; therefore, the system should support rather than replace human experts. Although multimodal training requires electricity, GPU resources, storage, and cooling, parameter-efficient adaptation, mixed-precision training, and early stopping can reduce resource consumption and improve computational sustainability. This thesis provides a foundation for future research in multimodal AI, explainable AI, edge deployment, and hardware-accelerated disaster-response systems.

Keywords: Disaster Classification; Multimodal Social Media; CrisisMMD; MEDIC; CLIP; Zero-Shot Learning; LoRA; Explainable AI; Humanitarian Information Classification.

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List of Abbreviations

Abbreviation	Full Form
AdamW	Adaptive Moment Estimation with Decoupled Weight Decay
AI	Artificial Intelligence
AMP	Automatic Mixed Precision
AUC	Area Under the Curve
BERT	Bidirectional Encoder Representations from Transformers
BLIP	Bootstrapping Language–Image Pre-training
CE	Cross-Entropy
CLIP	Contrastive Language–Image Pre-training
CNN	Convolutional Neural Network
CO	Course Outcome
CPU	Central Processing Unit
CrisisMMD	Crisis Multimodal Dataset
CSV	Comma-Separated Values
ECE	Electrical & Computer Engineering
ECCT	Electrical, Computer and Communication Technologies
FN	False Negative
FP	False Positive
FPGA	Field-Programmable Gate Array
FPR	False-Positive Rate
FT	Fine-Tuning
GELU	Gaussian Error Linear Unit
GPU	Graphics Processing Unit
GPT	Generative Pre-trained Transformer
Grad-CAM	Gradient-weighted Class Activation Mapping
IG	Integrated Gradients
LIME	Local Interpretable Model-Agnostic Explanations
LMM	Large Multimodal Model
LoRA	Low-Rank Adaptation
MEDIC	Multimodal Dataset for Emergency and Disaster Image Classification
MLP	Multi-Layer Perceptron
NLP	Natural Language Processing
NGO	Non-Governmental Organisation
OOD	Out-of-Distribution
PEFT	Parameter-Efficient Fine-Tuning
PO	Programme Outcome
RCNN	Recurrent Convolutional Neural Network
ReLU	Rectified Linear Unit
RGB	Red, Green, and Blue
ROC	Receiver Operating Characteristic
RUET	Rajshahi University of Engineering & Technology

Abbreviation	Full Form
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Over-sampling Technique
TN	True Negative
TP	True Positive
TPR	True-Positive Rate
TSV	Tab-Separated Values
URL	Uniform Resource Locator
VGG	Visual Geometry Group
ViT	Vision Transformer
XAI	Explainable Artificial Intelligence
XLM-R	Cross-lingual Language Model RoBERTa

List of Symbols

Symbol	Description
α	Gate scalar value in dynamic gated fusion
α_t	Class weight in focal loss
γ	Focusing parameter in focal loss
σ	Sigmoid activation function or standard deviation
η	Learning rate
η_u	Learning rate for unfrozen CLIP layers
η_h	Learning rate for the classification head
λ	Weight decay coefficient
ε	Small numerical-stability constant or label-smoothing value
τ	CLIP logit-scaling parameter
θ	Generic trainable model parameter
Θ_f	Frozen lower-layer parameters of CLIP
Θ_u	Unfrozen upper-layer parameters of CLIP
Θ_h	Parameters of the classification head
$\hat{m}_t$	Bias-corrected first moment estimate in AdamW
$\hat{v}_t$	Bias-corrected second moment estimate in AdamW
$\mathbf{g}$	Gradient vector
$g_{\max}$	Maximum gradient norm used in gradient clipping
$\mathbf{v}$	Visual embedding vector
$\mathbf{t}$	Textual embedding vector
$\mathbf{z}$	Fused multimodal representation
$H_{p,d}$	Activation value of patch p at feature dimension d
S_p	Saliency score of image patch p
$P(c)$	Predicted probability of class c
s_{ij}	Similarity score between image i and text j in CLIP
L	Generic training loss
L_{CE}	Cross-entropy loss
L_{LS}	Label-smoothing loss
L_{FL}	Focal loss
$q_{i,c}$	Smoothed target value for sample i and class c
y_i	True class label of sample i
p_y	Predicted probability of the true class
N	Number of samples
C	Number of classes
c	Class index
i, j	Sample indices
d	Feature-dimension index
p_{drop}	Dropout probability
r	LoRA rank

Symbol	Description
W_0	Original pretrained weight matrix
ΔW	Low-rank trainable weight update in LoRA

Publications

The following works arising from this thesis have been presented:

1. **Nabila Ferdous**, Hafsa Binte Kibria, Moloy Kumar Ghosh, and Humaira Tasnim Adiba, “Crisis-CLIP: A Resource-Efficient Multimodal Framework for Real-Time Disaster Response,” presented at the *International Conference on Electrical, Computer and Communication Technologies (ECCT 2026)*, Dhaka International University (DIU), Dhaka, Bangladesh, May 07–09, 2026. Presentation date: May 07, 2026. (Online Presentation)
2. **Nabila Ferdous**, Sadia Sultana Pinky, and Oyshi Joyti, “MANet: A Context-Aware Attention Network for Robust Zero-Day Ransomware Detection,” presented at the *International Conference on Electrical, Computer and Communication Technologies (ECCT 2026)*, Dhaka International University (DIU), Dhaka, Bangladesh, May 07–09, 2026. Presentation date: May 07, 2026. (Online Presentation)

Note: Both papers were presented online on May 07, 2026, at ECCT 2026, hosted by Dhaka International University (DIU), Dhaka, Bangladesh.

CO-PO Mapping

COs	POs	Sources
CO1	PO1	Chapter 1: Basic knowledge of disaster classification, multimodal AI, CLIP, image, and text analysis
CO2	PO2	Chapter 1: Analysis of noisy, imbalanced, and image–text disaster data
CO3	PO3	Chapters 2–3: Design of the proposed CLIP-based multimodal framework
CO4	PO4	Chapters 2 and 4: Experiments using CrisisMMD, MEDIC, CLIP, adapters, LoRA, and fusion
CO5	PO5	Chapter 3: Use of Python, PyTorch, Hugging Face, XAI tools, and Streamlit
CO6	PO6	Chapters 1 and 6: Humanitarian support for rescue, donation, affected people, and damage analysis
CO7	PO7	Chapters 6 and 8: Sustainability, efficient training, edge deployment, and FPGA scope
CO8	PO8	Chapter 6: Privacy, bias, misinformation, human supervision, and responsible AI use
CO9	PO9	Chapter 7: Stakeholder needs, emergency agencies, NGOs, volunteers, and disaster-response support
CO10	PO10	Thesis writing, publications, result presentation, XAI outputs, and Streamlit demonstration
CO11	PO11	Chapters 3–5: Planning, resource use, GPU memory, latency, and deployment feasibility
CO12	PO12	Chapter 8: Future work on improved models, cross-dataset testing, XAI, FPGA, and edge deployment

Chapter 1

Introduction

Disasters such as floods, earthquakes, cyclones, wildfires, landslides, and building collapses cause severe harm, and affected people, volunteers, and emergency organisations often share real-time photos, descriptions, and help requests on social media [1–4]. Since single-modality analysis misses context, this thesis builds a multimodal disaster-classification framework using CrisisMMD and MEDIC for informativeness, humanitarian-category, and damage-severity classification [7, 15]. Using CLIP ViT-B/32 [18, 43], it progresses from zero-shot prompting to parameter-efficient adaptation (adapters, LoRA, partial fine-tuning) with task-specific fusion layers, interpreted through an XAI suite (Grad-CAM, token attribution, modality-weight analysis). Models are evaluated on standard metrics and deployed via the Crisis-CLIP Streamlit application.

1.1 Overview

This research develops a computationally practical CLIP-based system for analysing multimodal disaster-related social-media posts affected by noisy text, ambiguous images, class imbalance, limited labels, overlapping categories, and image-text mismatch. CrisisMMD is used as the primary dataset for informativeness, humanitarian-category, and severity classification, while MEDIC supports additional training and out-of-distribution evaluation [7, 15]. Zero-shot CLIP baselines are established using prompt engineering [18, 21], followed by adapter and LoRA-based parameter-efficient fine-tuning, with partial fine-tuning tested for informativeness. Image and text features are combined through task-specific fusion methods, supported by dropout, class weighting, focal loss, early stopping, and mixed-precision training. Performance is evaluated using accuracy, F1-scores, confusion matrices, resource usage, and XAI methods such as Grad-CAM, Gradient×Input attribution, and modality ablation. Adapted models consistently outperformed zero-shot CLIP, with humanitarian accuracy improving from 66.49% to 90.58%, severity accuracy reaching 71.27%, and informativeness accuracy reaching 89.36%. The final models were deployed through the Crisis-CLIP Streamlit application, with informativeness used as the first classification gate.

1.2 Background and Motivation

Natural and human-induced disasters require fast, reliable, and organised information [1, 32]. Social media can provide early reports about damaged infrastructure, affected people, rescue activities, and urgent needs, but useful content is often mixed with irrelevant, repeated,

outdated, or unverified posts [2, 4]. Manual review is difficult during large-scale emergencies, while analysing only text or images may provide incomplete context. Therefore, this research develops a CLIP-based multimodal framework using both images and social-media text [12, 13, 18].

The study addresses three tasks: informativeness classification, humanitarian-category detection, and damage-severity assessment. CrisisMMD and MEDIC are used for model development and cross-dataset evaluation. Zero-shot prompt engineering, adapter fine-tuning, LoRA, and partial fine-tuning are compared using accuracy, precision, recall, macro-F1, weighted F1, confusion matrices, trainable parameters, and GPU memory. Task-specific fusion combines image and text features, while class weighting, focal loss, sampling, label smoothing, and regularisation address imbalance and overfitting.

In addition, the research incorporates visual and textual explainability techniques and implements a Streamlit-based web application to demonstrate interpretable, user-accessible, and deployment-oriented disaster classification.

1.3 Literature Review

The following papers represent the key prior works directly relevant to this research [1, 5–7, 10, 12, 13, 15]. Each summary is followed by a critical analysis identifying limitations that motivate the present study.

1.3.1 Case 1

Paper 1: Analysis of Social Media Data using Multimodal Deep Learning for Disaster Response

Authors: Ferda Ofli, Firoj Alam, Muhammad Imran [13].

This work proposes a multimodal deep learning framework that jointly analyzes textual and visual content from social media for disaster-response applications. Using the CrisisMMD dataset, containing tweet–image pairs collected from seven natural disasters, the study addresses two tasks: informativeness classification and humanitarian category classification. The authors use a 5-layer CNN with pre-trained 300-dimensional word2vec embeddings for the text and VGG16 pre-trained on ImageNet using transfer learning for the image branch. The high-level representations from the two branches are projected into equal-sized 1,000-dimensional hidden layers, which are concatenated via early feature fusion to produce a shared multimodal representation before the softmax classification. The multimodal model achieves an F1-score of 84.2% for the informativeness classification and 78.3% for the humanitarian classification, outperforming the text-only models with 80.9% and 67.7% F1, respectively, and the image-only models with 83.2% and 76.3% F1. The confusion matrix analysis also shows that multimodal learning can reduce the number of missed informative posts, as the number of false negatives is reduced from 155 in the text-only and 114

in the image-only models to 101 in the multimodal model, showing the benefit of combining complementary textual and visual evidence.

- Static word2vec embeddings and a shallow CNN text branch provide limited contextual understanding and may fail to capture ambiguity and long-range dependencies. (compared to transformer based models like BERT).
- VGG16 is computationally expensive and not on par with high-performance vision transformers and also efficient CNNs.
- Modality fusion is just simple hidden-feature concatenation with no attention, gating or cross-modal interaction to weight each modality on the fly.
- Only pairs with compatible labels are kept (conflicting, weakly aligned pairs are removed), halving the amount of the usable dataset.
- Severe class imbalance for minority humanitarian categories (e.g., affected), and hyperparameters for multimodal fusion were not tuned systematically.
- Explainable-AI methods such as LIME, SHAP, attention visualisation, and saliency analysis are not fully implemented. / use of explainable-AI method is limited [13].

The findings validate joint text–image learning as superior to unimodal baselines, but encourage future studies to implement smarter contextual embeddings, new visual encoders, attention-driven text-image fusion, imbalance handling, and XAI.

1.3.2 Case 2

Paper 2: Multimodal Learning with Augmentation Techniques for Natural Disaster Assessment

Authors: Adrian-Dinu Urse, Dumitru-Clementin Cercel, and Florin Pop [16].

This study proposes an advanced multimodal disaster assessment framework that investigates data augmentation across unimodal, multimodal, and multi-view learning settings using the imbalanced CrisisMMD dataset. The study addresses five humanitarian categories and tackles limited samples and severe class imbalance with modality-specific augmentation. Diffusion-based Real Guidance and DiffuseMix are used for images with targeted augmentation of minority classes like affected individuals, infrastructure damage, and rescue efforts. For text, multi-step back-translation, Mistral-7B-Instruct-based paraphrasing and BLIP-2 image-caption augmentation are explored [39]. Several modern architectures are evaluated: RoBERTa, DeBERTa, BERTweet, ResNet, EfficientNet, ViT and MambaViT. The best text baseline, RoBERTa-large, obtained 84.29% accuracy and 84.12% F1-score and improved to 84.71% accuracy and 84.55% F1 with back-translation and 84.61% accuracy and 84.69% F1 with paraphrasing. In the presented comparison, MambaViT achieved

around 82.4% accuracy and 82.2% F1-score for image classification, and augmentation benefits widely varied across architectures. For multimodal learning, text and image features were fused early and the RoBERTa-MambaViT model performed best on original data with 88.80% accuracy and 88.70% F1-score. Meanwhile, the combination of Real Guidance with back-translation improved RoBERTa-ViT from 87.02% accuracy and 86.82% F1 to 87.85% accuracy and 87.84% F1. The paper also proposes a multi-view architecture (see page 4) that jointly attends to original text, original images, augmented text, augmented images, and BLIP-2 captions with self-attention and cross-attention, but this more complex setup did not outperform the simpler multimodal baseline.

- Augmentation performance is very architecture-dependent: image models based on CNNs mostly benefit from augmentation, but both advanced ViT and MambaViT models in general degrade because the additional synthetic variation amounts to visual noise.
- Augmenting text using captions leads to poor performance since captions are employed during training but not during test time (train–test mismatch).
- The proposed multi-view architecture has worse performance than simpler multimodal fusion (RoBERTa-ViT: 86.07% F1 vs. 84.11% F1 and 82.28% F1 for cross-attention and self-cross-attention variants).
- Multi-view models are trained for only 10 epochs which is probably far too few for such attention-hungry architectures, and augmented views are replaced with empty placeholders at inference creating a train–inference inconsistency.
- DiffuseMix creates unnatural fractal artifacts, decreasing semantic fidelity and distracting attention-based vision models.
- The best multimodal result (88.7% F1) does not surpass the 91.8% F1 of Abavisani et al. without employing any dedicated XAI methods.

Results on imbalanced categories are improved by augmentation and modern encoders, but data complexity and multi-view fusion is not automatically better; adaptive augmentation selection and explainability remain open directions.

1.3.3 Case 3

Paper 3: Analysis of Multimodal Social Media Data Utilizing ViT Base 16 and GPT-2 for Disaster Response

Authors: Shilpa Gite, Shruti Patil, Biswajeet Pradhan, Madhuri Yadav, Sneha Basak, Arundarasi Rajendra, Abdullah Alamri, Kaustubh Raykar, and Ketan Kotecha [9].

This study introduces a multimodal classification method based on transformers to detect valuable and non-valuable disaster-related social media content by simultaneous analysis of

tweet text and images. The research utilizes the CrisisMMD dataset, which includes Twitter data from seven natural disasters such as hurricanes, wildfires, earthquakes, and floods. Although CrisisMMD provides tasks related to informativeness, humanitarian-category, and damage-severity, this research concentrates on classifying informativeness as a binary task. This dataset contains in total 18,082 records, 13,608 for training, 2,237 for validation and 2,237 for testing. The authors use GPT-2 for text classification. Preprocessing steps include converting text to lowercase, removing hashtags, URLs, HTML references, placeholders, non-letter characters, punctuation, Twitter handles and other noise elements. The text is then tokenized using Byte Pair Encoding. For image classification, the study uses an ImageNet-pretrained ViT Base 16 (ViT-B/16) model. Images are resized to 224×224 , horizontally and vertically flipped with a probability of 0.3, cropped, converted to tensors, and normalized. As shown in the proposed architecture in page 11, Figure 12, GPT-2 and ViT-B/16 are trained separately first, and their prediction outputs are combined through a Random Forest classifier to provide the final informative or non-informative prediction. The results of the experiments are shown in Table 5 (page 14). ViT-B/16 achieved an accuracy of 81.90% which performed better than ResNet-50 at 67.50%. GPT-2 achieved 80.70% accuracy which performed better than the CNN text model at 72.06%. The final ViT-B/16 + GPT-2 + Random Forest multimodal model achieved an accuracy of 84.66%, compared to 71.43% for the CNN + ResNet-50 baseline, an increase of 13.23 percentage-points. The final multimodal model also achieved a reported ROC-AUC of 0.87 which performed better than the image-only ViT model at 0.81 and text-only GPT-2 model at 0.70. Overall, the study suggests that ensemble learning by combining transformer-based visual and textual models can improve the classification of disaster-related social media informativeness.

- Fusion happens at the decision level via Random Forest on model predictions rather than learning deep feature-level interactions; cross-attention or multimodal transformers could capture relationships more effectively [40].
- Random Forest hyperparameters (100 estimators, max depth 2, 3 leaf nodes) were manually selected without systematic tuning or ablation.
- GPT-2 is a generative decoder-only model, arguably less suited to discriminative tweet classification than encoder models such as RoBERTa or DeBERTa.
- The multimodal model trades false negatives for more false positives (360 vs. 177 for image-only), and training data is imbalanced (9,638 informative vs. 3,970 non-informative).
- The decision-level ensemble discards rich hidden representations from GPT-2 and ViT; only binary informativeness is tested, leaving humanitarian and severity tasks unevaluated.
- No explainable-AI methods (SHAP, LIME) are implemented, and the model is not compared against CLIP-style joint vision-language architectures.

The dual-transformer plus Random Forest pipeline clearly improves on CNN baselines, but feature-level fusion, stronger text encoders, and explainability are identified as next steps.

1.3.4 Case 4

Paper 4: Crisis-DIAS: Towards Multimodal Damage Analysis – Deployment, Challenges and Assessment

Authors: Mansi Agarwal, Maitree Leekha, Ramit Sawhney, and Rajiv Ratn Shah [11].

This paper proposes Crisis-DIAS, an end-to-end gated multimodal deep learning framework for automatically identifying disaster-related damage and assessing its severity from social media posts. Using the CrisisMMD dataset, the framework jointly exploits textual and visual information through a sequential three-stage pipeline: Task 1 identifies informative versus non-informative posts, Task 2 detects infrastructural versus non-infrastructural damage, and Task 3 classifies severity into no/little, mild, or severe damage. Text is pre-processed by removing URLs, retweet-headers, punctuation and user-handles followed by lemmatization whereas images are resized to 299×299 and normalized. The linguistic branch uses an RCNN with 100-dimensional FastText embeddings to capture contextual dependencies and social-media specific morphological patterns whereas the visual branch uses ImageNet-pretrained Inception-v3. The authors compare decision fusion, feature fusion and attention fusion with attention fusion chosen for severity assessment because it learns feature-level importance across modalities. As illustrated by the architecture on page 4, outputs from earlier tasks act as binary gates, allowing informative posts to proceed to infrastructural-damage detection and relevant damaged posts to proceed to severity assessment. Using stratified five-fold cross-validation and SMOTE for linguistic class imbalance, Crisis-DIAS achieved 99% F1-score for informativeness detection, 99% F1-score for infrastructural damage detection, and 97% F1-score for severity assessment, with attention fusion reaching 0.98 recall and 0.98 AUC in Task 3. The overall system-level F1-score was reported to be 0.95 and the framework was able to process ~ 100 tweets/min, demonstrating strong predictive performance and practical responsiveness for disaster management.

- RCNN with static FastText embeddings may not capture deep contextual meaning, sarcasm, or long-range dependencies as well as transformer models (BERT, RoBERTa, DeBERTa); Inception-v3 could similarly be replaced by ViT, Swin, or Mamba-based vision models.
- The sequential gated architecture is vulnerable to error propagation — an incorrect early-stage prediction can prevent a relevant sample from reaching later damage-assessment stages.
- Severity analysis focuses on infrastructural/utility damage only, excluding other severe humanitarian categories such as affected individuals or vehicle damage.

- The dataset is highly imbalanced for infrastructural damage and mild/severe classes; SMOTE is applied only to the linguistic baseline, not to a comprehensive multimodal strategy.
- The framework assumes social-media content is credible, with no misinformation or authenticity verification, and may inherit geographic/socioeconomic bias from uneven social-media activity.
- No dedicated XAI (LIME, SHAP, Grad-CAM) is implemented, the system is not naturally robust to missing modalities, and broader cross-dataset validation is needed despite very high reported scores.

Crisis-DIAS shows that gated multimodal fusion is effective and fast enough for real-time use, but cascading errors, coverage, and interpretability remain key limitations.

1.3.5 Case 5

Paper 5: Disaster Image Classification by Fusing Multimodal Social Media Data

Authors: Zhiqiang Zou, Hongyu Gan, Qunying Huang, Tianhui Cai, and Kai Cao [14].

This study proposes a multimodal fusion framework to identify and classify useful disaster-related information from social media by jointly utilizing images and corresponding text. The study utilizes the CrisisMMD dataset, which contains data of seven major natural disasters, and composes the analysis into three tasks: Task 1 classifies posts as informative or non-informative; Task 2 categorizes informative disaster content, with related classes combined into people-related (P), damage-related (D), rescue-related (R), and other information (O); and Task 3 evaluates damage severity as severe, mild, or little/no damage. The proposed framework consists of three major modules: an image feature extractor, a text feature extractor, and a multimodal fusion module. As shown in the model diagram in page 7, visual features are extracted from an ImageNet-pretrained VGG16 network, where images are cropped and resized to 224×224 and the second-to-last fully connected layer is reduced from 1000 to 500 dimensions to decrease computational cost. Textual features are extracted using FastText after segmentation, cleaning, stop-word and special-character removal, stemming and lemmatization, producing a 500-dimensional text representation by averaging word embeddings. The visual and textual 500-dimensional vectors are then concatenated into a 1000-dimensional multimodal feature vector and passed through fully connected layers and a SoftMax classifier. The dataset is split into 70% training, 15% validation and 15% testing data. The experimental results shown in Table 9 on page 10 indicate the multimodal fusion achieved an accuracy of 87.6% and F1-score of 87.5% for Task 1, better than text-only accuracy of 85.2% and image-only accuracy of 83.3%. For Task 2 Step 1, the best performance was obtained by multimodal fusion with 92.6% accuracy and 92.6% F1-score. But for Task 2 Step 2, text-only classification achieved 92.2% accuracy surpassing the multimodal model with 91.25% accuracy, while Task 3 achieved only 68.9%

accuracy and 67.0% F1-score mainly due to severe class imbalance and fewer samples in the little/no-damage category. Overall, the study shows that the combination of visual and textual information can improve disaster-related social media classification especially for informativeness detection and broad disaster-content filtering.

- Averaged FastText embeddings lose word order and context and struggle with sarcasm and informal expressions; VGG16 is relatively old and computationally heavy compared with EfficientNet, ResNet, or Vision Transformer alternatives.
- Simple concatenation does not model cross-modal interaction or conflict, and multimodal fusion is not always better than unimodal models — text-only beat multimodal fusion on Task 2 Step 2 (92.2% vs. 91.25%).
- Severity classification is weak (68.9% accuracy, 67.0% F1) due to severe class imbalance (2,212 severe vs. 839 mild vs. 475 little/no-damage samples), with no strong imbalance-handling framework such as focal loss or synthetic minority generation.
- Only matching-label image–text pairs are retained, discarding potentially complementary or conflicting real-world cases; aggressive stemming and stop-word removal may strip useful context from short posts.
- No explainable-AI methods are implemented, and evaluation does not extend beyond CrisisMMD or address misinformation and duplicate content.

The framework performs well for informativeness and broad category filtering but shows that fusion is not universally superior and that severity assessment needs stronger imbalance handling and richer fusion.

1.3.6 Case 6

Paper 6: BanglaMM-Disaster: A Multimodal Transformer-Based Deep Learning Framework for Multiclass Disaster Classification in Bangla

Authors: Ariful Islam, Md Rifat Hossen, Md. Mahmudul Arif, Abdullah Al Noman, and Md Arifur Rahman [17].

This research proposes BanglaMM-Disaster, an end-to-end multimodal deep learning framework for multiclass disaster classification from Bangla social media posts by jointly analyzing textual captions and corresponding images. The authors construct a new dataset of 5,037 multimodal samples collected from public Bangla social media platforms and local news websites during major disaster events in Bangladesh from 2020–2023. Each sample contains a Bangla sentence and an associated image and is annotated into nine disaster classes: Agricultural Damage, Non Damage, Infrastructural Damage, Landslides, Damage to Natural Landscape, Floods, Fires, Economic Loss, and Others. Two trained disaster-management annotators independently label the samples, achieving Cohen’s $\kappa = 0.82$.

The dataset is divided into 70% training (3,526 samples), 10% validation (504), and 20% testing (1,007). The framework uses transformer-based encoders such as BanglaBERT, mBERT and XLM-RoBERTa for text processing and pretrained CNN architectures such as ResNet50, DenseNet169, ResNet101, MobileNetV2, and DenseNet201 for extracting visual information. The text representation is extracted from the contextual transformer features, particularly the [CLS] token embedding, and visual features are extracted from the CNNs after removing the classification heads and global average pooling. Figure 2 on page 3 depicts the framework diagram, showcasing the early fusion technique adopted by the model via feature concatenation, which merges the text and image representations into a single joint feature vector. This vector is then passed through fully connected layers, dropout and a final softmax classifier for nine-class prediction. The best text-only model, XLM-RoBERTa achieved 79.92% accuracy and 79.93% F1-score, while the best image-only model, ResNet50 achieved 66.85% accuracy and 66.85% F1-score. As shown in Table IV on page 6, the optimized mBERT + ResNet50 multimodal model achieved the best performance with 83.76% accuracy, 83.54% precision, 83.98% recall, and 83.76% F1-score. This represents a 3.84 percentage-point improvement over the best text-only model and 16.91 percentage points over the best visual-only model as reported. The class-wise error analysis shown in Figure 6 on page 6 further shows that the multimodal fusion consistently reduces errors across disaster classes; for example, Fires error reduced to 22.7%, while Economic Loss error reduced to 19.4%. The framework also reports an average inference time of 0.45 seconds and a 1.8 GB memory footprint, supporting potential real-time disaster-monitoring applications.

- Early fusion via direct concatenation does not model fine-grained word–region relationships; the authors themselves flag cross-attention and graph neural networks as future directions.
- The dataset (5,037 samples across nine classes) is relatively small for deep multimodal training, and class distribution is uneven (e.g., 937 ‘Others’ vs. only 300 ‘Fires’ samples).
- CNNs (ResNet50, DenseNet169) are used instead of modern vision transformers for computational efficiency, without a detailed accuracy–latency–memory ablation to justify the trade-off.
- English/Banglish content is machine-translated to Bangla via Google Translate, risking meaning loss for disaster terms, locations, or code-mixed expressions; text features rely only on the [CLS] token rather than pooled or layer-wise representations.
- Visual encoders are pretrained on ImageNet, creating a domain mismatch with disaster scenes, and the paper lacks detailed ablation on augmentation, translation, and fusion-layer choices.

- No explainable-AI methods (LIME, SHAP, Grad-CAM) are implemented, and evaluation is limited to one Bangla dataset from 2020–2023, limiting cross-event and cross-language generalisation claims.

BanglaMM-Disaster is a useful benchmark for a low-resource language, with the mBERT and ResNet50 fusion as the best performer, but attention-based fusion, larger balanced data, and explainability are needed for further progress.

1.4 Research Gap

1.4.1 Contributions

The contributions of this work are as follows:

- Development of a unified multimodal framework for informativeness, humanitarian-category, and damage-severity classification.
- Structured comparison of zero-shot prompting, adapters, LoRA, and partial fine-tuning using predictive and computational measures.
- Introduction of task-specific fusion methods, including concatenation, gated fusion, and symmetric image–text fusion.
- Parameter-efficient CLIP adaptation with bottleneck adapters and LoRA while preserving most pretrained knowledge.
- Integration of class-imbalance handling and XAI methods, including focal loss, label smoothing, visual saliency, token importance, occlusion, and ablation.
- Cross-dataset evaluation on MEDIC and deployment through the Crisis-CLIP Streamlit application, showing that the best adaptation method varies across tasks.

1.4.2 Impact of This Research

The proposed system can aid humanitarian information management by organizing large amounts of disaster-related social-media posts and prioritizing useful content for human review [1, 5, 32]. Humanitarian classification can identify reports of affected people, damage to infrastructure, rescue, volunteering, donations, and other relevant information. Damage-severity assessment can enable the initial review of visible damage, while informativeness classification can reduce the number of irrelevant posts requiring manual attention. These functions may improve situational awareness and help the response teams to locate urgent information. However, the system may also produce erroneous predictions due to the outdated, duplicated, incomplete, manipulated, or unclear nature of social-media content. Poor

image quality, class imbalance, cultural differences and unfamiliar disaster conditions may all further degrade performance. So the system should be an aid, not a replacement, for human decision making. Privacy and legal issues are also significant, since posts may contain identifiable people, locations, homes, phone numbers or requests for help that are sensitive. The environmental impact is mixed. Multimodal transformer models require electricity, GPU resources, storage, and cooling [30]. Using pretrained CLIP, adapters, LoRA, mixed-precision training, early stopping, checkpoint reuse, and efficient inference can reduce computational cost. Responsible deployment therefore requires human supervision, privacy protection, bias analysis, uncertainty communication, and sustainable resource use.

1.5 Objective

The main objective is to develop and evaluate a CLIP-based multimodal framework for classifying disaster-related social-media content using images and text. The specific objectives are:

1. To collect, organise, and preprocess disaster data from CrisisMMD and MEDIC.
2. To develop models for informativeness, humanitarian-category, and damage-severity classification.
3. To classify posts into Informative or Not Informative.
4. To classify humanitarian content into Infrastructure and Utility Damage, Other Relevant Information, Not Humanitarian, Rescue/Volunteering/Donation Effort and Affected Individuals.
5. To classify visible damage as Severe, Mild or None.
6. To compare zero-shot CLIP, adapter fine-tuning, LoRA, and partial fine-tuning.
7. To combine image and text features using concatenation, gated fusion or symmetric fusion.
8. To apply suitable loss functions, weights, sampling strategies and regularizers to tackle class imbalance and overfitting.
9. To compare the models on the accuracy, precision, recall, macro-F1, weighted F1, confusion matrices, trainable parameters and GPU memory.
10. To incorporate XAI and Streamlit deployment while investigating ethical, environmental, engineering and future FPGA-based development opportunities.

1.6 Thesis Outline

- **Chapter 1: Introduction** — Introduces the research problem, background, motivation, literature review, research gap, objectives, contributions, research impact, and thesis organisation.
- **Chapter 2: Methodology** — Describes the datasets, classification tasks, preprocessing steps, CLIP architecture, prompt engineering, adapter fine-tuning, LoRA, multimodal fusion, training strategy, evaluation metrics, and explainability methods.
- **Chapter 3: Design and Implementation** — Presents the software environment, data pipeline, mathematical formulation, task-specific model architectures, checkpointing strategy, XAI implementation, and Streamlit-based deployment.
- **Chapter 4: Results and Discussion** — Reports the experimental results for informativeness, humanitarian-category, and damage-severity classification, including comparisons among zero-shot CLIP, adapters, LoRA, and partial fine-tuning, along with class-level performance, confusion matrices, parameter efficiency, MEDIC evaluation, XAI outputs, and web-application results.
- **Chapter 5: Project Management and Finance** — Presents thesis planning, work schedule, timeline, equipment estimation, project estimation, and financial estimation.
- **Chapter 6: Social and Environmental Influence** — Discusses the humanitarian benefits, socioeconomic relevance, privacy issues, bias, misinformation risks, human supervision, ethical responsibility, and environmental sustainability of the proposed system.
- **Chapter 7: Complex Engineering Problems and Activities** — Explains how the proposed system relates to complex engineering knowledge, problem-solving, stakeholder needs, responsible engineering practice, and deployment-oriented engineering activities.
- **Chapter 8: Conclusion and Future Plan** — Summarises the major findings, limitations, and future directions, including FPGA acceleration, embedded processing, edge deployment, sensor integration, and wireless communication.

This chapter introduced a CLIP-based multimodal framework for classifying disaster-related social-media posts across informativeness, humanitarian category, and damage severity. It motivated the choice of CLIP ViT-B/32, zero-shot prompting, adapters, LoRA, partial fine-tuning, task-specific fusion, class-balancing, and Explainable AI, and outlined the objectives, contributions, social/environmental considerations, and thesis structure. The next chapter details the methodology: datasets, preprocessing, model selection, training strategy, evaluation measures, and explainability methods.

Chapter 2

Methodology

This chapter describes the methodology applied to develop the proposed CLIP-based multimodal disaster-classification framework. The methodology includes the overall research pipeline, the CrisisMMD and MEDIC datasets, target classes, preprocessing, model algorithms, multimodal fusion, training, evaluation, and explainability [7, 15, 18]. The study compares the proposed approach with previous state-of-the-art methods, including zero-shot prompting, adapter-based fine-tuning, LoRA, and partial fine-tuning, to identify appropriate methods for informativeness classification, humanitarian-category classification, and damage-severity assessment.

2.1 Overview

This chapter explains how disaster-related image–text records were converted into model-ready inputs and classified using CLIP ViT-B/32 [18, 20]. It includes dataset exploration, target mapping, missing-file management, text and image preprocessing, class-imbalance analysis, multimodal feature extraction, and task-specific model development. Zero-shot prompting is used as a training-free baseline, while adapters, LoRA, and partial fine-tuning provide different levels of task adaptation. The methodology allows performance, computational cost, and explainability to be evaluated together, supporting reproducible analysis, cross-dataset evaluation, efficient deployment, and future extension toward edge or FPGA-based implementation.

2.2 Case Study

This study focuses on multimodal classification of disaster-related social-media content using image and clean text pairs. During natural and human-induced disasters, people share photos, captions, short descriptions, and help requests through social media. These posts may contain useful information about infrastructure damage, affected people, rescue activities, donations, and physical destruction. However, such useful content is often mixed with irrelevant, duplicate, incomplete, or ambiguous posts. Therefore, this research studies disaster-related image–text pairs under different humanitarian categories.

CrisisMMD was used as the primary dataset. It contains paired disaster images, tweet text, and task-specific labels for informativeness, humanitarian category, and damage severity. This dataset is suitable because it supports the three main tasks of this thesis: identifying whether a post is informative, determining its humanitarian category, and assessing visible

damage severity. Its official training, development, and testing splits also support structured and reproducible experimentation [7, 8].

MEDIC was used for additional training and out-of-distribution evaluation. It includes disaster-related images labelled for informativeness, humanitarian category, damage severity, and disaster type. MEDIC helps examine whether the proposed model can generalise beyond CrisisMMD. For informativeness classification, original tweet text was used. For humanitarian and severity classification, neutral disaster-context prompts were used to reduce the risk of target leakage [15].

The case study includes three classification tasks. The first task is informativeness classification, where each post is classified as Informative or Not Informative. The second task is humanitarian-category classification, where posts are classified into Infrastructure and Utility Damage, Other Relevant Information, Not Humanitarian, Rescue/Volunteering/Donation Effort, and Affected Individuals. The third task is damage-severity classification, where visible damage is labelled as Severe, Mild, or None.

Class imbalance is a major challenge in this study. In humanitarian classification, the Not Humanitarian class has many more samples than Affected Individuals. Similarly, Severe damage is more common than Mild or None in severity classification. Therefore, accuracy alone is not sufficient, because a model may achieve high accuracy by predicting majority classes only. For this reason, macro-F1, class-wise recall, and confusion matrices were used for evaluation.

This case study also includes real-world challenges such as missing images, unsupported labels, noisy tweet text, weak image–text alignment, and dataset shift between CrisisMMD and MEDIC. Therefore, the study considers not only prediction performance but also pre-processing, robustness, explainability, computational efficiency, and deployment suitability.

2.3 Detailed Methodology

The proposed methodology follows a complete CLIP-based multimodal learning pipeline for classifying disaster-related social-media posts containing both images and text. The implementation was organised around three main classification tasks: informativeness classification, humanitarian-category classification, and damage-severity assessment. CrisisMMD was used as the primary dataset for model development, while MEDIC was used for additional retraining and out-of-distribution evaluation [7, 15]. The overall workflow included dataset loading, label cleaning, image validation, text preprocessing, CLIP-based image–text encoding, task-specific fusion, model adaptation, training, evaluation, explainability analysis, and deployment preparation.

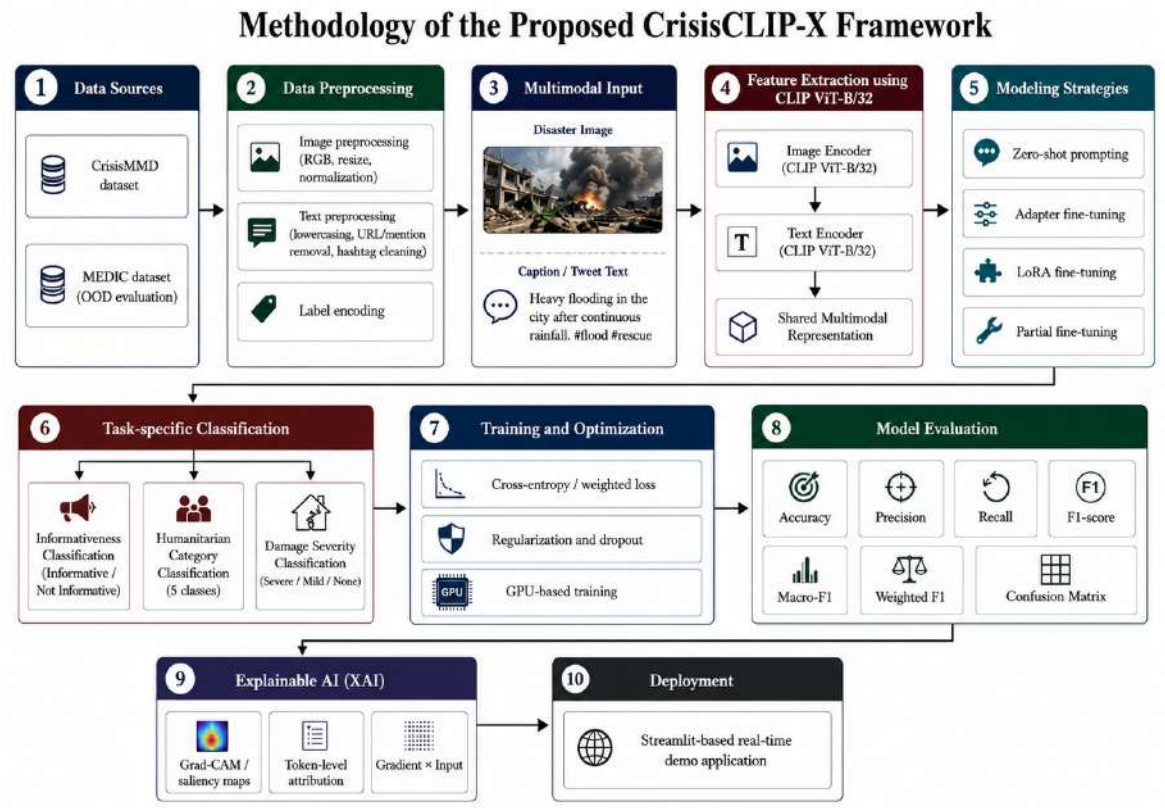


Fig. 2.1. Methodology of the Proposed CrisisCLIP-X Framework

2.3.1 Overall Methodological Pipeline

The complete methodology is summarised in Table 2.1. The pipeline starts from raw dataset files and ends with trained, evaluated, and interpretable task-specific models.

2.3.2 Dataset Loading and Cleaning

The methodology first loads CrisisMMD and MEDIC data from task-specific TSV files using `pandas`. CrisisMMD provides paired disaster images and tweet text with labels for informativeness, humanitarian category, and damage severity. MEDIC provides disaster images, text, disaster-type information, and corresponding task labels. Records with unsupported labels, missing image paths, unavailable files, or unreadable images are removed.

For MEDIC, list-like string labels are parsed and normalised before training or evaluation. Unsupported or empty labels are marked as invalid and excluded. Each valid image is opened with `PIL`, converted to RGB, and passed to the CLIP processor. A custom collation function removes invalid samples during batch creation so corrupted images do not interrupt training or testing.

Table 2.1. Overall Methodological Pipeline

Stage	Main Operation	Input	Output
Dataset loading	Load CrisisMMD and MEDIC task files	TSV files and image folders	Raw image-text records
Data inspection	Check labels, paths, class distribution, and invalid records	Raw records	Cleaned task-specific records
Label processing	Parse, normalise, and encode task labels	Original labels	Integer class labels
Image preprocessing	Validate image paths, open images, convert to RGB, and process using CLIP processor	Disaster images	Image tensors
Text preprocessing	Clean or preserve text depending on task requirements	Tweet text or neutral prompts	Tokenised text tensors
Feature extraction	Use CLIP ViT-B/32 visual and text encoders	Image and text tensors	512-dimensional embeddings
Model adaptation	Apply zero-shot prompting, adapters, LoRA, or partial fine-tuning	CLIP embeddings and labels	Task-adapted models
Fusion and classification	Combine image and text features using task-specific fusion	Visual and textual features	Class logits
Training and validation	Optimise model using suitable loss functions and checkpoint selection	Training and development sets	Best model checkpoint
Evaluation	Compute accuracy, precision, recall, F1-score, confusion matrix, and computational measures	Test/OOD data	Final performance results
Explainability	Generate saliency maps and text-token attribution	Trained model and input sample	Interpretable prediction evidence

2.3.3 Data Leakage Control

Several precautions were used to reduce target leakage. Tweet IDs, image IDs, file paths, event names, split indicators, and target-derived columns were not used as predictive features. They were used only for checking, loading, and descriptive analysis.

For MEDIC humanitarian and severity experiments, neutral disaster-context prompts were used where needed so that the text did not directly reveal the target category or severity label. Label harmonisation was also applied to keep CrisisMMD and MEDIC outputs consistent. This helped ensure that the model learned disaster-related image and text evidence rather than metadata or annotation shortcuts.

2.3.4 Task Formulation and Target Classes

The study defines three supervised classification tasks. The first task is informativeness classification, where each post is classified as either *Informative* or *Not Informative*. This

task works as the first filtering stage because non-informative posts do not need further humanitarian or severity analysis.

The second task is humanitarian-category classification. In this task, posts are classified into five classes: *Infrastructure and Utility Damage*, *Other Relevant Information*, *Not Humanitarian*, *Rescue/Volunteering/Donation Effort*, and *Affected Individuals*. The MEDIC label `affected_injured_or_dead_people` is harmonised with the thesis label *Affected Individuals* to maintain consistency with the CrisisMMD output structure.

The third task is damage-severity classification. In this task, visible damage is classified into *Severe*, *Mild*, or *None*. The original CrisisMMD label `little_or_no_damage` and corresponding MEDIC labels are mapped to the thesis label *None*. All target labels are encoded as integer indices before being passed to the loss function.

2.3.5 Text and Image Preprocessing

Preprocessing was task-specific. Informativeness text was cleaned more strongly using lowercasing, URL and mention removal, hashtag-symbol removal, unsupported-character removal, retweet-marker removal, and whitespace normalisation. For humanitarian and severity tasks, text was kept closer to its original form to preserve useful disaster context.

For MEDIC humanitarian and severity retraining, neutral disaster-context prompts such as “An image capturing the aftermath of a flood” were used to reduce target leakage.

Images were processed using the CLIP processor. Each valid image was converted to RGB, resized, centre-cropped, converted to tensor format, and normalised according to CLIP ViT-B/32 requirements [18]. Text was tokenised into `input_ids, attention_mask` with padding or truncation up to 77 tokens.

2.3.6 CLIP-Based Multimodal Feature Extraction

The proposed framework uses CLIP ViT-B/32 as the common multimodal backbone. CLIP contains a visual encoder and a text encoder that map images and text into a shared semantic embedding space [18, 43]. For each image–text pair, the visual encoder extracts image features and the text encoder extracts textual features. These embeddings are projected into a 512-dimensional space and normalised before fusion or classification.

This shared representation is useful because disaster posts often require both modalities for correct interpretation. Images may show damage, flooding, debris, or affected people, while text may provide event context, location, urgency, or clarification. Therefore, CLIP is used across all three classification tasks.

2.3.7 Model Adaptation Methods

The model-adaptation stage compares four approaches for adapting CLIP to disaster-specific tasks: zero-shot prompting, adapter fine-tuning, LoRA fine-tuning, and partial fine-tuning.

These methods differ in supervision level, computational cost, and adaptation strength.

Zero-shot prompting is used as the baseline because CLIP predicts classes without training on the target dataset. Adapter and LoRA fine-tuning are used as parameter-efficient methods, where most pretrained CLIP weights remain frozen while small task-specific updates are learned. Partial fine-tuning updates selected higher-level CLIP layers to test whether stronger adaptation improves performance.

Overall, these methods are compared in terms of predictive performance, trainable-parameter cost, efficiency, overfitting risk, and deployment suitability.

2.3.8 Task-Specific Fusion Strategy

Table 2.2. Task-Specific Methodological Design

Task	Fusion Strategy	Main Challenge	Preferred Metric
Informativeness	Image–text concatenation	Avoiding false rejection of useful posts	Accuracy and F1-score
Humanitarian category	Gated image–text fusion	Severe class imbalance and rare affected-person samples	Macro-F1 and rare-class recall
Damage severity	Gated or symmetric fusion	Distinguishing mild and no visible damage	Accuracy, macro-F1, and confusion matrix

The fusion strategy was selected according to task requirements. Informativeness classification uses image–text concatenation because either modality may indicate whether a post is useful.

Humanitarian classification uses gated fusion because some classes are more visual while others depend more on text. Severity classification uses gated or symmetric fusion because damage is mainly visual, but text can provide supporting context. Cross-modal fusion is important because image and text features may provide complementary or conflicting evidence [40].

2.3.9 Training Strategy

The training strategy was designed to compare adaptation methods fairly while handling class imbalance, limited minority samples, and overfitting. Training data were used to learn model parameters, development data were used for checkpoint selection, and test data were reserved for final evaluation where official splits were available.

Different losses were used according to task needs. Informativeness classification used a balanced objective to separate useful and non-informative posts. Humanitarian classification used class-sensitive learning because rare classes such as *Affected Individuals* are important. Severity classification focused on reducing confusion among *Severe*, *Mild*, and *None*.

Model selection was based on development performance rather than the final epoch. Macro-F1, class-wise recall, accuracy, F1-score, and confusion matrices were considered depending on the task. This strategy helped evaluate overall performance, minority-class recognition, overfitting risk, and deployment suitability.

2.3.10 Experimental Protocol

All experiments followed a fixed protocol using the same CLIP ViT-B/32 backbone and CLIP processor for fair comparison among zero-shot prompting, adapters, LoRA, and partial fine-tuning. CrisisMMD splits were used for training, checkpoint selection, and final testing, while MEDIC supported retraining and OOD evaluation.

2.3.11 Ablation and Comparative Analysis

The ablation analysis compared adaptation methods and task-specific fusion strategies to identify their contribution. Evaluation considered accuracy, precision, recall, macro-F1, weighted F1, confusion matrices, trainable parameters, memory use, latency, and deployment feasibility.

2.3.12 Out-of-Distribution Evaluation

MEDIC is used to evaluate out-of-distribution generalisation beyond CrisisMMD, since real-world models often face distribution shifts between training and deployment environments [42]. The informativeness model is tested on MEDIC OOD data using original tweet text, while the humanitarian and severity models use MEDIC training and development files for retraining and the OOD split for evaluation. Label harmonisation is applied so that MEDIC outputs match the thesis-level CrisisMMD label structure.

2.3.13 Explainability Analysis

Explainable AI methods are used to interpret model predictions. Visual saliency highlights important image regions, while token-level attribution and Gradient $\times$ Input identify important text tokens [27, 28]. Occlusion and modality-contribution analysis examine whether predictions depend mainly on image evidence, text evidence, or both.

These explanations are used as supporting evidence rather than performance metrics. They help verify whether the model focuses on meaningful disaster cues such as collapsed buildings, floodwater, debris, rescue activity, or affected people instead of irrelevant background regions or misleading words.

2.3.14 Deployment-Oriented Workflow

The trained task-specific models are connected to a Streamlit-based application. The deployed workflow begins with image upload and text input. The informativeness classifier is executed first. If the post is classified as informative, the humanitarian-category and damage-severity classifiers are applied. If the post is classified as not informative, downstream classification can be skipped unless the user enables an override option.

The application displays predicted labels, class probabilities, image-saliency outputs, text-token importance, and inference time. This deployment-oriented design demonstrates how the proposed model can be used as an interactive decision-support tool for disaster information screening. However, the system is intended to assist human analysts and should not replace official emergency response, field verification, or expert damage assessment.

2.3.15 Proposed Framework

The proposed framework first loads the CrisisMMD and MEDIC datasets, checks image paths, tweet text, labels, and task-related variables, and removes or normalises invalid records. Text and image data are then prepared for CLIP using task-specific text cleaning, RGB image conversion, resizing, centre-cropping, tensor conversion, and standard CLIP normalisation.

After preprocessing, each sample contains image tensors, tokenised text, attention masks, and encoded labels. Target labels are mapped consistently across training, testing, evaluation, and deployment. Identifier variables such as tweet IDs, image IDs, file paths, and split indicators are not used as predictive features to avoid target leakage.

The core model uses CLIP ViT-B/32, where the image encoder extracts visual embeddings and the text encoder extracts textual embeddings. The same CLIP backbone is used for all three tasks, but each task has its own fusion strategy and classification head. Informativeness uses image–text concatenation, humanitarian classification uses gated fusion, and severity classification uses fused image–text features.

The study compares zero-shot CLIP, adapters, LoRA, and partial fine-tuning. Class imbalance is handled using focal loss, weighted loss, sampling, label smoothing, dropout, early stopping, and class-aware model selection. Performance is evaluated using accuracy, precision, recall, macro-F1, weighted F1, confusion matrices, trainable parameters, memory use, and latency.

Explainability methods are used to interpret predictions through visual saliency, token-level attribution, Gradient $\times$ Input, modality contribution, and ablation analysis. Finally, the trained models are deployed in a Streamlit application, where informativeness classification runs first, followed by humanitarian and severity analysis only when the post is classified as informative.

Table 2.3. Framework Output Classes per Task

Task	Output Classes
Informativeness	Informative, Not Informative
Humanitarian category	Infrastructure and Utility Damage, Other Relevant Information, Not Humanitarian, Rescue/Volunteering/Donation Effort, Affected Individuals
Damage severity	Severe, Mild, None
Explainability	Image-region saliency and text-token attribution
Web application	Predicted classes, probabilities, explanations, latency

2.3.16 Framework Outputs

2.3.17 Data Visualisation

Visualisation was used to understand dataset composition, class imbalance, sample quality, model learning, prediction errors, and explanation patterns.

Class-Distribution Visualisation

Bar charts compare per-class sample counts: for class c , $P_c = (n_c/N) \times 100$. Grouped bars display training, development, and test counts together. The humanitarian distribution shows a strong long-tail pattern (Not Humanitarian dominant, Affected Individuals scarce), while Severe substantially exceeds Mild and None in the severity distribution — justifying class-sensitive losses, weighted sampling, macro-F1, and class-wise recall.

Missing-Value Visualisation

A missing-value bar chart or a record-by-variable missingness matrix can represent missing versus present observations, and should ideally be produced before invalid records are removed so it reflects original data quality. Since the final notebooks did not preserve these charts or exact totals, the methodology describes the required process without inventing values.

Image–Text Sample Grid

A random sample grid, showing the image, a shortened caption, the target class, the data split, and the disaster event or type, helps detect:

- incorrect image paths or unreadable, low-quality photographs;
- unrelated image–text pairs and misleading captions;
- uncertain labels and near-duplicate posts.

This visual inspection is important because numerical checks alone cannot confirm whether a caption and image are semantically consistent.

Image-Property Analysis

Original image characteristics examined before CLIP processing include:

- width, height, and aspect ratio;
- colour mode and file format;
- file size.

Although the CLIP processor standardises input dimensions, inspecting original properties helps identify corrupted files, unusual aspect ratios, and low-resolution samples.

Training Curves

Training loss, development loss, accuracy, macro-F1, and learning rate are plotted per epoch. The best checkpoint is selected from development performance to reduce overfitting risk.

Confusion-Matrix Visualisation

The confusion matrix C_{ij} counts samples with actual class i and predicted class j ; the row-normalised matrix $\hat{C}_{ij} = C_{ij} / \sum_j C_{ij}$ is especially useful for imbalanced data since each row is the recall distribution of one class. It reveals specific errors such as Affected Individuals mistaken for Not Humanitarian, Mild mistaken for Severe damage, or informative posts marked not informative.

ROC Visualisation

For binary informativeness, an ROC curve plots $\text{TPR} = \text{TP} / (\text{TP} + \text{FN})$ against $\text{FPR} = \text{FP} / (\text{FP} + \text{TN})$; multiclass tasks use one-versus-rest curves per class. The MEDIC evaluation notebooks generated ROC analysis from predicted probability arrays, but the final CrisisMMD comparison did not preserve all required arrays, so numerical AUC values are not reported in the results chapter.

Embedding Visualisation

t-SNE was used to project high-dimensional learned features into two dimensions to inspect whether samples form class-distinct clusters. Because t-SNE is sensitive to initialisation, perplexity, and distance scaling, it is used only for exploratory visualisation and not as formal performance evidence.

Explainability Visualisation

Explainability outputs — Grad-CAM-style saliency, token-gradient bar charts, Gradient $\times$ Input beeswarm-style plots, occlusion maps, word-deletion scores, and modality-contribution/ablation charts — help verify whether the model attends to meaningful evidence [27, 28, 33] (e.g., a severity model should focus on collapsed structures, damaged roads, debris, or flooding rather than unrelated background).

2.3.18 Correlation Matrix and Heatmap Analysis

Most dataset variables are images, text, categorical labels, identifiers, or file paths, so a standard Pearson correlation matrix is not appropriate for all of them; integer-encoded nominal labels (e.g., Infrastructure=0, Other Information=1, Not Humanitarian=2) do not carry numerical magnitude.

Suitable Numerical Variables

A numerical heatmap may instead use derived variables such as text length, word/hashtag/mention counts, image width/height/aspect ratio, missing-image indicators, image–text cosine similarity, predicted confidence, and correctness indicators, with Pearson correlation r_{xy} computed only when variables are genuinely numerical and approximately linear.

Ordinal Severity Correlation

Severity has an ordered structure (None=0, Mild=1, Severe=2), so Spearman rank correlation is more appropriate than Pearson: $\rho = 1 - 6 \sum d_i^2 / [n(n^2 - 1)]$, where d_i is the difference between ranks. A positive relationship between severity and visual-damage indicators would mean stronger damage features tend to occur with higher severity.

Cramér’s V for Categorical Variables

Associations among categorical variables can be measured using the following statistic:

$$V = \sqrt{\frac{\chi^2}{n \cdot \min(r - 1, c - 1)}} \quad (2.1)$$

where χ^2 is the chi-square statistic, n is the sample size, and r and c are the contingency-table dimensions. Cramér’s V ranges from 0 (weak association) to 1 (strong association). Possible categorical comparisons include:

- event name vs. humanitarian category;
- disaster type vs. severity;
- informativeness vs. humanitarian category or severity.

Image–Text Informativeness Agreement

CrisisMMD contains separate `label_image` and `label_text` informativeness annotations. Their agreement can be examined using a contingency table, an agreement percentage, and Cohen’s kappa, $\kappa = (p_o - p_e)/(1 - p_e)$, where p_o is observed agreement and p_e is expected chance agreement; a high kappa indicates agreement beyond chance. However, because the main informativeness target is created from `label_image`, correlation between the encoded target and `label_image` is perfect by construction and provides no meaningful discovery.

Correlation Limitations

The notebooks did not preserve final correlation coefficients, so exact values are not reported without rerunning the analysis. Throughout, identifier variables are excluded, arbitrary label codes are not treated as continuous, target annotations are not used as predictive features, correlation is not interpreted as causation, and confusion matrices are not described as correlation matrices.

2.3.19 Theoretical Foundations and Differences of the Models

CLIP as the Common Foundation

CLIP was used as the common multimodal foundation because it learns a shared semantic space for images and text [18]. This makes it suitable for disaster-related posts, where visual evidence and textual descriptions often provide complementary information. In this thesis, CLIP ViT-B/32 was selected because it supports both modalities, provides a strong zero-shot baseline, produces compact visual and textual representations, and can be adapted through parameter-efficient transfer-learning methods.

Why One Identical Model Was Not Used for All Tasks

One identical model was not used because the three tasks have different requirements. Informativeness is a binary filtering task, humanitarian classification is a five-class imbalanced task, and severity assessment depends strongly on visible damage. Therefore, the framework uses a shared CLIP backbone with task-specific fusion and classification heads.

Zero-Shot Prompt Engineering

Zero-shot prompting was used as a training-free baseline. It tests whether pretrained CLIP can recognise disaster-related classes without supervised training on CrisisMMD or MEDIC. Although it has low computational cost, its performance depends strongly on prompt wording and CLIP’s pretraining knowledge. Therefore, it was used as a baseline rather than the final model.

Adapter Fine-Tuning

Adapter fine-tuning was used as a parameter-efficient method that learns task-specific transformations while keeping most CLIP parameters frozen. It can improve disaster-specific classification without full backbone training. However, adapters add extra modules, and their performance depends on bottleneck size and placement.

LoRA Fine-Tuning

LoRA fine-tuning was used to evaluate compact transformer adaptation [22]. It updates only a small number of additional parameters while keeping most pretrained weights frozen, making it suitable for storage- and memory-efficient deployment. However, its performance depends on rank, target layers, and task complexity.

Partial Fine-Tuning

Partial fine-tuning was tested as a stronger adaptation method. It updates selected higher-level CLIP layers for the disaster task and can improve learning capacity. However, it also increases training cost, memory use, checkpoint size, and overfitting risk. Therefore, it was used as a comparative method rather than the default strategy.

Comparison of Adaptation Methods

Table 2.4 summarises the methodological differences between zero-shot prompting, adapter fine-tuning, LoRA, and partial fine-tuning.

Table 2.4. Comparison of Adaptation Methods

Characteristic	Zero-shot	Adapter	LoRA	Partial FT
Task-specific training	No	Yes	Yes	Yes
Backbone update	None	Mostly frozen	Mostly frozen	Selected layers updated
New parameters	None	Bottleneck modules	Low-rank matrices	Limited direct updates
Adaptation capacity	Lowest	Moderate	Moderate	Highest
Training cost	Lowest	Low–moderate	Low–moderate	Highest
Main strength	Training-free baseline	Flexible adaptation	Compact attention update	Stronger task learning
Main limitation	Domain gap	Extra modules	Rank sensitivity	Cost and overfitting

Differences Among Task Models

Table 2.5 summarises the conceptual differences among the three task-specific models.

Table 2.5. Differences Among Task-Specific Models

Component	Humanitarian	Severity	Informativeness
Classes	5	3	2
Main fusion	Modality gate	Symmetric/feature gate	Concatenation
Imbalance	Very high	Moderate-high	Lower
Main methodological concern	Rare-class recognition	Damage-level separation	Useful-post filtering
Preferred evidence	Image and text jointly	Mainly image with text support	Image or text evidence
Key metric	Macro-F1, rare-class recall	Accuracy & macro-F1	Accuracy & F1

2.4 Justification of the Study

This study is justified by the need for fast and structured disaster-information management. During crises, many image-text posts are shared on social media, and some contain useful evidence about damage, affected people, rescue activities, and urgent needs. Manual review is slow; therefore, a multimodal classification system can support early screening, categorisation, and prioritisation for human review.

Multimodal learning is suitable because disaster posts often require both image and text understanding. Images may show floods, debris, damaged buildings, or affected people, while text may provide location, urgency, and context. CLIP ViT-B/32 is selected because it represents images and text in a shared embedding space and supports zero-shot classification, adapters, and LoRA-based adaptation.

The study compares zero-shot prompting, adapter fine-tuning, LoRA, and partial fine-tuning because each method has different strengths. Task-specific fusion is also justified because informativeness, humanitarian-category, and damage-severity classification require different image-text contributions. CrisisMMD is used as the primary dataset because it provides paired disaster images and text with task-specific labels. MEDIC is used for retraining and out-of-distribution evaluation. Since both datasets are imbalanced, the study applies focal loss, weighted loss, sampling strategies, macro-F1, class-wise recall, and confusion matrices.

Explainability and deployment also support the study. Visual saliency, token attribution, and modality-weight analysis help users understand model decisions. The Streamlit workflow applies informativeness classification first and then performs humanitarian and severity analysis only when needed. Overall, the study integrates multimodal learning, efficient adaptation, imbalance handling, explainability, and deployment while keeping human supervision essential.

2.4.1 Model-Selection Justification

Model selection was not based on accuracy alone. The study considered macro-F1, minority-class recall, confusion-matrix errors, trainable parameters, GPU memory, inference cost, checkpoint compatibility, and explanation quality.

For humanitarian classification, macro-F1 and *Affected Individuals* recall were prioritised because accuracy may hide weak minority-class performance. For severity classification, macro-F1 was considered with accuracy due to class imbalance. For informativeness classification, informative-class recall was important because a false rejection could stop later analysis.

LoRA checkpoints were used in deployment because they were parameter-efficient and compatible with the application pipeline. However, LoRA was not the best in all experiments: adapters performed best for humanitarian classification, while partial fine-tuning performed best for informativeness. Therefore, final deployment also considered storage, checkpoint availability, compatibility, and application design.

2.4.2 Methodological Limitations

- The informativeness experiment reused the test loader during development, and preprocessing was not fully consistent across tasks.
- The zero-shot method used image-to-prompt classification rather than full image-text fusion.
- CrisisMMD and MEDIC followed different training protocols, so their results are not directly comparable.
- The MEDIC humanitarian OOD split lacked Other Relevant Information samples, and duplicate or missing-value counts were not fully recorded.
- LoRA settings differed between experiments and deployment, and severe class imbalance made macro-F1 and class-wise recall essential.

This chapter presented the methodology of the proposed CLIP-based multimodal disaster-classification framework. CrisisMMD and MEDIC were used for dataset preparation, preprocessing, model adaptation, fusion, training, evaluation, and explainability. Overall, the methodology provides the basis for the implementation, deployment, and system-design discussion in the next chapter.

Chapter 3

Design and Implementation

This chapter presents the architecture and implementation of the proposed multimodal disaster-classification system [7, 15, 18]. The system follows a single integrated pipeline for assessing disaster-related social-media posts containing both an image and accompanying text, producing informativeness, humanitarian-category, and damage-severity predictions. Zero-shot prompting, adapters, and LoRA are examined as task-adaptation strategies on top of the pretrained CLIP multimodal backbone [18, 21, 22]. CrisisMMD was used mainly for model development, while MEDIC supported retraining and cross-dataset evaluation. The selected models were combined with explainable-AI (XAI) methods and integrated into a Streamlit web application [27, 28].

3.1 Design Overview

The proposed system is designed as a sequential pipeline for multimodal classification. The input is a disaster-related image together with its caption or social-media text; a shared CLIP processor converts both modalities into model-ready tensors, which are then passed to task-specific classifiers [18].

The overall system workflow is:



Fig. 3.1. Sequential inference workflow for informative disaster posts.

If a post is classified as informative, the pipeline continues to the humanitarian-category and damage-severity classifiers in parallel:

If informative $\rightarrow$ {Humanitarian-Category Classifier, Damage-Severity Classifier} $\rightarrow$
Prediction + Explanation

The informativeness classifier therefore acts as the first stage of the pipeline. The humanitarian and severity classifiers are not executed if a post is judged not informative, unless the user activates an override option; the Streamlit application follows this same sequential decision logic.

3.1.1 Modern Tools Used

The implementation used Python, PyTorch, Hugging Face Transformers, and PEFT for model development, together with pandas, NumPy, Pillow, scikit-learn, Matplotlib, and Seaborn for data analysis, and Streamlit for the front-end interface. The pipeline covers dataset preparation, training, evaluation, explainability, and web deployment.

3.1.2 Software Implementation

All experiments were carried out on a Windows PC using a single Anaconda environment (`tweet_project`). PyTorch automatically used the available CUDA-enabled GPU where possible, falling back to CPU execution otherwise [24]. Mixed-precision training with gradient scaling was used to conserve GPU memory and speed up training. Table 3.1 summarises the software and experimental environment in which the thesis implementation was carried out.

Table 3.1. Software and Experimental Environment Specification

Component	Specification
Operating system	Microsoft Windows
Python environment	Anaconda (<code>tweet_project</code>)
CUDA device	GPU if available, otherwise CPU
CPU	12th Gen Intel(R) Core(TM) i5-12500H
GPU	NVIDIA GeForce RTX 3050 Laptop GPU
GPU memory	4 GB
System memory	16 GB
Storage	NVMe Samsung MZVL2512HCJQ-00B07, 512 GB
Deep-learning framework	PyTorch
Multimodal framework	Hugging Face Transformers
Parameter-efficient fine-tuning	PEFT
Application framework	Streamlit

The processing device is selected dynamically at runtime:

```
device = "cuda" if torch.cuda.is_available() else "cpu"
```

The Streamlit web application follows the same automatic device-selection mechanism.

3.1.3 Hardware Design

No separate hardware implementation was developed in this thesis. The proposed system is a software-based multimodal disaster-classification framework, implemented and deployed using deep-learning tools and a Streamlit web application. Therefore, this section describes the computational environment used to train, test, and deploy the proposed system rather than a physical hardware circuit or embedded hardware design.

All experiments were carried out on a Windows PC using a single Anaconda environment (`tweet_project`). PyTorch automatically used the available CUDA-enabled GPU where possible, falling back to CPU execution otherwise [24]. The software deployment was completed through a Streamlit application, where users can upload an image and provide a caption or tweet text for prediction.

Thus, the thesis does not include hardware implementation; instead, it focuses on software implementation, model deployment, and interactive prediction through the Streamlit application.

3.1.4 Simulation

No circuit-based or physical hardware simulation was performed in this thesis because the project is a software-based multimodal disaster-classification system. The experimental simulation was limited to dataset processing, model training, model evaluation, explainability analysis, and Streamlit-based inference. Future work may extend the system towards FPGA, embedded, or edge-device implementation.

3.1.5 Task Output Classes

Table 3.2 summarises the output classes for each of the three classification tasks.

Table 3.2. Output Classes for the Three Classification Tasks

Task	Classes
Informativeness	Informative, Not Informative
Humanitarian category	Infrastructure and Utility Damage, Other Relevant Information, Not Humanitarian, Rescue/Volunteering/Donation Effort, Affected Individuals
Damage severity	Severe, Mild, None

To keep labels consistent across the two datasets, CrisisMMD’s `little_or_no_damage` category was mapped to the thesis label *None*, and MEDIC’s Affected Individuals category was mapped to `affected_injured_or_dead_people` so that both datasets share the same five-class humanitarian and three-class severity label spaces.

3.2 Mathematical Formulation

The multimodal disaster dataset is formally written as

$$\mathcal{D} = \{(I_i, T_i, y_i)\}_{i=1}^N \quad (3.1)$$

where I_i is the i -th image from a social-media post, T_i is its associated text, and y_i is the task-specific label.

3.2.1 Multimodal Feature Extraction

The CLIP vision encoder f_v and text encoder f_t produce image and text features,

$$\mathbf{v}_i = f_v(I_i), \quad \mathbf{t}_i = f_t(T_i) \quad (3.2)$$

which are then L_2 -normalised,

$$\hat{\mathbf{v}}_i = \mathbf{v}_i / \|\mathbf{v}_i\|_2, \quad \hat{\mathbf{t}}_i = \mathbf{t}_i / \|\mathbf{t}_i\|_2. \quad (3.3)$$

Both projected CLIP embeddings are 512-dimensional.

3.2.2 Zero-Shot Prompt Classification

For zero-shot classification, a set of text prompts representing class c is encoded with the CLIP text encoder and averaged into a single normalised class prototype $\mathbf{q}_c$. The similarity between an image and a class prototype is computed using CLIP’s learned logit scale τ ,

$$s_{i,c} = \tau \cdot \frac{\hat{\mathbf{v}}_i \cdot \mathbf{q}_c}{\|\hat{\mathbf{v}}_i\|_2 \|\mathbf{q}_c\|_2}, \quad (3.4)$$

and the predicted class is

$$\hat{y}_i = \arg \max_c s_{i,c}. \quad (3.5)$$

The class prototype itself is computed by averaging the normalised text embeddings of multiple prompt templates per class.

3.2.3 Adapter-Based Fine-Tuning

A bottleneck adapter transforms a hidden representation $\mathbf{h}$ as

$$\mathbf{h}' = \mathbf{h} + W_{\text{up}} [\sigma(W_{\text{down}} \mathbf{h} + b_{\text{down}})] + b_{\text{up}}, \quad (3.6)$$

where W_{down} reduces the dimensionality, W_{up} restores it, and σ is a nonlinear activation. The residual connection preserves the original pretrained CLIP representation while the adapter

branch learns a task-specific correction.

3.2.4 LoRA-Based Fine-Tuning

In LoRA, a low-rank update to a pretrained weight matrix W_0 is decomposed into two low-rank matrices,

$$W' = W_0 + \Delta W, \quad \Delta W = (\alpha/r) BA, \quad (3.7)$$

where $A \in \mathbb{R}^{r \times d}$ and $B \in \mathbb{R}^{k \times r}$, r is the LoRA rank, and α is the scaling factor. Only the low-rank matrices and the task-specific layers are updated; the rest of the CLIP backbone remains frozen.

3.2.5 Informativeness Fusion

For informativeness classification, the image and text embeddings are concatenated,

$$\mathbf{z}_i^{\text{inf}} = [\hat{\mathbf{v}}_i; \hat{\mathbf{t}}_i] \in \mathbb{R}^{1024}, \quad (3.8)$$

and passed through the classifier to obtain the two output logits,

$$\mathbf{o}_i^{\text{inf}} = g_{\text{inf}}(\mathbf{z}_i^{\text{inf}}). \quad (3.9)$$

3.2.6 Humanitarian Gated Fusion

The image and text features are first projected independently,

$$\mathbf{v}_i^p = W_v \hat{\mathbf{v}}_i + b_v, \quad \mathbf{t}_i^p = W_t \hat{\mathbf{t}}_i + b_t, \quad (3.10)$$

and a gating network assigns each modality a relative weight,

$$[\alpha_i^v, \alpha_i^t] = \text{Softmax}(W_g[\hat{\mathbf{v}}_i; \hat{\mathbf{t}}_i] + b_g). \quad (3.11)$$

The fused representation is

$$\mathbf{z}_i^{\text{hum}} = \text{LayerNorm}(\alpha_i^v \mathbf{v}_i^p + \alpha_i^t \mathbf{t}_i^p), \quad (3.12)$$

and the humanitarian logits are

$$\mathbf{o}_i^{\text{hum}} = g_{\text{hum}}(\mathbf{z}_i^{\text{hum}}). \quad (3.13)$$

3.2.7 Severity Gated Fusion

For severity classification, the projected image and text features are concatenated,

$$\mathbf{c}_i = [\mathbf{v}_i^p; \mathbf{t}_i^p], \quad (3.14)$$

and a feature-level sigmoid gate is computed as

$$\mathbf{g}_i = \sigma(W_{g2} \text{ReLU}(W_{g1} \mathbf{c}_i + b_{g1}) + b_{g2}). \quad (3.15)$$

The fused severity representation is the element-wise product

$$\mathbf{z}_i^{\text{sev}} = \mathbf{g}_i \odot \mathbf{c}_i, \quad (3.16)$$

and the three severity logits are

$$\mathbf{o}_i^{\text{sev}} = g_{\text{sev}}(\mathbf{z}_i^{\text{sev}}). \quad (3.17)$$

3.2.8 Classification Probability and Loss

For each task, the classifier produces a logit vector $\mathbf{o}_i = [o_{i,1}, o_{i,2}, \dots, o_{i,C}]$ for sample i , where C is number of task-specific classes. The logits are converted into class probabilities using softmax,

$$P(y_i = c \mid I_i, T_i) = \frac{\exp(o_{i,c})}{\sum_{j=1}^C \exp(o_{i,j})}. \quad (3.18)$$

The predicted class is therefore

$$\hat{y}_i = \arg \max_{c \in \{1, \dots, C\}} P(y_i = c \mid I_i, T_i). \quad (3.19)$$

For one-hot target vector $\mathbf{y}_i$, the standard cross-entropy loss is

$$L_{\text{CE}} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C y_{i,c} \log P(y_i = c \mid I_i, T_i). \quad (3.20)$$

For imbalanced tasks, class weights can be added to penalise errors in rare classes more strongly:

$$L_{\text{WCE}} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C w_c y_{i,c} \log P(y_i = c \mid I_i, T_i), \quad (3.21)$$

where w_c is the weight assigned to class c . Label smoothing was used in selected experiments to reduce overconfident predictions. The smoothed target is

$$q_{i,c} = \begin{cases} 1 - \varepsilon, & c = y_i, \\ \varepsilon / (C - 1), & c \neq y_i, \end{cases} \quad (3.22)$$

and the corresponding loss is

$$L_{LS} = -\frac{1}{N} \sum_{i=1}^N \sum_{c=1}^C q_{i,c} \log P(y_i = c). \quad (3.23)$$

For the highly imbalanced humanitarian task, focal loss was considered to reduce the dominance of easy majority-class examples:

$$L_{FL} = -\alpha_y (1 - p_y)^\gamma \log(p_y), \quad (3.24)$$

where p_y is the predicted probability of the true class, α_y is the class-specific weight, and γ is the focusing parameter. These losses support the three classifiers while keeping the final prediction rule consistent across informativeness, humanitarian-category, and damage-severity tasks.

3.2.9 Algorithms

The framework combines several algorithmic components spanning data preparation, feature extraction, adaptation, fusion, optimisation, checkpoint selection, and explainability.

Algorithm 1: Multimodal Data Preparation

Purpose: validate and prepare disaster image–text records. **Input:** a TSV file, image directory, task label map, and CLIP processor. **Procedure:** read the TSV file; parse task labels and retain only supported classes; construct the full image path and verify the image exists; open and convert the image to RGB; obtain original or generated text and apply task-specific cleaning; process image and text with `CLIPProcessor`; truncate or pad text to 77 tokens; encode the target label; and remove invalid samples through the custom collate function. **Output:** `pixel_values`, `input_ids`, `attention_mask`, and an integer label.

Algorithm 2: CLIP Feature Extraction

CLIP contains a visual encoder f_v and a text encoder f_t . For image I_i , $\mathbf{v}_i = f_v(I_i)$; for text T_i , $\mathbf{t}_i = f_t(T_i)$. Both projected features are L_2 -normalised and have 512 dimensions.

$$\hat{\mathbf{v}}_i = \mathbf{v}_i / \|\mathbf{v}_i\|_2 \quad \hat{\mathbf{t}}_i = \mathbf{t}_i / \|\mathbf{t}_i\|_2 \quad (3.25)$$

Input: processed image and tokenised text. **Output:** normalised image and text embeddings.

Algorithm 3: Zero-Shot Prompt-Ensemble Classification

Zero-shot classification uses class descriptions without updating CLIP. For class c , a prompt set $\mathcal{P}_c = \{p_{c,1}, \dots, p_{c,M}\}$ is encoded and normalised, and the class prototype is the nor-

malised mean of the prompt embeddings:

$$\mathbf{q}_c = \frac{(1/M) \sum_m \mathbf{e}_{c,m}}{\|(1/M) \sum_m \mathbf{e}_{c,m}\|_2} \quad (3.26)$$

Image–class similarity uses CLIP’s learned logit-scale τ , $s_{i,c} = \exp(\tau) \cdot \hat{\mathbf{v}}_i^\top \mathbf{q}_c$, and the predicted class is $\hat{y}_i = \arg \max_c s_{i,c}$. **Input:** disaster image and class prompts. **Output:** similarity scores and predicted class. **Trainable parameters:** none. This zero-shot branch performs image-to-prompt classification and does not fuse the original tweet with the image.

Algorithm 4: Bottleneck Adapter Fine-Tuning

A bottleneck adapter modifies hidden state $\mathbf{h}$, with a residual connection preserving the original representation:

$$A(\mathbf{h}) = W_{\text{up}}[\sigma(W_{\text{down}} \cdot \text{LN}(\mathbf{h}))] \quad \mathbf{h}' = \mathbf{h} + A(\mathbf{h}) \quad (3.27)$$

Here W_{down} reduces dimension, W_{up} restores it, σ is GELU or another nonlinearity, and LN is layer normalisation; the up-projection may be initialised near zero so the adapter initially behaves as identity. **Trainable components:** adapter layers, fusion layers, and the classification head; **frozen:** most pretrained CLIP parameters. **Input:** intermediate transformer representation. **Output:** adapted task-specific representation.

Algorithm 5: Low-Rank Adaptation (LoRA)

LoRA assumes task-specific changes can be represented by a low-rank update to a pretrained matrix W_0 , which itself remains frozen:

$$W' = W_0 + \Delta W, \quad \Delta W = (\alpha/r) \cdot BA, \quad A \in \mathbb{R}^{r \times d}, \quad B \in \mathbb{R}^{k \times r} \quad (3.28)$$

Here r is the LoRA rank and α the scaling parameter. **Trainable components:** LoRA matrices, task-specific fusion, and the classification head. **Input:** attention-layer representation. **Output:** attention output modified by the low-rank update. The final task-specific configurations, which were tuned independently rather than reused across tasks, were as follows:

- informativeness LoRA: rank 8, alpha 16, applied to query, key, value, and output;
- humanitarian enhanced LoRA: rank 128, alpha 256, applied to query and value;
- deployed severity LoRA: rank 32, alpha 64, applied to query and value;
- MEDIC humanitarian/severity LoRA: rank 8, alpha 8, applied to all four projections.

Algorithm 6: Informativeness Classification

The informativeness model concatenates normalised image and text features, $\mathbf{h}_i^{\text{inf}} = [\hat{\mathbf{v}}_i; \hat{\mathbf{t}}_i] \in \mathbb{R}^{1024}$, passed through a classifier structured as $1024 \rightarrow 512 \rightarrow 128 \rightarrow 2$ with linear layers, batch normalisation, ReLU, and dropout. **Input:** image and tweet text. **Output:** Informative/Not Informative logits. Direct concatenation was chosen because informativeness is binary and either modality may provide sufficient evidence.

Algorithm 7: Humanitarian Gated Fusion

Image and text features are separately projected, $\mathbf{v}'_i = g_v(\hat{\mathbf{v}}_i)$ and $\mathbf{t}'_i = g_t(\hat{\mathbf{t}}_i)$, and a softmax gate computed from the concatenated raw features determines their relative contribution:

$$[\alpha_i^t, \alpha_i^v] = \text{Softmax}(W_2 \cdot \sigma(W_1[\hat{\mathbf{t}}_i; \hat{\mathbf{v}}_i])), \quad \alpha_i^t + \alpha_i^v = 1 \quad (3.29)$$

The fused feature is $\mathbf{h}_i^{\text{hum}} = \text{LN}(\alpha_i^t \mathbf{t}'_i + \alpha_i^v \mathbf{v}'_i)$, passed to a five-class classifier. **Input:** 512-D image and text embeddings. **Output:** five-class humanitarian prediction plus modality weights — useful because humanitarian evidence may be text-dominant, image-dominant, or balanced depending on the post.

Algorithm 8: Severity Symmetric Fusion

The deployed severity model uses mutual image–text verification via scaled dot-product cross-attention, $\text{Attention}(Q, K, V) = \text{Softmax}(QK^\top / \sqrt{d})V$, where text queries the image ($\mathbf{a}_t = \text{Attention}(\mathbf{t}, \mathbf{i}, \mathbf{i})$) and the image queries text ($\mathbf{a}_i = \text{Attention}(\mathbf{i}, \mathbf{t}, \mathbf{t})$). The attended representations are concatenated, $\mathbf{u} = [\mathbf{a}_t; \mathbf{a}_i]$, gated by $\mathbf{g} = \sigma(W_g \mathbf{u} + b_g)$, and combined with a learnable-scale residual:

$$\mathbf{h} = \text{LN}(\mathbf{u}_{\text{original}} + \gamma(\mathbf{g} \odot \mathbf{u})) \quad (3.30)$$

The classifier structure is $1024 \rightarrow 512 \rightarrow 128 \rightarrow 3$. **Input:** image and text embeddings. **Output:** Severe, Mild, or None.

Algorithm 9: MEDIC Severity Feature-Level Fusion

The MEDIC severity retraining model projects image and text features ($\mathbf{v}_i^p = W_v \hat{\mathbf{v}}_i + b_v$, $\mathbf{t}_i^p = W_t \hat{\mathbf{t}}_i + b_t$), concatenates them ($\mathbf{c}_i = [\mathbf{v}_i^p; \mathbf{t}_i^p]$), and applies a 1,024-dimensional sigmoid gate $\mathbf{g}_i = \sigma(W_{g2} \cdot \text{ReLU}(W_{g1} \mathbf{c}_i + b_{g1}) + b_{g2})$, giving the gated feature $\mathbf{h}_i^{\text{sev}} = \mathbf{g}_i \odot \mathbf{c}_i$. **Input:** image and neutral disaster-context text. **Output:** three-class severity prediction.

Algorithm 10: Partial CLIP Fine-Tuning

Partial fine-tuning updates selected upper CLIP layers. Splitting parameters into frozen lower layers Θ_f , unfrozen upper layers Θ_u , and the classifier Θ_h , only the latter two are updated:

$$\Theta_u \leftarrow \Theta_u - \eta_u \nabla_{\Theta_u} L \quad \Theta_h \leftarrow \Theta_h - \eta_h \nabla_{\Theta_h} L \quad (3.31)$$

The lower layers remain unchanged. The informativeness implementation unfreezes the final visual and textual transformer [35] layers and uses a smaller learning rate for CLIP than for the classifier. **Input:** image and tweet text. **Output:** binary informativeness logits.

Algorithm 11: Cross-Entropy and Label Smoothing

Normal cross-entropy is $L_{CE} = -(1/N) \sum \log p_{i,y_i}$. With label smoothing, the soft target is $q_{i,c} = 1 - \varepsilon$ for $c = y_i$ and $\varepsilon/(C - 1)$ for $c \neq y_i$, giving the smoothed loss $L_{LS} = -(1/N) \sum_i \sum_c q_{i,c} \log p_{i,c}$. Label smoothing reduces extreme confidence and may improve generalisation [26].

Algorithm 12: Focal Loss

Focal loss is $L_{FL} = -\alpha_y(1 - p_y)^\gamma \log p_y$ [25]. The factor $(1 - p_y)^\gamma$ reduces the contribution of easy examples, which is useful for the highly imbalanced humanitarian task.

Algorithm 13: AdamW Optimisation

AdamW updates parameters through [23] $\theta_{t+1} = \theta_t - \eta \cdot \hat{m}_t / (\sqrt{\hat{v}_t} + \varepsilon) - \eta \lambda \theta_t$, where η is the learning rate, $\hat{m}_t$ and $\hat{v}_t$ are the bias-corrected first and second moments, and λ is weight decay. AdamW separates weight decay from the adaptive gradient update.

Algorithm 14: Grad-CAM-Style Vision Saliency

Let $H_{p,d}$ be activation dimension d for image patch p . For selected class c , the patch importance is:

$$S_p = \text{ReLU} \left(\sum_d \left[\frac{\partial P(c)}{\partial H_{p,d}} \right] \cdot H_{p,d} \right) \quad (3.32)$$

The class token is excluded, remaining patch scores are reshaped into a spatial grid, normalised, and resized to the original image as a heatmap overlay. **Input:** trained model, image, text, and selected class. **Output:** image-region saliency map.

Mixed-Precision Training

Mixed-precision training uses lower-precision arithmetic for selected operations: a forward pass under automatic casting, scaled loss calculation, backward pass, gradient unscaling, optimiser update, and scale adjustment. This reduces GPU-memory use and may increase training speed.

Gradient Clipping

Gradient clipping limits the gradient norm, $\|\mathbf{g}\|_2 \leq g_{\max}$; the informativeness pipeline used $g_{\max} = 1.0$, reducing unstable parameter updates.

Checkpoint Selection

After every epoch: training loss is calculated; the model is evaluated on development data; the selection metric is calculated; performance is compared with the best previous score; and an improved checkpoint is saved. For humanitarian classification, macro-F1 is the preferred selection criterion; for severity, accuracy and macro-F1 should be examined together; for informativeness, checkpoint selection should use the official development set.

Main Training Configuration

Table 3.3. Main Training Configuration by Task

Configuration	Humanitarian	Severity	Informativeness
Backbone	CLIP ViT-B/32	CLIP ViT-B/32	CLIP ViT-B/32
Classes	5	3	2
Batch size	16	16	32
Text length	77	77	77
Optimiser	AdamW	AdamW	AdamW
Main loss	Focal / weighted CE	Label-smoothed CE	Label-smoothed CE
Main fusion	Modality gate	Symmetric/gated fusion	Concatenation
Checkpoint criterion	Dev macro-F1	Dev accuracy & F1	Dev accuracy/F1
Precision	Mixed	Mixed	Standard/mixed

MEDIC humanitarian/severity retraining used the following common configuration:

- batch size 32, 5 epochs;
- learning rate 5×10^{-5} , weight decay 0.01;
- AdamW optimiser with automatic mixed precision [23, 24].

LoRA configurations remained task-specific rather than universal:

- informativeness LoRA: rank 8, alpha 16, on query, key, value, and output;
- enhanced humanitarian LoRA: rank 128, alpha 256, on query and value;
- deployed severity LoRA: rank 32, alpha 64, on query and value;
- MEDIC humanitarian/severity LoRA: rank 8, alpha 8, on all four projections.

These differences reflect the fact that each task was tuned independently rather than reused wholesale from another task’s configuration.

3.3 Implementation

3.3.1 Dataset Selection

Reason for Selecting CrisisMMD

CrisisMMD was selected as the primary dataset because it provides paired social-media images and tweet text with annotations for informativeness, humanitarian category, and damage severity [7,8]. These features make it suitable for CLIP-based multimodal disaster classification. Only raw images and tweet text were used as model inputs. Informativeness text was cleaned more strongly, while humanitarian and severity text was passed more directly to the CLIP processor.

Reason for Selecting MEDIC

MEDIC was used as a secondary dataset for retraining, cross-dataset testing, and out-of-distribution evaluation [15]. Original image and tweet text were used for informativeness evaluation. For humanitarian and severity retraining, neutral disaster-context prompts, such as “An image capturing the aftermath of a flood”, were generated to provide general context without revealing the target class.

3.3.2 Dataset Description

Two disaster datasets were used in this study. CrisisMMD was used for primary model development and comparison, while MEDIC was used for additional retraining and OOD evaluation. Both datasets contain disaster images and task-specific annotations, but MEDIC used neutral disaster-context prompts for humanitarian and severity retraining.

CrisisMMD Dataset

CrisisMMD is a multimodal disaster dataset containing social-media image–text pairs with labels for informativeness, humanitarian category, and damage severity [7,8]. The official train, development, and test splits were preserved where available. The dataset is suitable

for this thesis because it supports multimodal classification, task-specific evaluation, and explainability analysis.

CrisisMMD Variables

Table 3.4. CrisisMMD Dataset Variables

Variable	Type	Description	Use
image	String/path	Relative or full image path	Loads the disaster image
tweet_text	Text	Associated tweet or caption	Input to the CLIP text encoder
label	Categorical	Humanitarian or severity target	Main task label
label_image	Categorical	Image informativeness annotation	Main informativeness target
label_text	Categorical	Text informativeness annotation	Auxiliary/annotation comparison
event_name	Categorical	Disaster-event identifier	Descriptive meta-data
tweet_id / image_id	Identifier	Unique post/image identifier	Record checking only
encoded label	Integer	Numeric class index	Input to loss function

The main predictive inputs were $X = \{\text{image pixels, tweet text}\}$; identifier fields and target annotations were not used as model features.

CrisisMMD Data Splits

Table 3.5. CrisisMMD Data Splits by Task

Task	Training	Development	Testing	Total
Informativeness	13,608	2,237	2,237	18,082
Humanitarian category	6,126	998	955	8,079
Damage severity	2,468	529	529	3,526

The humanitarian task uses an agreed-label subset where image and text annotations were made consistent after category merging. Although a development file exists for informativeness, parts of the experimental notebook reused the test loader during development; a strict final evaluation should use the development set for checkpoint selection and reserve the test set for a single final evaluation.

Humanitarian Target Classes

The humanitarian task contains five classes, mapped from the original CrisisMMD labels as follows.

Table 3.6. Humanitarian Target Class Mapping

Original Label	Code	Thesis Label	Description
infrastructure_and_utility_damage	0	Infrastructure and Utility Damage	Damage to buildings, roads, bridges, power, or communication systems
other_relevant_information	1	Other Relevant Information	Disaster-related information not covered by another category
not_humanitarian	2	Not Humanitarian	Content without useful humanitarian information
rescue_volunteering_or_donation_effort	3	Rescue/Volunteering/Donation Effort	Rescue, evacuation, volunteering, donation, or relief activity
affected_individuals	4	Affected Individuals	Displaced, injured, trapped, missing, dead, or otherwise affected people

Humanitarian-Class Distribution

Table 3.7. CrisisMMD Humanitarian-Class Distribution

Class	Training	Development	Testing	Total	Train %
Infrastructure and Utility Damage	612	80	81	773	9.99%
Other Relevant Information	1,279	239	235	1,753	20.88%
Not Humanitarian	3,252	521	504	4,277	53.09%
Rescue/Volunteering/Donation	912	149	126	1,187	14.89%
Affected Individuals	71	9	9	89	1.16%
Total	6,126	998	955	8,079	100%

The distribution is strongly imbalanced: over half the training samples fall under Not Humanitarian, while Affected Individuals represents only 1.16%. The imbalance ratio $IR = n_{\max}/n_{\min} = 3252/71 \approx 45.80$ — the largest class has almost 46 times as many training records as the smallest. Accuracy alone is therefore not an appropriate selection measure; macro-F1, class-wise recall, and confusion matrices are required.

Damage-Severity Target Classes

The deployment label *None* represents the original *Little or No Damage* category.

Table 3.8. Damage-Severity Target Class Mapping

Original Label	Code	Thesis Label	Description
severe_damage	0	Severe	Major destruction, collapse, or unusable infrastructure
mild_damage	1	Mild	Partial or limited physical damage
little_or_no_damage	2	None	No visible damage or only minor effects

Damage-Severity Distribution

Table 3.9. CrisisMMD Damage-Severity Distribution

Class	Training	Development	Testing	Total	Train %
Severe	1,548	332	332	2,212	62.72%
Mild	587	126	126	839	23.78%
None	333	71	71	475	13.49%
Total	2,468	529	529	3,526	100%

The severity imbalance ratio is $1548/333 \approx 4.65$ — smaller than the humanitarian task but still substantial, with Severe as the majority class and None the smallest.

Informativeness Target Classes

The main informativeness target was created from `label_image: not_informative` (0) $\rightarrow$ Not Informative, and `informative` (1) $\rightarrow$ Informative. The `label_text` field was retained only for annotation-level analysis and was not used as an input to the deployable multimodal model.

MEDIC Dataset

MEDIC contains disaster-related images and labels for informativeness, humanitarian category, damage severity, and disaster type [15]. In the supplied implementation, MEDIC was used in three ways: the CrisisMMD-trained informativeness model was evaluated directly on the MEDIC OOD test set; the humanitarian model was retrained using MEDIC training and development data; and the severity model was retrained using MEDIC data and evaluated on the OOD split. The MEDIC files used were `MEDIC_train.tsv`, `MEDIC_dev.tsv`, and `MEDIC_test_OOD_only.tsv`; the out-of-distribution test split contained 10,846 valid samples after task-specific filtering.

Table 3.10. MEDIC Dataset Variables

Variable	Type	Description	Use
image_path	String/path	Relative image location	Image loading
tweet_text	Text	Original associated text	Informativeness OOD evaluation
informative	Categorical/list-like	Informativeness annotation	Informativeness target
humanitarian	Categorical/list-like	Humanitarian annotation	Humanitarian target
damage_severity	Categorical/list-like	Severity annotation	Severity target
disaster_types	Categorical/list-like	Disaster category	Builds neutral prompts
label_clean	Derived categorical	Normalised target label	Filtering and encoding

MEDIC Variables

Some MEDIC annotations were stored as strings representing lists; the implementation used safe parsing to extract the first class value, converting unsupported values into an `unknown` class that was then removed.

MEDIC Text Strategy

The original `tweet_text` was used for informativeness evaluation. For humanitarian and severity retraining, a neutral text prompt was generated from `disaster_types`, $T_i =$ “An image capturing the aftermath of a [disaster type].” — for example, “An image capturing the aftermath of a flood.” This provides general event context without directly revealing whether the image belongs to infrastructure damage, affected individuals, rescue activity, or a particular severity class, reducing the possibility of target leakage from strongly descriptive captions.

MEDIC Target Mapping

The project harmonised MEDIC humanitarian labels with the five-class CrisisMMD output structure. The affected/injured/dead class is broader than the CrisisMMD Affected Individuals category; the thesis label Affected Individuals is used for consistency, but this difference should be acknowledged during interpretation.

MEDIC Training and Test Usage

After label and image filtering, 49,353 MEDIC samples were reported for humanitarian and severity training.

Table 3.11. MEDIC Target Label Mapping

Task	Original MEDIC Label	Thesis Label
Informativeness	not_informative / informative	Not Informative / Informative
Humanitarian	affected_injured_or_dead_people	Affected Individuals
Humanitarian	infrastructure_and_utility_damage	Infrastructure and Utility Damage
Humanitarian	not_humanitarian	Not Humanitarian
Humanitarian	other_relevant_information	Other Relevant Information
Humanitarian	rescue_volunteering_or_donation_effort	Rescue/Volunteering/Donation Effort
Severity	little or none / mild / severe	None / Mild / Severe

Table 3.12. MEDIC Training and Test Usage Summary

Task	Training Method	Development Usage	OOD Size	Test
Informativeness	CrisisMMD-trained model evaluated directly	Not used for retraining	10,846	
Humanitarian category	Retrained on MEDIC training data	Used for checkpoint selection	10,846	
Damage severity	Retrained on MEDIC training data	Used in development/evaluation	10,846	

MEDIC OOD Class Support

Table 3.13. MEDIC OOD Test Class Support

Task	Class	OOD Samples
Informativeness	Not Informative / Informative	5,561 / 5,285
Humanitarian	Affected/Injured/Dead	521
Humanitarian	Infrastructure Damage	3,916
Humanitarian	Not Humanitarian	6,138
Humanitarian	Other Relevant Information	0
Humanitarian	Rescue/Donation	271
Severity	None / Mild / Severe	6,823 / 1,097 / 2,926

The informativeness OOD set is relatively balanced. The humanitarian OOD set contains zero Other Relevant Information samples, so five-class macro-F1 must be interpreted cautiously; the severity OOD distribution is also imbalanced, with None as the largest category.

Comparison of CrisisMMD and MEDIC

Table 3.14. Comparison of CrisisMMD and MEDIC Datasets

Characteristic	CrisisMMD	MEDIC
Main role	Primary model development	Retraining and OOD evaluation
Image / text variable	image / tweet_text	image_path / tweet_text or neutral prompt
Informativeness target	label_image	informative
Humanitarian / severity target	label	humanitarian / damage_severity
Split structure	Official train/dev/test	Train/dev/OOD-test
Main challenge	Strong class imbalance	Distribution shift, missing OOD classes
Use in deployment	Source of final checkpoints	External/retraining analysis

Missing and Invalid Data

The notebooks did not print a complete variable-by-variable missing-value table, so exact missing counts are not reported; only the applied handling rules are described:

- missing or unsupported labels were converted to `unknown/missing` and removed;
- missing image paths were removed before loading;
- images that could not be found or opened were excluded as invalid;
- empty text was converted to a safe string where appropriate;
- invalid MEDIC list labels were parsed, normalised, or removed;
- empty batches were protected by custom collation logic.

The missing rate for variable j is $\text{Missing Rate}_j = (m_j/N) \times 100$, where m_j is the number of missing values in variable j .

3.3.3 Preprocessing Implementation

Duplicate Removal

Duplicate or near-duplicate posts across training and testing splits can cause data leakage. Possible exact-duplicate keys include `tweet_id`, `image_id`, `tweet_text`, `image path`, or a combined hash $D_i = \text{hash}(\text{normalised text}_i, \text{image identifier}_i)$. The provided implementation used the official dataset splits without an additional `drop_duplicates()` step,

so no project-level duplicate-removal count is reported. Therefore, the methodology does not claim a specific removal count unless the code produced it. Future verification could use `df.duplicated(subset=["tweet_id", "image_id"])` or perceptual image hashing for near-duplicates.

Missing-Value Handling

Unsupported target labels, missing cleaned labels, missing image paths, unavailable/corrupted images, and invalid MEDIC list values were all removed or normalised, and empty batches were protected by custom collation logic, formally:

$$\mathcal{D}' = \mathcal{D} \setminus \{x_i : \text{label_clean}_i = \text{NaN} \vee \text{image_path}_i = \text{NaN}\} \quad (3.33)$$

The dataset loader attempted to open every image and returned an invalid sample on failure; the custom collate function then dropped invalid samples before stacking tensors, preventing a single corrupted file from halting training.

Class-Imbalance Handling

Class imbalance was addressed differently for each task.

Humanitarian Focal Loss: $\text{FL}(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t)$, where p_t is the probability of the true class, α_t is the class weight, and γ controls emphasis on difficult examples. When p_t is high, $(1 - p_t)^\gamma$ becomes small, so easy examples contribute less to the total loss and rare or difficult classes receive greater emphasis.

Weighted Cross-Entropy: MEDIC humanitarian training used inverse-frequency class weights, $w_c = 1/n_c$, giving the weighted loss $L_{\text{WCE}} = -(1/N) \sum w_{y_i} \log p_{i,y_i}$, which increases the penalty for errors in smaller classes.

Weighted Sampling: the severity pipeline also used weighted sampling in selected experiments, with $s_c = q_c/n_c$, where q_c is the desired sampling proportion and n_c the observed count — changing the probability that a training record appears in a batch.

Label Smoothing: informativeness and severity experiments used label smoothing, where the soft target q_c is $1 - \varepsilon$ for $c = y$ and $\varepsilon/(C - 1)$ otherwise, reducing overconfidence and potentially improving generalisation.

Macro-F1 Model Selection: $\text{macro-F1} = (1/C) \sum F1_c$ was used when minority-class performance was important, preventing the majority class from completely controlling the evaluation.

Feature Removal

Only fields available at real inference time were used as predictive inputs; tweet/image IDs, split indicators, unrelated event identifiers, raw path strings, target labels, and target-derived variables were excluded, giving the following per-model input sets:

$$X_{\text{CrisisMMD}} = \{\text{image}, \text{tweet_text}\} \quad (3.34)$$

$$X_{\text{MEDIC-info}} = \{\text{image_path}, \text{tweet_text}\} \quad (3.35)$$

$$X_{\text{MEDIC-H/S}} = \{\text{image_path}, \text{neutral disaster prompt}\} \quad (3.36)$$

An annotation-based Random Forest baseline using `label_image` and `label_text` contained target leakage (since its target derives from `label_image`) and was not treated as a deployable model.

Text Preprocessing

Text preprocessing differed by task rather than following one fixed pipeline. The informativeness notebook applied a fairly aggressive cleaning procedure:

- lowercasing;
- HTML, URL, and mention removal;
- hashtag-symbol stripping;
- bracketed-placeholder and unsupported-character removal;
- English stop-word removal;
- whitespace normalisation.

In contrast, the humanitarian and severity notebooks passed text more directly to the CLIP processor, with little additional cleaning. This distinction matters because CLIP was pre-trained on naturally occurring captions, so aggressive cleaning risks removing punctuation, numbers, hashtags, and emergency expressions that carry meaning.

The Streamlit application applies a moderate cleaning procedure instead:

- lowercasing;
- URL, mention, hashtag-symbol, and retweet-marker removal;
- unsupported-character removal;
- whitespace normalisation.

A controlled comparison of raw versus cleaned text on the development set is recommended for future work.

Image Preprocessing

Each valid image was loaded from its path, converted to RGB, passed to `CLIPProcessor`, resized and cropped according to CLIP's configuration, converted into a tensor, and normalised using CLIP statistics. CLIP ViT-B/32 uses an input resolution of approximately 224×224 ; with a patch size of 32×32 , the image is divided into $7 \times 7 = 49$ visual patches, in addition to the class token. No manual rotation, flipping, colour jitter, or random-crop augmentation was part of the main comparative implementation, so these operations are not claimed.

CLIP Tokenisation

The CLIP processor tokenises text and produces `input_ids` and an `attention_mask`. Text is padded or truncated to $L_{\max} = 77$ tokens, and the `attention_mask` identifies real tokens versus padded positions. Special tokens are removed during explanation visualisation so that the plots show only meaningful user text.

Label Encoding

Categorical labels were mapped into integer indices, $y_i = g(l_i)$, where l_i is the original string label, g is the label-mapping function, and $y_i \in \{0, \dots, C - 1\}$. The same mapping must be used for training, development, testing, confusion matrices, classification reports, and Streamlit output — changing the label order would display incorrect class names even if the underlying logits were correct.

Statistical Analysis

Before training, descriptive statistics were checked for the following aspects of the data:

- dataset size;
- class distribution;
- text length;
- missing data;
- image availability and dimensions;
- duplicate records;
- event distribution;
- annotation agreement;
- class association;

- model uncertainty.

For a numerical variable x , the mean is $\bar{x} = (1/n) \sum x_i$ and the sample standard deviation is $s = \sqrt{\sum (x_i - \bar{x})^2 / (n - 1)}$. For skewed variables such as text length, the median and interquartile range should also be reported.

Preprocessing Summary

After preprocessing, each sample contains `input_ids`, `attention_mask`, `pixel_values`, and `label`; for explainability, the system may additionally retain raw text, the original image, and the image path, used only for visualisation and not supplied as hidden predictive metadata. The preprocessing objectives were to preserve valid image–text pairs, exclude unsupported labels, handle corrupted files safely, control class imbalance, maintain consistent mappings, avoid target leakage, and produce CLIP-compatible tensors.

3.3.4 Model Selection

Selection Criteria

Model comparison and selection were guided by the following criteria:

- ability to process image and text together;
- availability of pretrained multimodal knowledge;
- support for zero-shot, adapter, and LoRA-based adaptation;
- computational and memory efficiency;
- robustness against overfitting and class imbalance;
- explainability of predictions;
- compatibility with Streamlit-based deployment; and
- generalisation to out-of-distribution MEDIC data.

Since the humanitarian and severity datasets were imbalanced, macro-F1 was given greater importance than accuracy alone, along with weighted F1-score and class-wise performance.

Selected Model

CLIP ViT-B/32 (`openai/clip-vit-base-patch32`) was selected as the main multimodal backbone. Task-specific fusion and classification heads were added for informativeness, humanitarian-category, and severity classification. LoRA was selected as the main parameter-efficient method because it updates only small low-rank matrices while keeping most CLIP parameters frozen.

The task-specific LoRA settings were:

- informativeness: rank 8, alpha 16, dropout 0.1;
- CrisisMMD humanitarian model: rank 128, alpha 256, dropout 0.05;
- MEDIC humanitarian and severity models: rank 8, alpha 8, dropout 0.1.

Comparison of Adaptation Approaches

Zero-shot CLIP requires no training but depends strongly on prompt quality. Full fine-tuning offers stronger adaptation but needs more memory, time, and trainable parameters. Adapter tuning adds useful bottleneck modules but increases model complexity. LoRA keeps the original CLIP structure and adds lightweight low-rank updates, producing smaller checkpoints and lower training cost. Therefore, CLIP ViT-B/32 with LoRA and task-specific fusion was selected for its balance of accuracy, efficiency, explainability, and deployment suitability.

3.3.5 Model Description

The system architecture is organised into five layers: the user-input layer, the preprocessing layer, the CLIP feature-extraction layer, the task-specific fusion and classification layer, and the application and explanation layer.

User-Input Layer

The user-input layer accepts an image I and a caption or tweet text T through the Streamlit interface.

CLIP Feature-Extraction Layer

The selected backbone is `openai/clip-vit-base-patch32` [18, 20]. The image I_i and text T_i are first converted into CLIP-compatible tensors by the CLIP processor:

$$X_i^v = \text{Processor}_v(I_i), \quad X_i^t = \text{Processor}_t(T_i). \quad (3.37)$$

The image tensor is passed through the CLIP vision encoder f_v , and the tokenised text is passed through the CLIP text encoder f_t :

$$\mathbf{v}_i = f_v(X_i^v; \theta_v), \quad \mathbf{t}_i = f_t(X_i^t; \theta_t), \quad (3.38)$$

where θ_v and θ_t are the visual and textual encoder parameters. The projected image and text embeddings are then L_2 -normalised:

$$\hat{\mathbf{v}}_i = \frac{\mathbf{v}_i}{\|\mathbf{v}_i\|_2 + \epsilon}, \quad \hat{\mathbf{t}}_i = \frac{\mathbf{t}_i}{\|\mathbf{t}_i\|_2 + \epsilon}. \quad (3.39)$$

In CLIP ViT-B/32, both projected embeddings are 512-dimensional. These normalised features are the shared inputs to the zero-shot baseline, adapter/LoRA fine-tuned models, task-specific fusion layers, and explainability modules.

Fusion and Classification Layer

Table 3.15 summarises the fusion method and classification-head structure used for each task.

Table 3.15. Fusion Method and Classification Structure by Task

Task	Fusion Method	Classification Structure
Informativeness	Image–text concatenation	$1024 \rightarrow 512 \rightarrow 128 \rightarrow 2$
Humanitarian	Softmax-gated modality fusion	$512 \rightarrow 512 \rightarrow 5$
Damage severity	Sigmoid-gated fusion (feature level)	$1024 \rightarrow 512 \rightarrow 128 \rightarrow 3$

For informativeness classification, the two 512-dimensional CLIP embeddings are concatenated into a 1,024-dimensional vector,

$$\mathbf{z}_i^{\text{inf}} = [\hat{\mathbf{v}}_i; \hat{\mathbf{t}}_i] \in \mathbb{R}^{1024}, \quad (3.40)$$

and the classifier produces two logits:

$$\mathbf{o}_i^{\text{inf}} = g_{\text{inf}}(\mathbf{z}_i^{\text{inf}}). \quad (3.41)$$

For humanitarian-category classification, image and text embeddings are first projected into a common hidden space,

$$\mathbf{v}_i^p = W_v \hat{\mathbf{v}}_i + b_v, \quad \mathbf{t}_i^p = W_t \hat{\mathbf{t}}_i + b_t. \quad (3.42)$$

A softmax gate learns the relative contribution of both modalities:

$$[\alpha_i^v, \alpha_i^t] = \text{Softmax}(W_g[\hat{\mathbf{v}}_i; \hat{\mathbf{t}}_i] + b_g), \quad \alpha_i^v + \alpha_i^t = 1. \quad (3.43)$$

The fused humanitarian feature and logits are

$$\mathbf{z}_i^{\text{hum}} = \text{LayerNorm}(\alpha_i^v \mathbf{v}_i^p + \alpha_i^t \mathbf{t}_i^p), \quad \mathbf{o}_i^{\text{hum}} = g_{\text{hum}}(\mathbf{z}_i^{\text{hum}}). \quad (3.44)$$

For damage-severity classification, a feature-level gate is used after concatenating projected image and text features:

$$\mathbf{c}_i = [\mathbf{v}_i^p; \mathbf{t}_i^p], \quad \mathbf{g}_i = \sigma(W_{g2} \text{ReLU}(W_{g1} \mathbf{c}_i + b_{g1}) + b_{g2}). \quad (3.45)$$

The gated severity representation and logits are

$$\mathbf{z}_i^{\text{sev}} = \mathbf{g}_i \odot \mathbf{c}_i, \quad \mathbf{o}_i^{\text{sev}} = g_{\text{sev}}(\mathbf{z}_i^{\text{sev}}). \quad (3.46)$$

Thus, the implementation keeps one common CLIP feature-extraction backbone but uses task-specific mathematical fusion functions and task-specific output heads.

Application and Explanation Layer

This layer loads the selected model checkpoints, converts the classification logits into probabilities, and returns the results to the user interface. The explanation layer provides:

- Grad-CAM-style image saliency;
- text-token importance;
- Gradient $\times$ Input text attribution;
- image-region occlusion;
- word-deletion importance;
- multimodal-fusion contribution; and
- modality-ablation analysis.

Image explanations are rendered as heatmap overlays on the uploaded image, and text explanations are shown as token-importance bar charts and beeswarm-style contribution plots.

Informativeness Model

The informativeness classifier determines whether a multimodal social-media post contains disaster-relevant information. A 512-D image embedding and a 512-D text embedding are normalised and concatenated into a 1,024-D vector, which is passed through the classification head

$$1024 \rightarrow 512 \rightarrow 128 \rightarrow 2$$

implemented as `Linear(1024, 512)`, batch normalisation, `ReLU`, `Dropout(p=0.3)`, `Linear(512, 128)`, `ReLU`, and a final `Linear(128, 2)` output layer, producing the Not Informative and Informative classes.

Humanitarian-Category Model

The humanitarian model characterises the type of humanitarian information present in an informative disaster post. Image and text features are independently mapped to 512 dimensions and passed to a gating network that outputs two weights, one per modality, summing to one. The fused representation then passes through the classification head

$$512 \rightarrow 512 \rightarrow 5$$

consisting of batch normalisation, a nonlinear activation, and dropout, followed by a final layer with five outputs corresponding to Infrastructure and Utility Damage, Other Relevant Information, Not Humanitarian, Rescue/Volunteering/Donation Effort, and Affected Individuals.

Damage-Severity Model

The severity model estimates the level of visible physical damage in a disaster image. Image and text features are projected separately to 512 dimensions and concatenated into a 1,024-dimensional vector; a sigmoid gating network learns a separate weight for each feature dimension, and the gated vector is passed to the classification head

$$1024 \rightarrow 512 \rightarrow 128 \rightarrow 3$$

producing the three output classes Severe, Mild, and None. The MEDIC retraining of this model used neutral disaster-context text so that the severity prediction relies primarily on the image rather than on descriptive text.

3.3.6 Model Implementation

Dataset Implementation

Custom `PyTorch Dataset` classes were built for the three CrisisMMD tasks and the three MEDIC tasks. Each dataset class performs the following operations:

- reads the corresponding TSV file;
- cleans and maps the task label;
- removes unsupported classes;
- removes samples with a missing image path;

- builds the full image path;
- opens the image and converts it to RGB;
- obtains the original or generated text;
- applies `CLIPProcessor`;
- returns image tensors, token IDs, attention mask, and label.

A custom collation function ignores samples that could not be loaded and dynamically pads text sequences within each batch.

Zero-Shot Implementation

The zero-shot implementation defines a set of text-description templates for each class. For class c , the prompt set is written as

$$\mathcal{P}_c = \{p_{c,1}, p_{c,2}, \dots, p_{c,K}\}, \quad (3.47)$$

where K is the number of prompt templates for that class. Each prompt is encoded using the CLIP text encoder and then normalised:

$$\mathbf{e}_{c,k} = \frac{f_t(p_{c,k})}{\|f_t(p_{c,k})\|_2 + \epsilon}. \quad (3.48)$$

The class prototype is obtained by averaging the prompt embeddings and normalising the result:

$$\mathbf{q}_c = \frac{\frac{1}{K} \sum_{k=1}^K \mathbf{e}_{c,k}}{\left\| \frac{1}{K} \sum_{k=1}^K \mathbf{e}_{c,k} \right\|_2 + \epsilon}. \quad (3.49)$$

At inference time, the image embedding $\hat{\mathbf{v}}_i$ is compared with each class prototype using CLIP's logit scale τ :

$$s_{i,c} = \exp(\tau) \hat{\mathbf{v}}_i^\top \mathbf{q}_c. \quad (3.50)$$

The predicted label is selected by maximum similarity:

$$\hat{y}_i = \arg \max_c s_{i,c}. \quad (3.51)$$

No model parameter is updated in this branch. Therefore, zero-shot prompting acts as a mathematical training-free baseline before adapter, LoRA, and partial fine-tuning are applied.

Adapter Implementation

Bottleneck adapter modules were added to the pretrained CLIP model while most original CLIP parameters remained frozen. For a transformer hidden state $\mathbf{h}$, the adapter first normalises and down-projects the representation, applies a nonlinear transformation, and then restores the dimensionality:

$$A(\mathbf{h}) = W_{\text{up}} \sigma(W_{\text{down}} \text{LN}(\mathbf{h}) + b_{\text{down}}) + b_{\text{up}}. \quad (3.52)$$

The adapted hidden representation is obtained through a residual connection:

$$\mathbf{h}' = \mathbf{h} + A(\mathbf{h}). \quad (3.53)$$

If d is the original hidden dimension and b is the adapter bottleneck dimension, then $W_{\text{down}} \in \mathbb{R}^{b \times d}$ and $W_{\text{up}} \in \mathbb{R}^{d \times b}$. The set of trainable parameters is

$$\Theta_{\text{train}} = \{W_{\text{down}}, W_{\text{up}}, b_{\text{down}}, b_{\text{up}}, \Theta_{\text{fusion}}, \Theta_{\text{head}}\}, \quad (3.54)$$

while the major pretrained CLIP parameters remain fixed. This makes the adapter implementation a parameter-efficient alternative to full CLIP fine-tuning while still allowing task-specific nonlinear transformations.

LoRA Implementation

LoRA was implemented using the PEFT library's `LoraConfig` and `get_peft_model` utilities. Instead of updating the full pretrained matrix $W_0 \in \mathbb{R}^{k \times d}$, LoRA represents the task-specific update as a low-rank decomposition:

$$W' = W_0 + \Delta W, \quad \Delta W = \frac{\alpha}{r} B A, \quad (3.55)$$

where $A \in \mathbb{R}^{r \times d}$, $B \in \mathbb{R}^{k \times r}$, r is the LoRA rank, and α is the scaling parameter. During training, W_0 remains frozen and only A , B , the task-specific fusion layer, and the classification head are updated:

$$\Theta_{\text{train}} = \{A, B, \Theta_{\text{fusion}}, \Theta_{\text{head}}\}. \quad (3.56)$$

Depending on the task, LoRA was applied to attention projection layers such as `q_proj`, `k_proj`, `v_proj`, and `out_proj`. The implemented task-specific configurations were:

- informativeness: rank 8, alpha 16, dropout 0.1, applied to query, key, value, and output projections;
- CrisisMMD humanitarian model: rank 128, alpha 256, dropout 0.05, applied to query and value projections;

- MEDIC humanitarian and severity retraining: rank 8, alpha 8, dropout 0.1, applied to query, key, value, and output projections.

This formulation keeps the main CLIP backbone largely frozen while allowing attention-level adaptation for disaster-specific classification.

Training Implementation

The general model-training process was:

1. instantiate a CLIP model with the required adaptation layers for the target classification task;
2. load the training and development datasets;
3. compute the mini-batch loss and update the trainable parameters with AdamW;
4. evaluate the checkpoint on the development set after every epoch and save the best-performing model.

For a mini-batch $\mathcal{B}$, the model produces logits $\mathbf{o}_i = g_\theta(I_i, T_i)$ and probabilities

$$p_{i,c} = \frac{\exp(o_{i,c})}{\sum_{j=1}^C \exp(o_{i,j})}. \quad (3.57)$$

The standard supervised objective is cross-entropy,

$$L_{\text{CE}}(\theta) = -\frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \log p_{i,y_i}. \quad (3.58)$$

For label smoothing with smoothing value ε , the soft target $q_{i,c}$ is

$$q_{i,c} = \begin{cases} 1 - \varepsilon, & c = y_i, \\ \varepsilon/(C - 1), & c \neq y_i, \end{cases} \quad (3.59)$$

and the smoothed loss becomes

$$L_{\text{LS}}(\theta) = -\frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \sum_{c=1}^C q_{i,c} \log p_{i,c}. \quad (3.60)$$

For the humanitarian task, focal loss was used to reduce the influence of easy majority-class examples:

$$L_{\text{FL}}(\theta) = -\frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \alpha_{y_i} (1 - p_{i,y_i})^\gamma \log(p_{i,y_i}). \quad (3.61)$$

AdamW was used for parameter updates. With gradient $\mathbf{g}_t = \nabla_{\theta} L_t$, first moment $\hat{\mathbf{m}}_t$, second moment $\hat{\mathbf{v}}_t$, learning rate η , and weight decay λ , the update can be written as

$$\theta_{t+1} = \theta_t - \eta \frac{\hat{\mathbf{m}}_t}{\sqrt{\hat{\mathbf{v}}_t} + \epsilon} - \eta \lambda \theta_t. \quad (3.62)$$

Gradient clipping constrains unstable updates using

$$\|\mathbf{g}\|_2 \leq g_{\max}, \quad (3.63)$$

where the informativeness pipeline used $g_{\max} = 1.0$. The task-specific training configurations were as follows.

Informativeness LoRA: batch size 32, 5 epochs, learning rate 1×10^{-4} , AdamW optimiser, weight decay 0.01, label smoothing ($\epsilon = 0.1$), and gradient clipping (max norm 1.0).

CrisisMMD enhanced humanitarian model: batch size 16, up to 15 epochs, with separate discriminative learning rates for the CLIP encoder, fusion module, classifier head, and adapter components; focal loss with per-class weight tuning; checkpoint selection on the development set.

MEDIC humanitarian retraining: batch size 32, 5 epochs, learning rate 5×10^{-5} , AdamW optimiser, weight decay 0.01, class-weighted cross-entropy loss, mixed-precision training.

MEDIC severity retraining: batch size 32, 5 epochs, learning rate 5×10^{-5} , AdamW optimiser, weight decay 0.01, cross-entropy loss, mixed-precision training.

Checkpoint Management

Checkpoint management was used to preserve the best-performing model during training. After each epoch e , the model was evaluated on the development set and assigned a selection score S_e . The general checkpoint rule was

$$e^* = \arg \max_e S_e, \quad \theta^* = \theta_{e^*}. \quad (3.64)$$

For humanitarian classification, the selection score was mainly macro-F1 because minority classes such as Affected Individuals are easily hidden by overall accuracy:

$$S_e^{\text{hum}} = \text{MacroF1}_e. \quad (3.65)$$

For severity classification, both accuracy and macro-F1 were examined:

$$S_e^{\text{sev}} \in \{\text{Accuracy}_e, \text{MacroF1}_e\}. \quad (3.66)$$

For informativeness classification, accuracy and F1-score were considered while maintaining the informativeness gate required for deployment:

$$S_e^{\text{inf}} \in \{\text{Accuracy}_e, F1_e\}. \quad (3.67)$$

The deployed Streamlit application loads the selected task-specific checkpoints only once per session through resource caching. This reduces repeated loading time and supports more efficient CPU/GPU inference.

Training Configuration

Table 3.16. Main Training Configuration by Task

Configuration	Humanitarian	Severity	Informativeness
Backbone	CLIP ViT-B/32	CLIP ViT-B/32	CLIP ViT-B/32
Classes	5	3	2
Batch size	16	16	32
Text length	77	77	77
Optimiser	AdamW	AdamW	AdamW
Main loss	Focal / weighted CE	Label-smoothed CE	Label-smoothed CE
Main fusion	Modality gate	Symmetric/gated fusion	Concatenation
Checkpoint criterion	Dev macro-F1	Dev accuracy & F1	Dev accuracy/F1
Precision	Mixed	Mixed	Standard/mixed

MEDIC humanitarian/severity retraining used the following common configuration:

- batch size 32, 5 epochs;
- learning rate 5×10^{-5} , weight decay 0.01;
- AdamW optimiser with automatic mixed precision [23, 24].

LoRA configurations remained task-specific rather than universal:

- informativeness LoRA: rank 8, alpha 16, on query, key, value, and output;
- enhanced humanitarian LoRA: rank 128, alpha 256, on query and value;
- deployed severity LoRA: rank 32, alpha 64, on query and value;
- MEDIC humanitarian/severity LoRA: rank 8, alpha 8, on all four projections.

These differences reflect the fact that each task was tuned independently rather than reused wholesale from another task’s configuration.

Model Evaluation Implementation

Trained models were evaluated using accuracy, precision, recall, macro F1-score, weighted F1-score, confusion matrices, receiver operating characteristic curves with area under the curve, and t-SNE feature visualisation. The CrisisMMD test files were used for in-distribution evaluation, while `MEDIC_test_OOD_only.tsv` was used for cross-dataset out-of-distribution evaluation.

For each class c , true positives, false positives, false negatives, and true negatives are denoted by TP_c , FP_c , FN_c , and TN_c . Accuracy is computed as

$$\text{Accuracy} = \frac{\sum_{c=1}^C TP_c}{N}. \quad (3.68)$$

Class-wise precision, recall, and F1-score are

$$\text{Precision}_c = \frac{TP_c}{TP_c + FP_c + \epsilon}, \quad \text{Recall}_c = \frac{TP_c}{TP_c + FN_c + \epsilon}, \quad (3.69)$$

$$F1_c = \frac{2 \cdot \text{Precision}_c \cdot \text{Recall}_c}{\text{Precision}_c + \text{Recall}_c + \epsilon}. \quad (3.70)$$

Macro-F1 gives equal importance to all classes:

$$\text{MacroF1} = \frac{1}{C} \sum_{c=1}^C F1_c. \quad (3.71)$$

Weighted F1 considers class support n_c :

$$\text{WeightedF1} = \sum_{c=1}^C \frac{n_c}{N} F1_c. \quad (3.72)$$

The confusion matrix $\mathbf{C}$ records class-wise prediction errors:

$$C_{a,b} = |\{i : y_i = a, \hat{y}_i = b\}|. \quad (3.73)$$

For binary informativeness and one-versus-rest multiclass analysis, the true positive rate and false positive rate are

$$\text{TPR} = \frac{TP}{TP + FN}, \quad \text{FPR} = \frac{FP}{FP + TN}. \quad (3.74)$$

Latency and throughput were used to assess deployment feasibility:

$$\text{Throughput} = \frac{N}{\sum_{i=1}^N t_i}, \quad (3.75)$$

where t_i is the inference time for sample i . Macro-F1 and class-wise recall were emphasised for imbalanced humanitarian and severity tasks, while informativeness was assessed with

accuracy and F1-score because false rejection can prevent downstream analysis.

Explainable AI Implementation

Explainable AI methods were used to study how image and text information contribute to model predictions.

Image saliency: patch-level activations and gradients were extracted from a layer of the vision-transformer encoder immediately before the patch embeddings for the deployed models. Class-specific gradient $\times$ activation scores were computed over an $H \times W$ patch grid and resized to the original image dimensions to produce a saliency overlay.

Text explanation: the second-to-last CLIP text-transformer layer was used, since the final layer mainly pools information into the end-of-text token, leaving near-zero gradients elsewhere. Gradient-norm token importance and signed Gradient $\times$ Input contributions were computed, with padding and special tokens removed before display as token-importance bar plots and beeswarm-style contribution plots.

For the MEDIC severity model, additional explanation methods were implemented:

- **Image occlusion:** the image is divided patch by patch, and the relevance of each patch is the resulting drop in the predicted-class probability when it is masked.
- **Word-deletion importance:** one word is deleted at a time, and the drop in the predicted-class probability is recorded.
- **Fusion analysis:** the average contribution of the image and text components of the fusion representation is computed.
- **Modality ablation:** the text is replaced with a generic sentence, or the image is heavily blurred, to evaluate how much the prediction relies on each modality.

Streamlit Deployment

The trained models were deployed as a Streamlit application named **Crisis-CLIP**. Users upload an image together with a caption, and the application returns predicted classes and probabilities along with image-saliency and text-token explanations. The application includes:

- automatic CPU/CUDA device selection;
- three independent checkpoint pipelines, one per classification task;
- an informativeness-gate override;
- optional explainability output;

- probability visualisation for each task; and
- total inference-time reporting.

Streamlit’s resource caching loads the CLIP processor and the trained model checkpoints only once per session, so the models are not reloaded on every user interaction with the interface.

Streamlit Web Application Implementation

The web application is organised into four files, summarised in Table 3.17.

Table 3.17. Streamlit Application File Structure

File	Function
<code>app.py</code>	User interface, input processing, prediction pipeline, and output display
<code>models.py</code>	Model architectures, label mappings, LoRA configurations, and checkpoint loading
<code>xai.py</code>	Image and text explanation functions
<code>style.py</code>	Interface styling, including probability-bar and result-card formatting

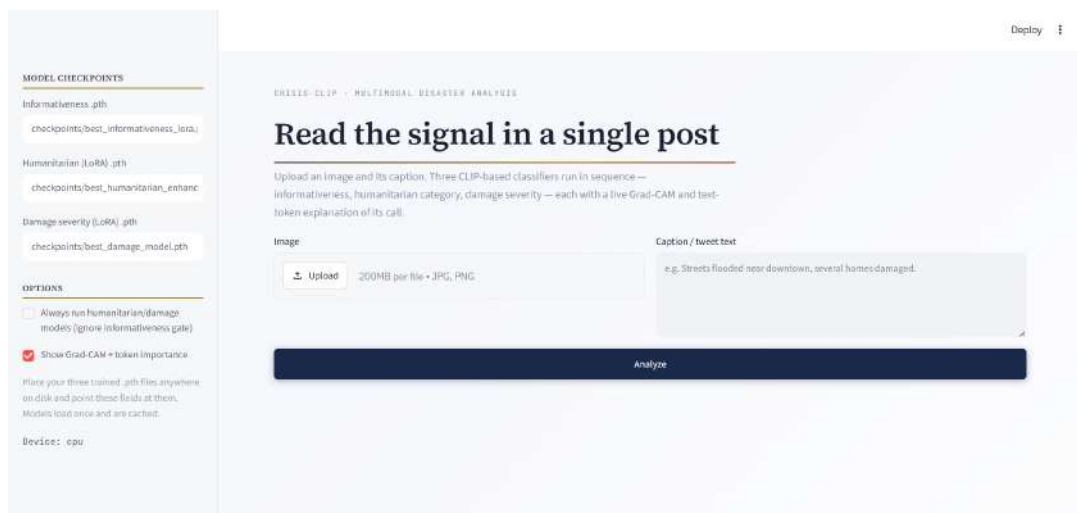


Fig. 3.2. Streamlit application input interface. *The interface allows users to upload an image and enter a caption or tweet text.*

The application page uses a wide layout with a disaster-themed icon, an image uploader, a caption/tweet-text field, checkpoint-path settings, an informativeness-gate override toggle, and an XAI display toggle, under the title “Crisis-CLIP.” The three deployed checkpoints are `best_informativeness_lora.pth`, `best_humanitarian_enhanced_lora.pth`,

and `best_damage_model.pth`; each model is loaded only once, since the processor and model-loading functions are cached as application resources.

The live text-cleaning function applied by the web app performs the following steps: convert to lowercase; remove URLs; remove user mentions; remove the hashtag symbol while preserving the hashtag word; remove the retweet marker (RT) and unsupported characters; and normalise whitespace. The cleaned text and RGB image are then passed to the CLIP processor, and the resulting tensors are moved to the active device.

The forward pass of the prediction function applies softmax to the logits,

$$\mathbf{p} = \text{Softmax}(\mathbf{o}), \quad \hat{y} = \arg \max_c p_c. \quad (3.76)$$

The Streamlit inference sequence proceeds as follows:

1. read the image and text, clean the text, convert the image to RGB, and process both with CLIP;
2. run the informativeness classifier and display the predicted class;
3. if the post is Not Informative, stop unless the override is enabled; otherwise, run the humanitarian and severity classifiers and display their predicted classes and probabilities;
4. if explainability is enabled, compute and display the XAI outputs;
5. report the total inference time.

Each task output is displayed as a separate visualisation card with the predicted label, confidence scores, saliency map, text-token importance, Gradient×Input contribution, and processing latency.

This chapter described the design and implementation of the proposed CrisisCLIP-X system, including architecture, preprocessing, model design, checkpoint handling, explainability, and Streamlit deployment. The next chapter presents the results and discussion.

Chapter 4

Results and Discussion

This chapter presents the experimental results of the proposed CLIP-based multimodal disaster-classification framework [7, 15, 18]. It compares zero-shot prompting, adapters, LoRA, and partial fine-tuning across informativeness, humanitarian-category, and damage-severity tasks.

The evaluation includes standard performance metrics, confusion matrices, computational cost, MEDIC OOD testing, explainability analysis, and Streamlit application results.

4.1 Overview

The experiments test whether task-specific adaptation improves pretrained CLIP for disaster-related social-media classification. Zero-shot prompting served as the baseline, since it uses only pretrained knowledge without training on CrisisMMD [18, 21]; adapter and LoRA fine-tuning were compared against it, and partial fine-tuning was additionally evaluated for informativeness.

Because the datasets are imbalanced, both overall and class-level performance are examined — accuracy alone can hide weak performance on minority classes such as Affected Individuals, Mild Damage, and Little/No Damage [34]. Computational efficiency (trainable parameters, GPU memory, inference cost) is also considered, alongside MEDIC cross-dataset [42] evaluation, confidence intervals, SHAP, and gradient-based explanations.

4.1.1 Result Parameters

The models are evaluated using the standard classification metrics defined below.

Accuracy

Accuracy represents the proportion of correctly classified samples:

$$\text{Accuracy} = \frac{\text{Number of correctly classified samples}}{\text{Total number of samples}} \quad (4.1)$$

For binary classification this may also be written as

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (4.2)$$

where TP , TN , FP , and FN denote true positives, true negatives, false positives, and false negatives. Accuracy gives a direct measure of overall performance but can be misleading on

imbalanced data, since a model may score highly by favouring the majority class.

Precision and Recall

Precision measures the proportion of samples predicted as a class that truly belong to it:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4.3)$$

High precision means fewer false-positive predictions, which matters here because incorrect emergency classifications waste the time of humanitarian analysts.

Recall measures the proportion of actual samples of a class that are correctly identified:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (4.4)$$

Recall is especially important for urgent categories — a low recall for Affected Individuals, Rescue/Donation, or Severe Damage means important disaster information may be missed.

F1-Score

The F1-score is the harmonic mean of precision and recall:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4.5)$$

Macro-F1 gives equal weight to every class:

$$F1_{\text{macro}} = \frac{1}{C} \sum_{i=1}^C F1_i \quad (4.6)$$

This is useful when minority-class performance matters. Weighted F1 instead weights each class's F1 by its sample count relative to the total, representing performance according to the observed class distribution.

Confusion Matrix

A confusion matrix records, for every pair of classes (i, j) , the number of samples with actual class i and predicted class j . Diagonal entries represent correct predictions, while off-diagonal entries represent classification errors. A row-normalised confusion matrix — where each row is expressed as a proportion of the true class — is particularly informative for imbalanced classification, since it shows how each actual class was distributed across the predicted classes.

ROC Curve and AUC

The Receiver Operating Characteristic (ROC) curve plots the true-positive rate ($TPR = TP/(TP + FN)$) against the false-positive rate ($FPR = FP/(FP + TN)$) at varying classification thresholds. The Area Under the Curve (AUC) summarises ranking ability, with 0.5 indicating random ranking and values near 1.0 indicating strong discrimination; for multiclass problems, one-vs-rest curves are computed per class and then macro- or micro-averaged.

The supplied experimental outputs did not retain the softmax probability arrays required to reconstruct ROC curves, so no numerical ROC-AUC value is reported here. Estimating or inventing an AUC value would not be technically valid; the evaluation would need to be rerun while explicitly saving the class probabilities.

4.2 Outcome of Methodology

4.2.1 CLIP Model

This subsection presents the CLIP-based visual outputs, numerical results, and resource statistics for Clip Model. The relevant screenshots and plots are organised as LaTeX-ready figures and tables.

Detailed Informativeness Experiment Outputs

Table 4.1. Detailed informativeness-model comparison from the experimental run

Method	Trainable Params	Total Params	Trainable %	Accuracy	F1	Precision	Recall	Latency (ms)	Peak GPU Mem. (MB)
Zero-shot prompting	0	151,277,313	0.000	0.7631	0.7631	0.7633	0.7631	3.381	678.58
Adapter fine-tuning	2,605,442	153,882,755	1.693	0.7792	0.7792	0.7792	0.7792	5.219	1,315.31
LoRA fine-tuning	1,574,786	152,852,099	1.030	0.8717	0.8715	0.8729	0.8717	5.774	1,916.39
Partial fine-tuning	21,072,258	151,869,059	13.875	0.8936	0.8936	0.8937	0.8936	4.974	2,755.35

The detailed run confirms that partial fine-tuning reached the highest informativeness accuracy, while LoRA provided the strongest parameter-efficient improvement over zero-shot prompting.

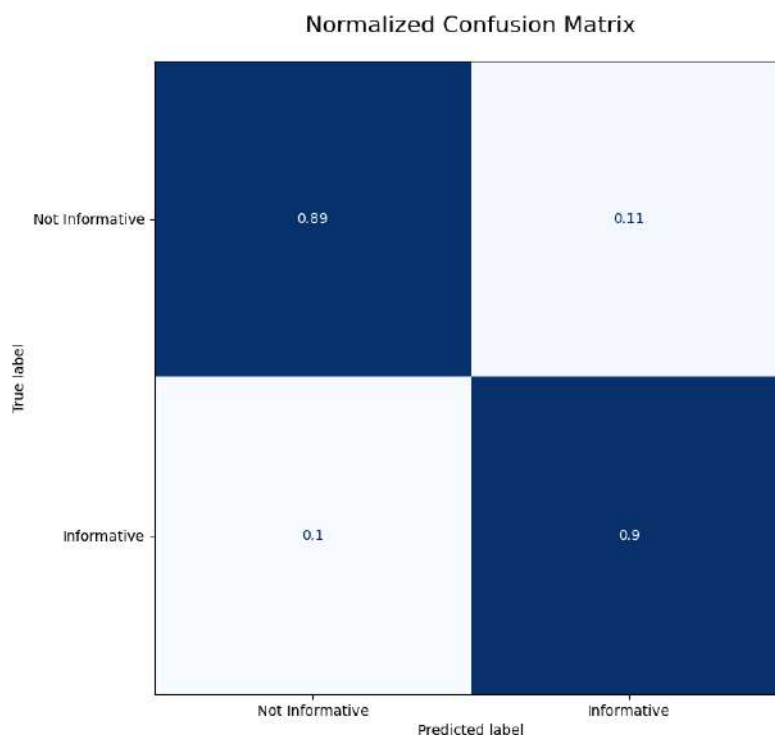


Fig. 4.1. Normalized confusion matrix for informativeness classification. *Most samples are concentrated on the diagonal, indicating strong separation between informative and not-informative posts.*

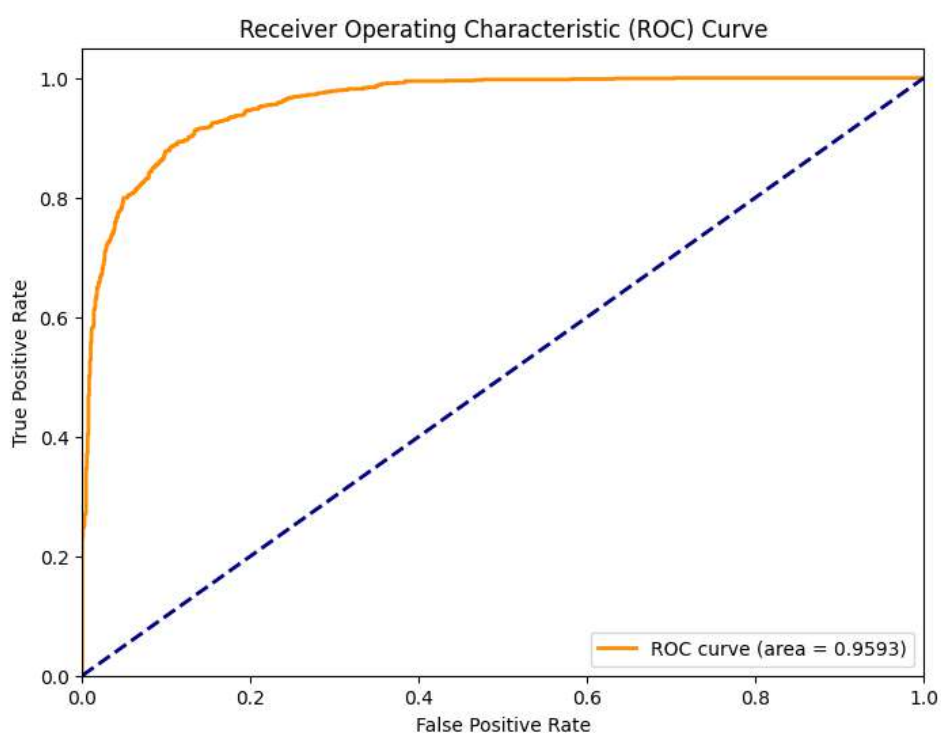


Fig. 4.2. ROC curve for binary informativeness classification. *The curve stays close to the upper-left corner, showing strong discrimination between the two classes.*

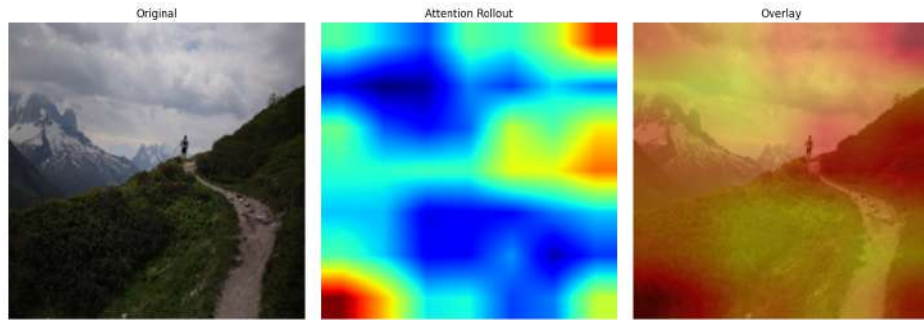


Fig. 4.3. Attention rollout visualization for an input image. *The original image, attention map, and overlay show which visual regions influenced the model. Warmer areas indicate stronger attention during the prediction.*

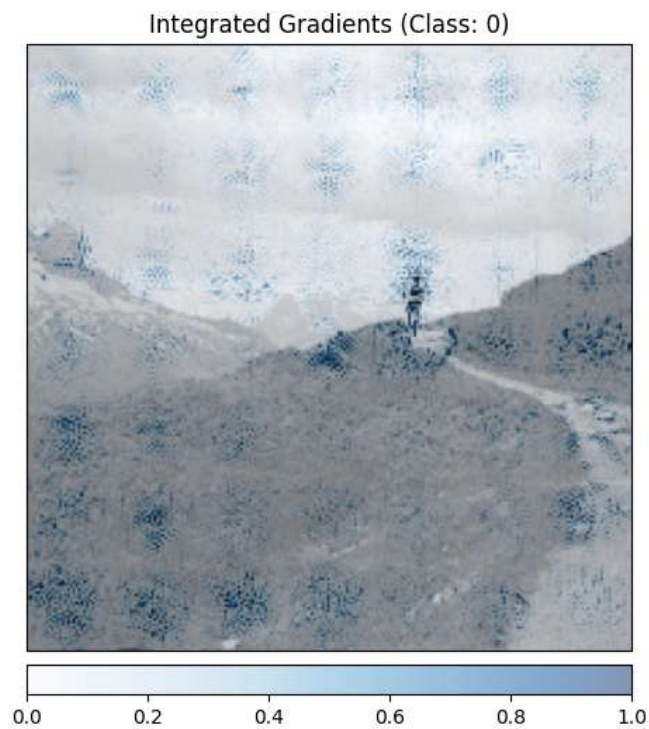


Fig. 4.4. Integrated gradients image attribution for class 0(non-informative). *This heatmap highlights image regions that contributed to the selected class prediction. Stronger shading indicates areas with greater attribution.*

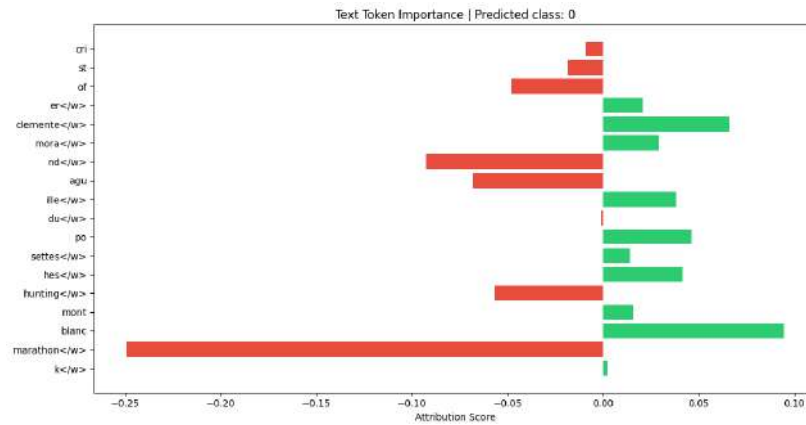


Fig. 4.5. Text token importance for the predicted class(0)(non-informative). The bar chart shows positive and negative token contributions to the model output. Tokens with larger bars had stronger influence on the prediction.

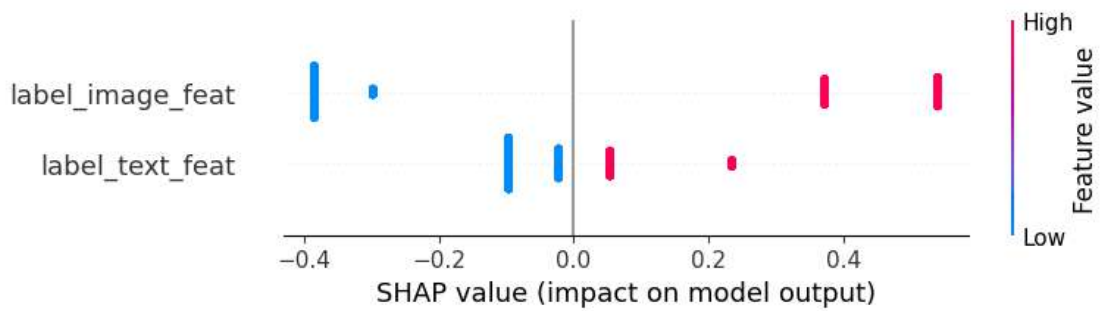


Fig. 4.6. SHAP summary for annotation-derived image and text features. The plot compares the influence of image and text label features on the model output. Wider spread indicates a stronger contribution to prediction changes.

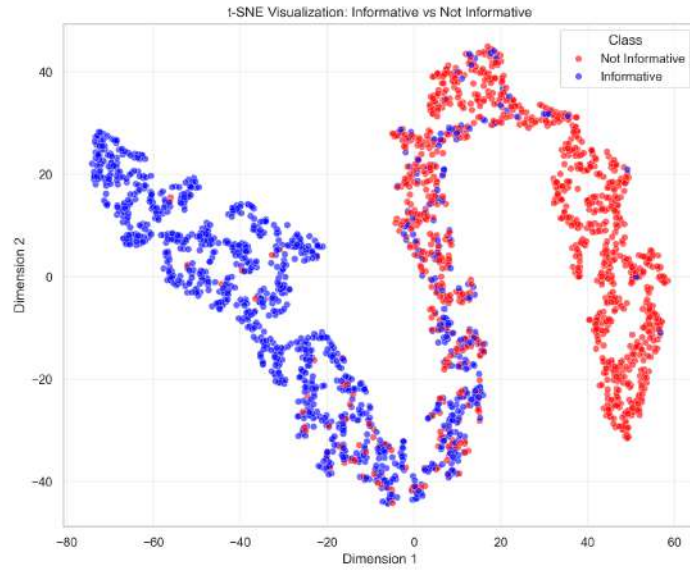


Fig. 4.7. T-SNE Visualization: Informative vs Not Informative. *The t-SNE projection separates many informative and not-informative samples into distinct regions*

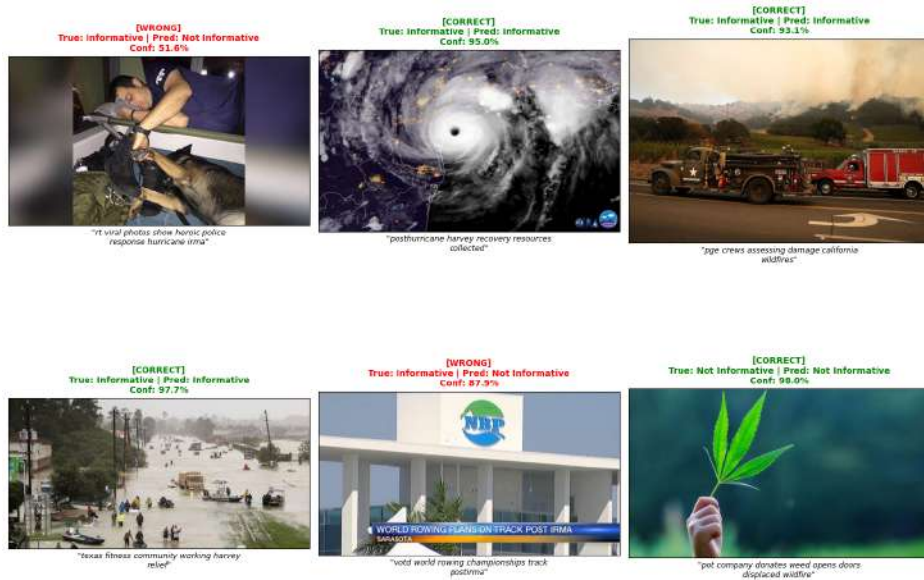


Fig. 4.8. Sample informativeness predictions. *The examples show true and predicted informative/not-informative labels with confidence scores. Correct and incorrect cases illustrate how the model responds to diverse disaster-related posts.*

Detailed Damage-Severity Experiment Outputs

Table 4.2. Detailed damage-severity resource and performance profile

Method	Test Acc.	Trainable Params	Total Params	Trainable %	Avg. Latency (ms)	Latency Std. (ms)	Infer GPU Mem. (MB)	Sec./Train Step	Train GPU Mem. (MB)
Zero-shot prompting	0.6011	0	151,277,313	0.000	54.999	11.189	588.13	0.000	0.00
Adapter fine-tuning	0.7127	3,628,035	154,905,348	2.342	31.329	3.806	1,251.97	0.064	1,347.43
LoRA fine-tuning	0.7089	9,509,379	162,431,751	5.854	38.512	3.983	1,943.20	0.080	2,069.58

For damage severity, the adapter achieved the best accuracy, while LoRA produced slightly stronger minority-class balance at a higher parameter and memory cost.



Fig. 4.9. Grad-CAM explanation for a severe damage prediction. The original flood-damage image is shown beside the Grad-CAM overlay. The visualization indicates which areas supported the model’s severe-damage decision.

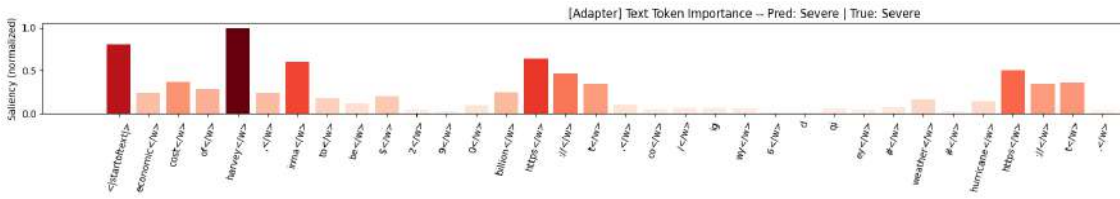


Fig. 4.10. Text token importance for a severe damage prediction. The chart ranks caption tokens according to saliency for the adapter model.

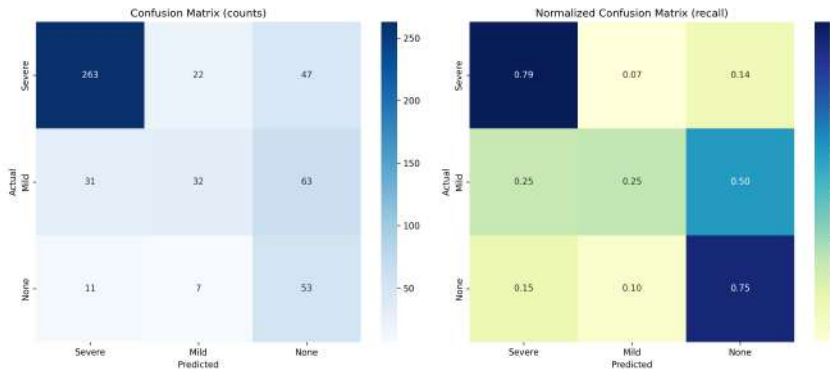


Fig. 4.11. Confusion matrices for damage-severity classification. The count matrix and normalized recall matrix show how severe, mild, and no-damage cases were classified.

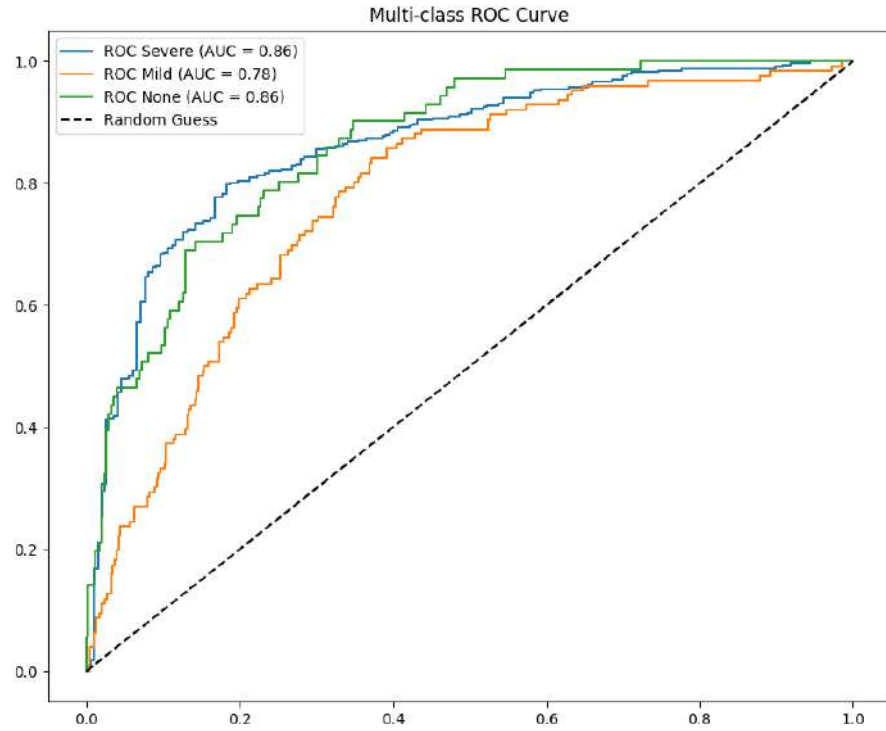


Fig. 4.12. Multiclass ROC curve for damage-severity assessment. *The one-vs-rest curves compare the model's ranking ability for severe, mild, and none classes.*

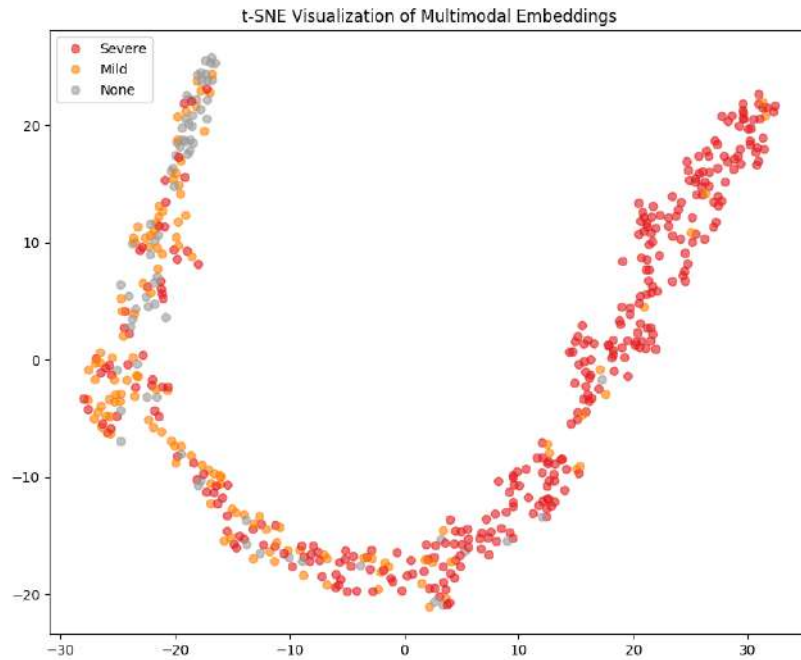


Fig. 4.13. T-SNE Visualization: Severe v/s mild v/s none. *The t-SNE projection visualises severe, mild, and none damage embeddings and shows stronger overlap among mild and none samples than in the binary informativeness task.*

Detailed Humanitarian-Category Experiment Outputs

Table 4.3. Humanitarian-category model summary from the supplied run

Method	Accuracy	Macro-F1	Trainable Params	Trainable %
Zero-shot prompting	0.6649	0.5617	0	0.00
Adapter fine-tuning	0.9058	0.8324	5,302,279	3.39
LoRA fine-tuning	0.8702	0.8117	8,919,815	5.57

Table 4.4. Humanitarian inference resource profile for a single forward pass at batch size 16

Metric	Zero-shot	Adapter	LoRA
Latency mean (ms)	47.24	88.34	92.58
Latency std. (ms)	0.41	0.71	0.95
Latency min. (ms)	46.64	86.99	91.35
Latency max. (ms)	48.44	90.04	94.93
Throughput (samples/s)	338.72	181.11	172.82
Peak VRAM (MB)	2,012.86	2,014.99	2,014.99
RAM delta (MB)	0.01	0.00	0.00
RAM after-call (MB)	1,855.57	1,855.58	1,855.58
Total parameters	151,277,313	156,579,592	160,197,128
Trainable parameters	151,277,313	5,302,279	8,919,815
Frozen parameters	0	151,277,313	151,277,313
Trainable %	100.00	3.39	5.57

Table 4.5. Per-class mean modality-gate weights for humanitarian classification

Class	Text Weight	Image Weight	Dominant Modality
Infrastructure	0.520	0.480	Text
Other Relevant Information	0.516	0.484	Text
Not Humanitarian	0.524	0.476	Text
Rescue/Donation	0.555	0.445	Text
Affected Individuals	0.510	0.490	Text

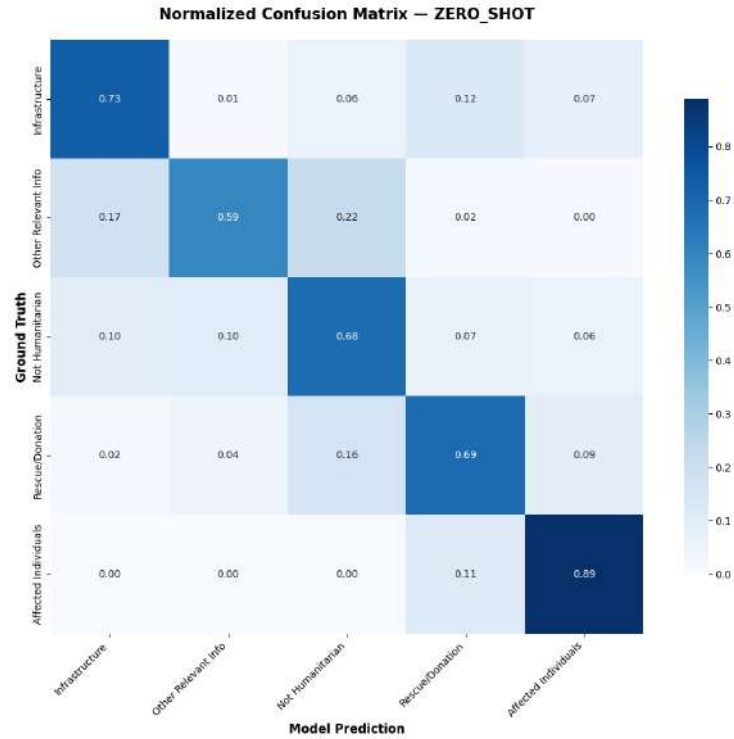


Fig. 4.14. Zero-shot normalized confusion matrix for humanitarian classification. *This matrix shows baseline performance before task-specific training.*

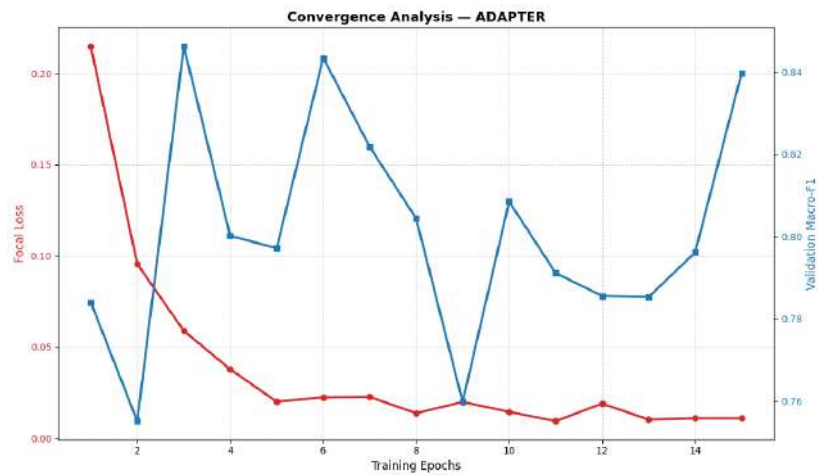


Fig. 4.15. Adapter convergence analysis for humanitarian classification. *Training loss decreases sharply over early epochs while validation macro-F1 fluctuates.*

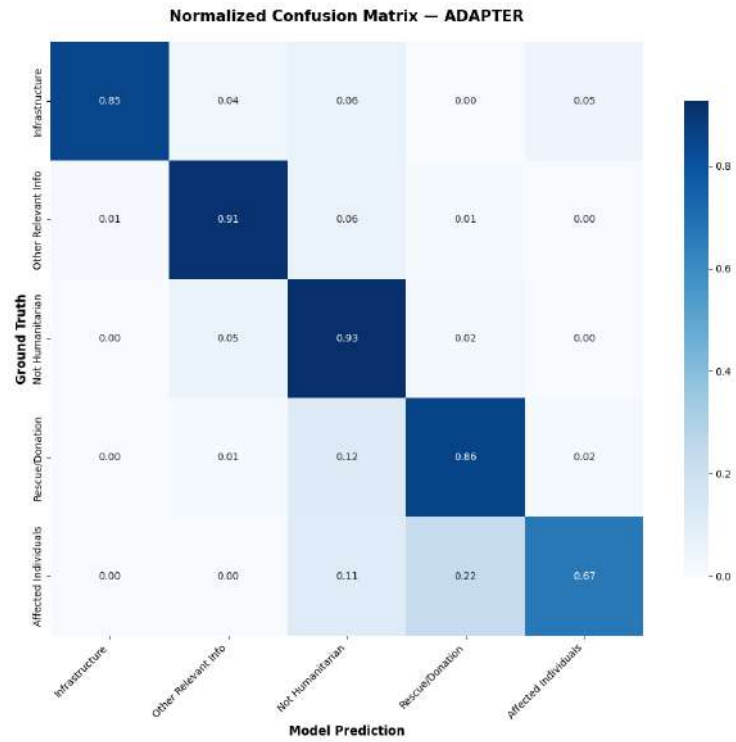


Fig. 4.16. Adapter normalized confusion matrix for humanitarian classification. *The adapter model improves diagonal accuracy across most classes compared with zero-shot.*

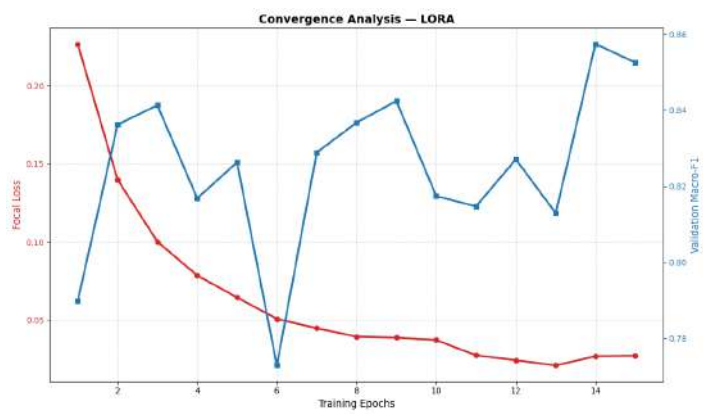


Fig. 4.17. LoRA convergence analysis for humanitarian classification. *The focal loss decreases across epochs, while validation macro-F1 varies before improving near the end.*

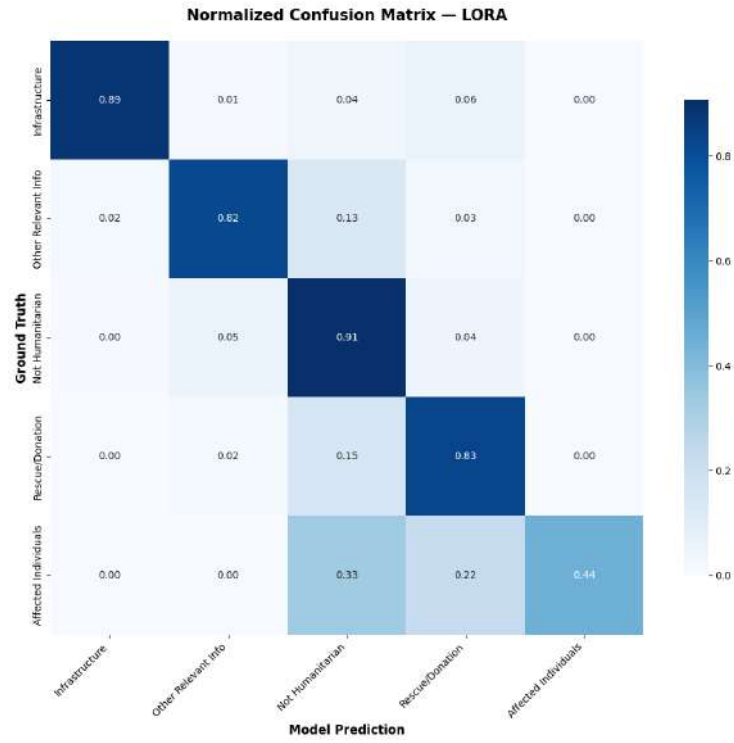


Fig. 4.18. LoRA normalized confusion matrix for humanitarian classification. *LoRA performs strongly on infrastructure, not humanitarian, and rescue/donation classes.*



Fig. 4.19. Qualitative humanitarian-category predictions. *The examples display per-class multimodal predictions for infrastructure, other information, not humanitarian, rescue/donation, and affected individuals.*

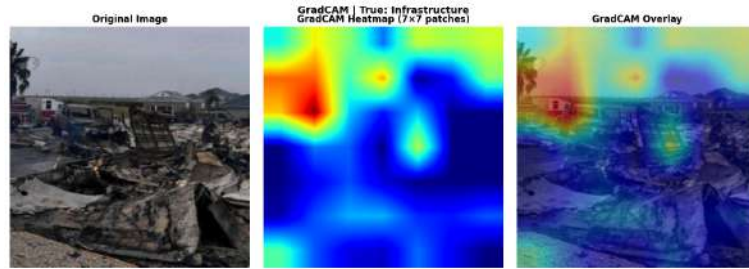


Fig. 4.20. Grad-CAM explanation for a rescue/donation example. *The Hurricane Harvey relief poster is shown with a Grad-CAM heatmap and overlay.*

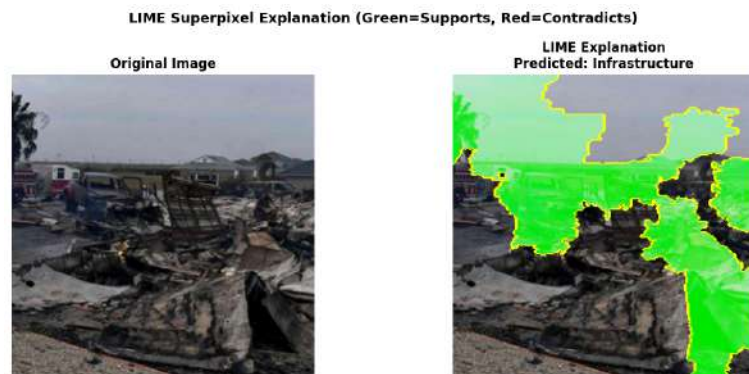


Fig. 4.21. LIME superpixel explanation for rescue/donation prediction. *Green regions support the predicted rescue/donation class, while red regions contradict it.*

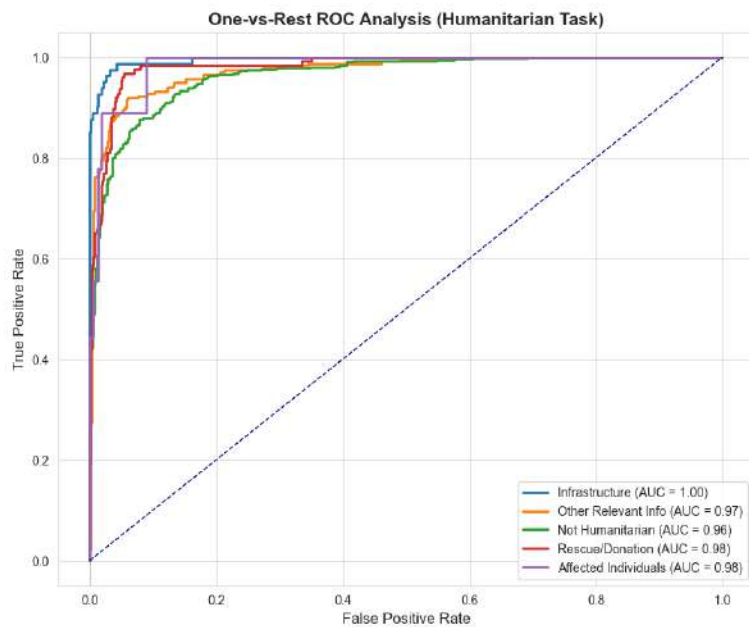


Fig. 4.22. SHAP summary for the text modality. *The plot ranks disaster-related words by their impact on the model output. Terms such as Irma, Harvey, hurricane, and earthquake contribute to the explanation.*

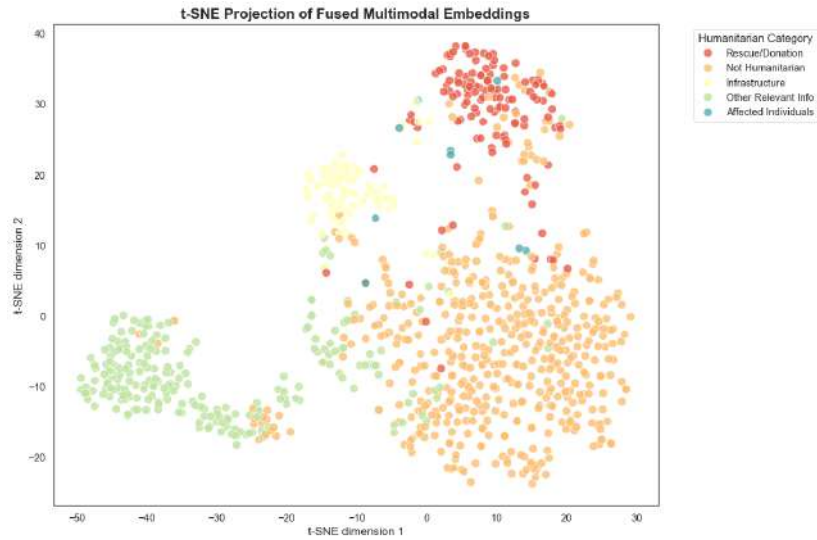


Fig. 4.23. One-vs-rest ROC analysis for the humanitarian task. *The curves show strong class-level discrimination across the five humanitarian categories.*

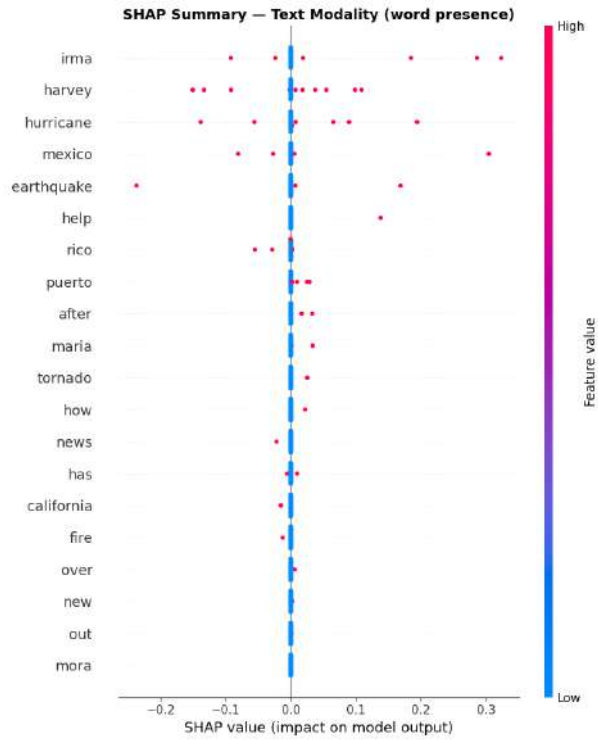


Fig. 4.24. SHAP summary for the text modality. *The plot ranks disaster-related words by their impact on the model output. Terms such as Irma, Harvey, hurricane, and earthquake contribute to the explanation.*

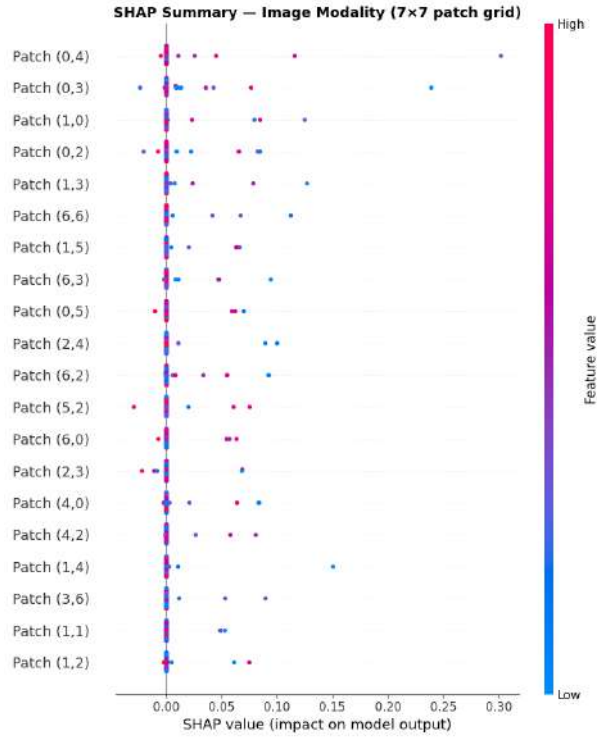


Fig. 4.25. SHAP summary for the image modality. *The patch-grid plot shows which visual regions influence predictions across samples.*

4.2.2 Web Application Results

The trained models were integrated into a Streamlit application, Crisis-CLIP, which accepts an uploaded disaster image and caption/tweet text and processes it through three sequential stages: informativeness classification, humanitarian-category classification, and damage-severity assessment. The informativeness model acts as a gate — Not Informative inputs skip the remaining stages unless the user enables an override.

Deployed Models

The default checkpoints are:

- `best_informativeness_lora.pth`;
- `best_humanitarian_enhanced_lora.pth`;
- `best_damage_model.pth`.

i.e., the deployed system uses the parameter-efficient LoRA models rather than the single highest-scoring configuration for every task.

The deployed informativeness model trails partial fine-tuning by 2.19 points but uses $\sim 1.03\%$ trainable parameters instead of 13.88%. The deployed humanitarian LoRA model trails the

Table 4.6. Performance of Deployed Configurations

Stage	Deployed Configura- tion	Accuracy	Macro/F1	Weighted F1
Informativeness	CLIP + LoRA	87.17%	87.15% F1	~87.15%
Humanitarian	CLIP + enhanced LoRA, gated fusion	87.02%	81.17% macro-F1	Not retained
Damage severity	CLIP + LoRA, symmet- ric fusion	70.89%	58% macro-F1	69%

adapter by 3.56 points in accuracy and 2.07 in macro-F1, but matches the saved enhanced-LoRA checkpoint for parameter-efficient inference. The deployed severity LoRA model trails the adapter by only 0.38 points in accuracy while giving better macro/weighted F1 — a reasonable deployment trade-off. In short, the best experimental model and the deployed model are not always identical; checkpoint compatibility and efficiency are weighed alongside raw performance.

Input, Prediction, and Explainability

Input images (JPG/JPEG/PNG) and caption text are preprocessed (lowercasing, URL/mention removal, hashtag-symbol stripping, retweet-marker removal, character/space normalisation) before CLIP tokenisation. For each completed stage the app shows the predicted class, per-class probabilities as sorted bars, an image explanation, and a text explanation.

Visual explanations use Grad-CAM-style saliency from the final vision-transformer layer; text explanations use token-gradient importance and signed $\text{Gradient} \times \text{Input}$ values from the second-to-last text-transformer layer (chosen because the final layer pools mainly through the end-of-text token, giving near-zero gradients elsewhere). A cosine-similarity comparison of token-importance vectors across stages indicates whether different task-specific classifiers attend to similar or different textual evidence.

Models and processor load via cached Streamlit functions, so they initialise once and are reused across reruns; total latency is timed from the Analyze click and reported per stage or for the full pipeline. The supplied files do not include a repeated latency benchmark, separate CPU/GPU timings, or a usability study, so no average response time or user-satisfaction score is reported, and no fabricated example prediction is included.

Streamlit Application Visual Outputs

The following images document the deployed application workflow: input submission, sample-post preparation, informativeness prediction, token attribution, $\text{Gradient} \times \text{Input}$ attribution, and cross-stage attribution comparison.

Table 4.7. Streamlit Application Functional Results

Component	Result
Image / text input	JPG, JPEG, PNG images; caption or tweet text accepted
Missing-input validation	Warning shown when image or text is absent
Informativeness gate	Downstream stages skipped for Not Informative unless overridden
Humanitarian / severity classification	5-class / 3-class prediction and probabilities displayed
Explanations	Grad-CAM overlay; token-importance and Gradient×Input attribution
Cross-stage check	Cosine similarity of token importance across stages
Device / caching	CUDA when available, else CPU; models cached after first load
Latency	Total processing time displayed per request

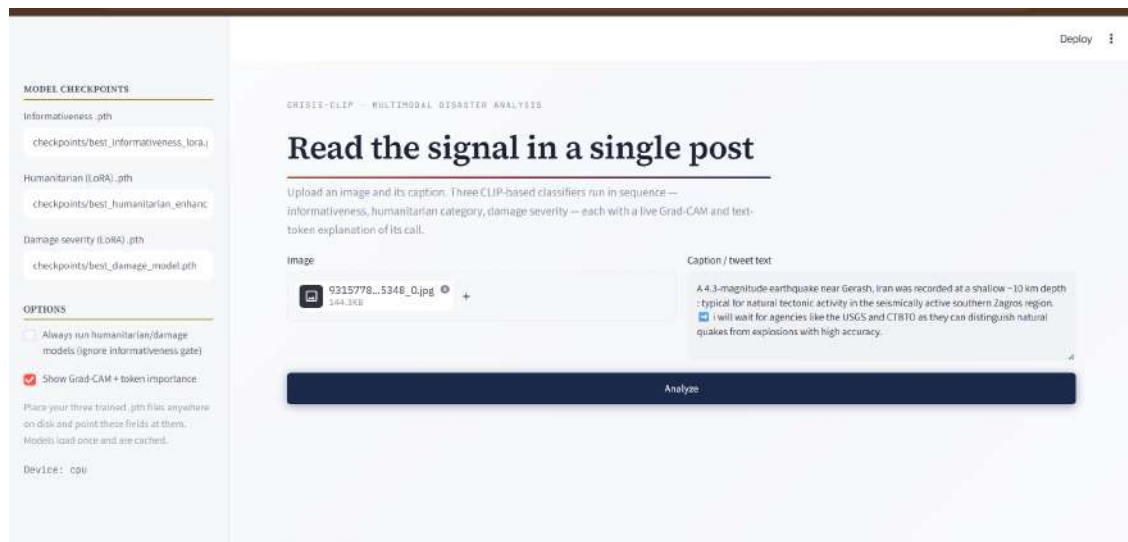


Fig. 4.26. Streamlit application with a sample post prepared for analysis. *The screenshot shows a disaster-related image file and caption text loaded into the app.*

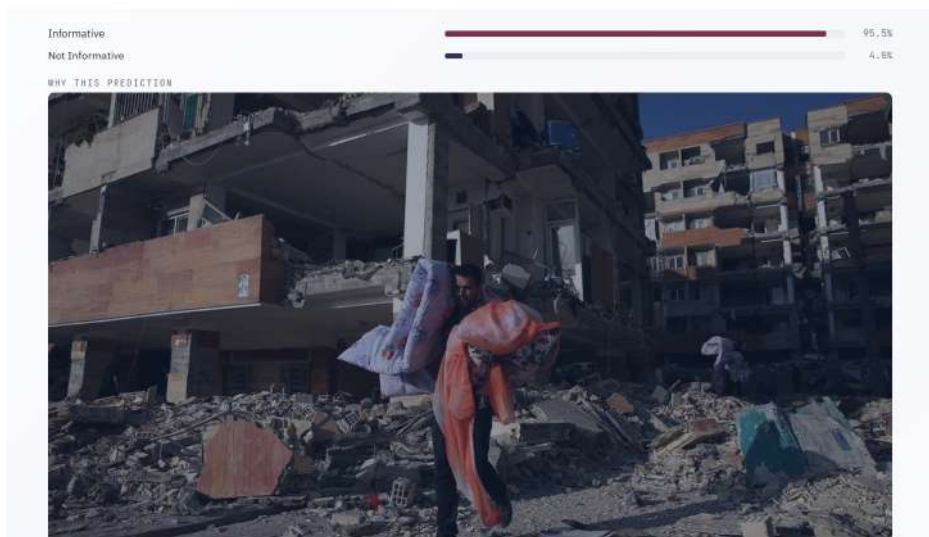


Fig. 4.27. Streamlit informativeness prediction output. *The application predicts the post as informative with high confidence and displays the uploaded disaster image.*

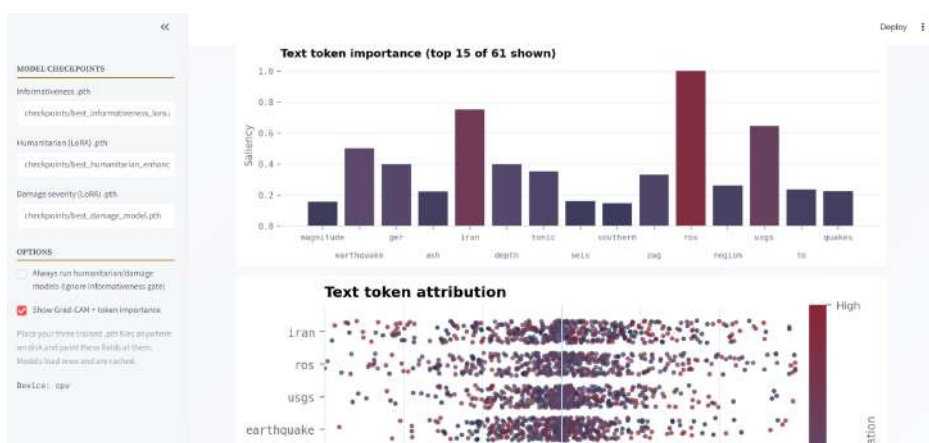


Fig. 4.28. Streamlit text-token importance and attribution view. *The app highlights the most influential words and shows token-level attribution for the prediction.*

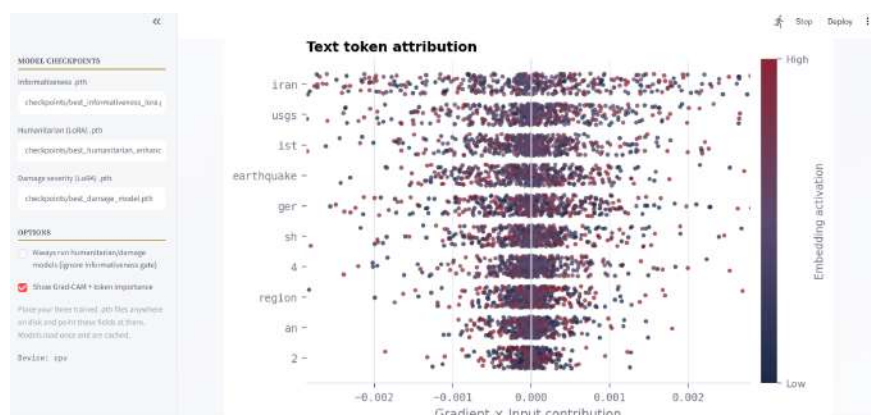


Fig. 4.29. Gradient-times-input attribution for text tokens. *The visualization shows signed token contributions using Gradient $\times$ Input values.*

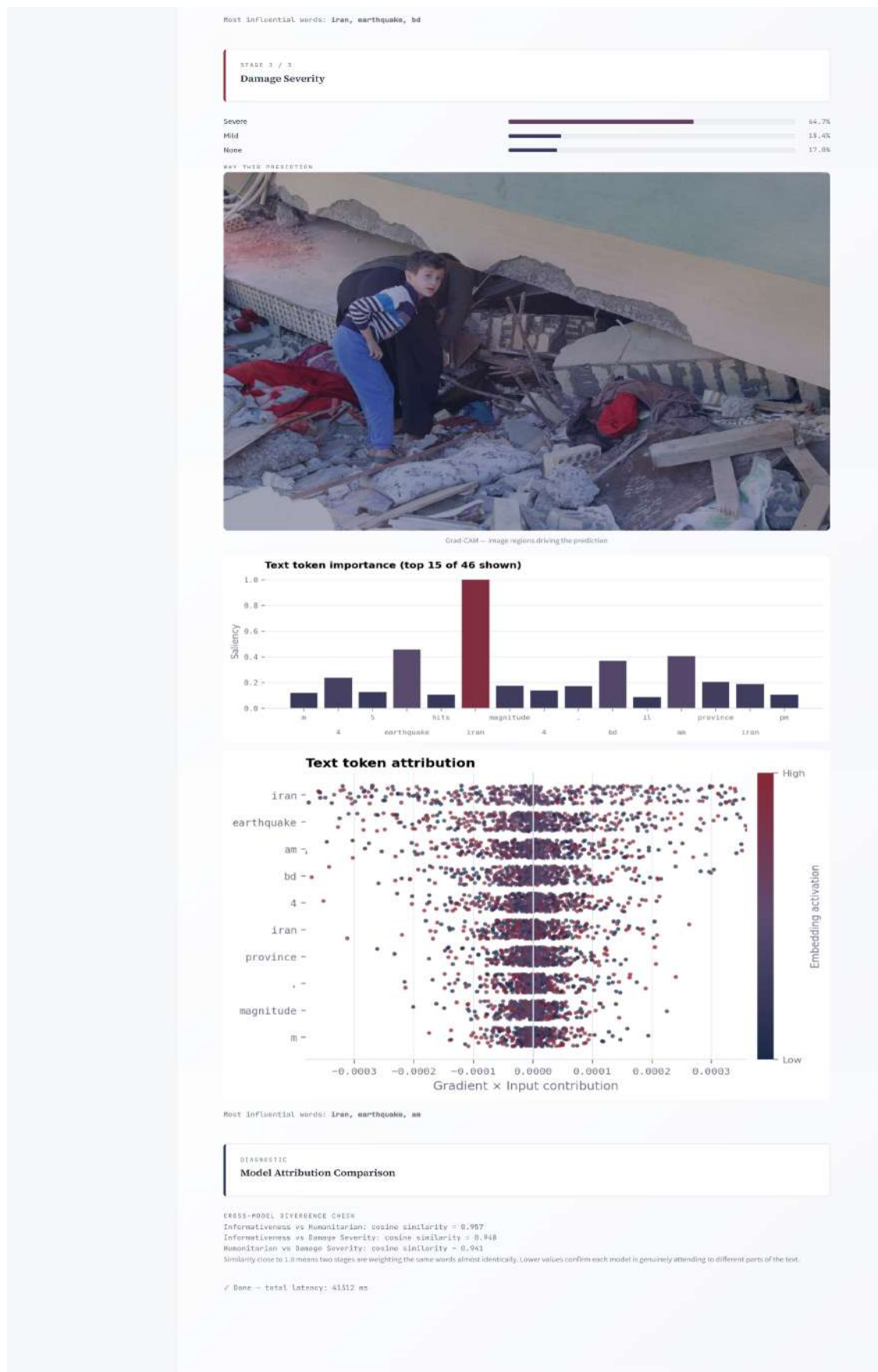


Fig. 4.30. Model attribution comparison and cross-stage divergence check. *The diagnostic panel compares token importance across informativeness, humanitarian, and severity stages. High cosine similarity indicates that multiple stages attend to similar textual evidence.*

4.3 Comparison

4.3.1 Model Performance

The model-performance comparison is based on the available CLIP-model results, plots, and resource statistics.

Overall Model-Performance Comparison

Table 4.8. Humanitarian-Category Classification Performance

Method	Accuracy	Macro-F1	Trainable Params	Trainable %
Zero-shot prompting	66.49%	56.17%	0	0.00%
Adapter fine-tuning	90.58%	83.24%	5,302,279	3.39%
LoRA fine-tuning	87.02%	81.17%	8,919,815	5.57%

Humanitarian-Category Classification The adapter model gave the strongest humanitarian result, improving accuracy by 24.09 points and macro-F1 by 27.07 points over zero-shot, and exceeding LoRA by 3.56 points in accuracy and 2.07 points in macro-F1. The smaller macro-F1 gap suggests both methods improved minority-class recognition, though the adapter was more balanced overall.

Table 4.9. Damage-Severity Classification Performance

Method	Accuracy	Macro-F1	Weighted F1	Trainable Params	Inference GPU Mem
Zero-shot	60.11%	38%	54%	0	588.13 MB
Adapter	71.27%	56%	68%	3,628,035	1,251.97 MB
LoRA	70.89%	58%	69%	9,509,379	1,943.20 MB

Damage-Severity Assessment The adapter achieved the highest accuracy (only 0.38 points above LoRA), while LoRA produced slightly stronger macro-F1 and weighted F1 using nearly three times as many trainable parameters. The adapter offered the better balance between accuracy, parameters, and memory; LoRA distributed performance more evenly across Severe, Mild, and None, while the adapter benefited from strong performance on the majority Severe class.

Informativeness Classification *Note:* Peak GPU memory usage for zero-shot prompting, adapter fine-tuning, LoRA fine-tuning, and partial fine-tuning was 678.58 MB, 1,315.31 MB, 1,916.39 MB, and 2,755.35 MB, respectively.

Partial fine-tuning achieved the highest informativeness accuracy of 89.36%, outperforming LoRA by 2.19 percentage points. However, LoRA required only about 1.03%

Table 4.10. Informativeness-Classification Performance

Method	Accuracy	Precision	Recall	F1	Trainable %
Zero-shot	76.31%	76.33%	76.31%	76.31%	0.00%
Adapter	77.92%	77.92%	77.92%	77.92%	1.69%
LoRA	87.17%	87.29%	87.17%	87.15%	1.03%
Partial fine-tuning	89.36%	89.37%	89.36%	89.36%	13.88%

trainable parameters, compared with 13.88% for partial fine-tuning. Therefore, LoRA provides a more parameter-efficient configuration, while partial fine-tuning provides the highest accuracy.

These results should be interpreted carefully because part of the informativeness-model development used the test loader during notebook-based experimentation. A stricter final evaluation should repeat checkpoint selection using the official development set and reserve the test set only for final reporting.

Graphical Comparison of Model Performance and Resources

The following figures provide a visual comparison of model performance and computational efficiency across the main adaptation methods. These visual results complement the numerical tables by showing accuracy, macro-F1, latency, GPU memory, throughput, and trainable-parameter differences among zero-shot prompting, adapter fine-tuning, LoRA fine-tuning, and partial fine-tuning.

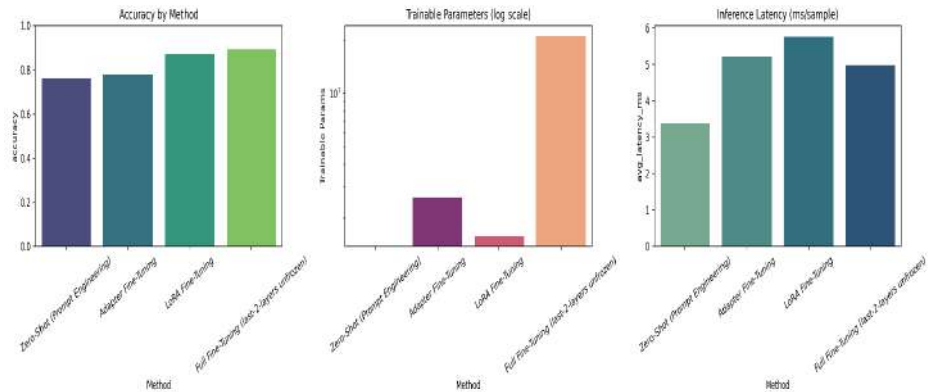


Fig. 4.31. Informativeness-model comparison based on accuracy, trainable parameters, and inference latency.

Figure 4.31 shows that partial fine-tuning achieved the highest informativeness accuracy, while LoRA provided a strong parameter-efficient alternative with much fewer trainable parameters than partial fine-tuning.

Figure 4.32 compares the damage-severity models. Adapter fine-tuning achieved the highest accuracy, while LoRA showed competitive performance with higher trainable parameters and GPU memory usage.

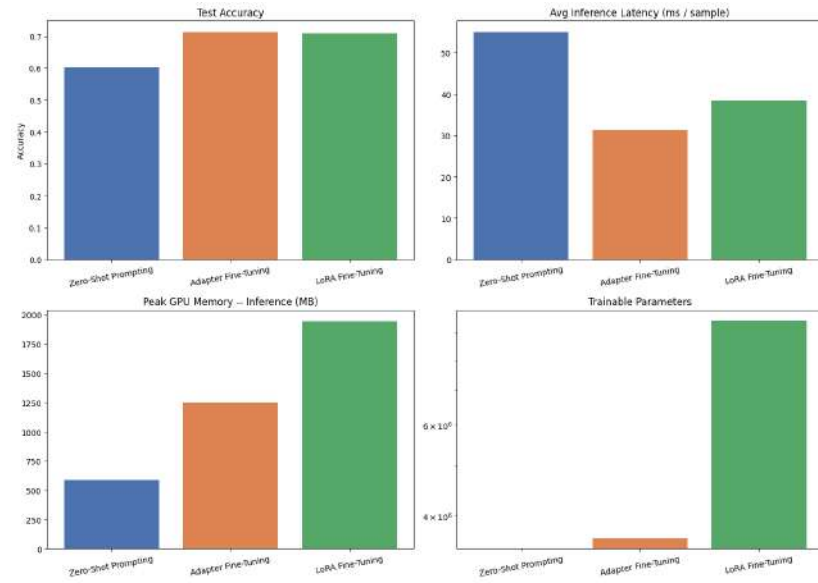


Fig. 4.32. Damage-severity comparison based on accuracy, inference latency, GPU memory, and trainable parameters.

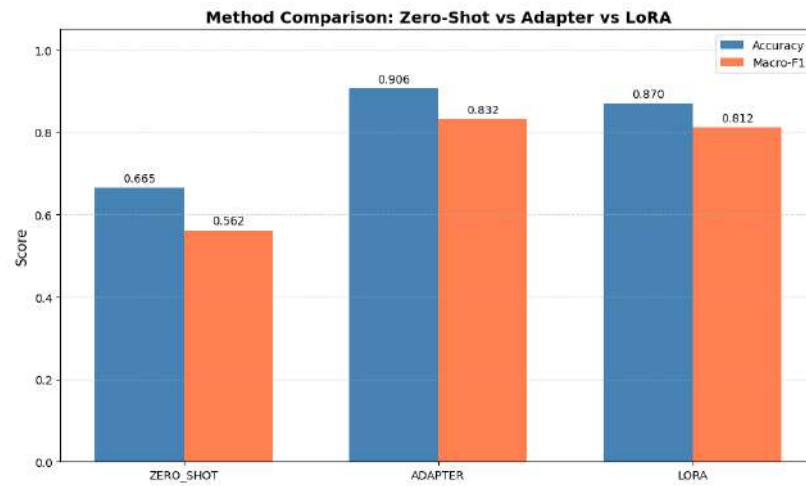


Fig. 4.33. Humanitarian-category method comparison among zero-shot prompting, adapter fine-tuning, and LoRA fine-tuning.

Figure 4.33 shows that adapter fine-tuning achieved the best humanitarian-category performance in both accuracy and macro-F1. LoRA also improved substantially over zero-shot prompting, confirming the benefit of task-specific adaptation.

Figure 4.34 shows that zero-shot prompting had the lowest computational adaptation cost but also the weakest classification performance. Adapter fine-tuning achieved the best performance while using 5,302,279 trainable parameters, whereas LoRA used 8,919,815 trainable parameters and remained a strong parameter-efficient alternative.

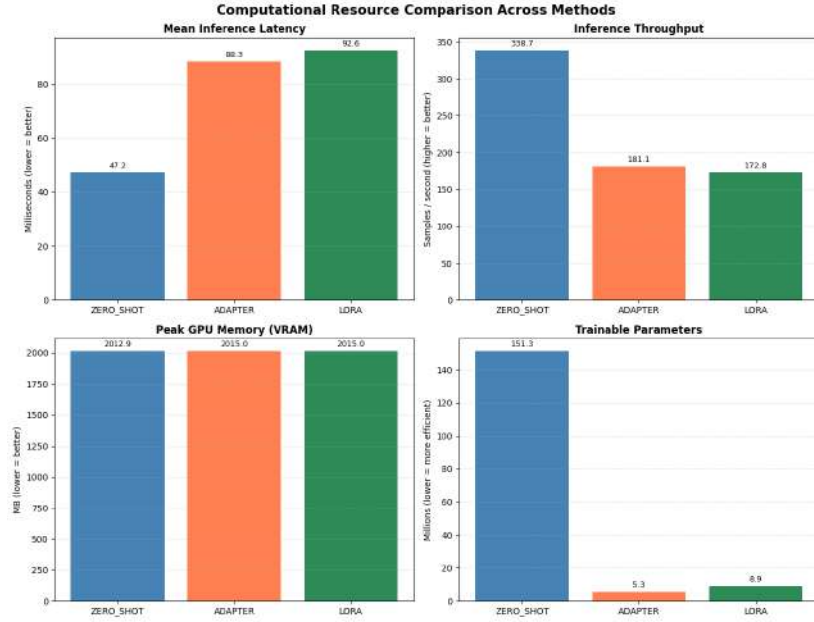


Fig. 4.34. Computational-resource comparison across humanitarian-category classification methods.

MEDIC Cross-Dataset Evaluation

Additional MEDIC results provide evidence from a different disaster-data distribution. The informativeness model was evaluated in an out-of-distribution setting by testing the CrisisMMD-trained model on MEDIC. For humanitarian-category and damage-severity classification, the models were retrained on MEDIC data before evaluation. Therefore, the three protocols cannot be compared directly and should be interpreted separately.

Table 4.11. Reported MEDIC OOD Performance

Task	Evaluation Setting	Accuracy	Weighted F1
Informativeness	CrisisMMD-trained, tested on MEDIC OOD	87%	Not retained
Humanitarian category	MEDIC-retrained, tested on MEDIC OOD	88%	89%
Damage severity	MEDIC-retrained, tested on MEDIC OOD	86%	85%

The 87% cross-dataset informativeness accuracy shows the classifier retained substantial predictive power on MEDIC, though label construction differs between datasets. The humanitarian and severity models performed well after retraining, but these values are dataset-specific results rather than direct evidence of improvement over CrisisMMD, since MEDIC has a different class distribution (notably many Little/None cases).

Class-Level Performance

Table 4.12. Class-Level Humanitarian Results (Precision / Recall / F1 / Support)

Model	Class	Prec.	Rec.	F1	Supp.
Zero-shot	Infrastructure	0.39	0.73	0.51	81
Zero-shot	Other Relevant Info	0.71	0.59	0.65	235
Zero-shot	Not Humanitarian	0.82	0.68	0.74	504
Zero-shot	Rescue/Donation	0.64	0.69	0.67	126
Zero-shot	Affected Individuals	0.14	0.89	0.24	9
Adapter	Infrastructure	0.95	0.85	0.90	81
Adapter	Other Relevant Info	0.88	0.91	0.90	235
Adapter	Not Humanitarian	0.93	0.93	0.93	504
Adapter	Rescue/Donation	0.89	0.86	0.87	126
Adapter	Affected Individuals	0.50	0.67	0.57	9
LoRA	Infrastructure	0.92	0.89	0.91	81
LoRA	Other Relevant Info	0.87	0.82	0.85	235
LoRA	Not Humanitarian	0.89	0.91	0.90	504
LoRA	Rescue/Donation	0.76	0.83	0.79	126
LoRA	Affected Individuals	1.00	0.44	0.62	9

Humanitarian Classes Zero-shot showed high recall (0.89) but very low precision (0.14) for Affected Individuals — many unrelated samples were misclassified into this category. Adapter fine-tuning raised precision to 0.50 with recall 0.67, while LoRA reported perfect precision (1.00) but recall of only 0.44, missing over half the true cases. With only nine test samples, this class remained sensitive to individual predictions. The adapter’s normalised confusion matrix showed 85–93% correct classification across Infrastructure, Other Relevant Information, Not Humanitarian, and Rescue/Donation, with most errors involving confusion with Not Humanitarian.

Table 4.13. Class-Level Damage-Severity Results (Precision / Recall / F1 / Support)

Model	Class	Prec.	Rec.	F1	Supp.
Zero-shot	Severe	0.71	0.86	0.78	332
Zero-shot	Mild	0.21	0.03	0.06	126
Zero-shot	None	0.25	0.38	0.31	71
Adapter	Severe	0.76	0.94	0.84	332
Adapter	Mild	0.49	0.33	0.40	126
Adapter	None	0.67	0.34	0.45	71
LoRA	Severe	0.77	0.90	0.83	332
LoRA	Mild	0.46	0.40	0.43	126
LoRA	None	0.74	0.37	0.49	71

Damage-Severity Classes Zero-shot almost entirely failed to detect Mild Damage (3% recall), over-predicting the majority Severe class. Both adapter and LoRA substantially improved the Mild and None categories; LoRA gave better recall for Mild/None while the adapter gave better recall for Severe — explaining why the adapter led on accuracy while LoRA led on macro/weighted F1. Severity remained the hardest task because the boundary between Severe, Mild, and None is visually gradual.

Table 4.14. Class-Level Informativeness Results

Class	Precision	Recall	F1	Support
Not Informative	0.9013	0.8748	0.8879	1,086
Informative	0.8850	0.9096	0.8972	1,151
Macro average	0.8932	0.8922	0.8925	2,237
Weighted average	0.8929	0.8927	0.8926	2,237

Informativeness Classes (Partial Fine-Tuning)

Statistical Comparison with the Zero-Shot Baseline

Table 4.15. Accuracy Improvement over the Zero-Shot Baseline

Task	Baseline	Proposed	Abs. Improvement	Relative	Error Reduction
Humanitarian	66.49%	90.58% (Adapter)	24.09 pts	36.23%	71.89%
Severity	60.11%	71.27% (Adapter)	11.15 pts	18.55%	27.96%
Informativeness	76.31%	87.17% (LoRA)	10.86 pts	14.24%	45.85%

Classification-error reduction was calculated as

$$\text{Error Reduction} = \frac{(1 - A_b) - (1 - A_p)}{1 - A_b} \times 100 \quad (4.7)$$

where A_b is the baseline accuracy and A_p is the proposed-model accuracy.

The adapter cut humanitarian classification error by $\sim 72\%$, the largest reduction of the three tasks, suggesting pretrained CLIP needed substantial task-specific adaptation for these specialised labels. The severity adapter reduced error by $\sim 28\%$, with remaining errors concentrated in the Mild and None classes. LoRA cut informativeness error by $\sim 46\%$ while updating only $\sim 1.03\%$ of parameters.

Non-overlapping or minimally overlapping intervals indicate practical improvement over zero-shot, though this does not replace a paired significance test. McNemar’s test [34], given by

Table 4.16. Approximate 95% Confidence Intervals (Wilson)

Task / Method	Accuracy	95% CI
Humanitarian zero-shot	66.49%	63.44%–69.41%
Humanitarian adapter	90.58%	88.56%–92.27%
Severity zero-shot	60.11%	55.88%–64.20%
Severity adapter	71.27%	67.27%–74.96%
Informativeness zero-shot	76.31%	74.50%–78.02%
Informativeness LoRA	87.17%	85.72%–88.49%

$$\chi^2 = \frac{(|b - c| - 1)^2}{b + c} \quad (4.8)$$

where b and c are the counts of samples correctly classified by only one of two paired classifiers, is suitable for comparing two classifiers on the same test set. However, the notebooks did not preserve complete paired prediction-disagreement tables, so a valid p -value cannot be reconstructed from aggregate accuracy values alone.

A 1,000-sample bootstrap for the partial fine-tuned informativeness model gave a mean accuracy of 89.26% (SD 0.68%, 95% CI 88.02%–90.61%), suggesting reasonably stable performance across resampled subsets — though this analysis reused development-phase test data and should be repeated with a strict train/test separation.

Evaluating Fairness with SHAP Values

SHAP (SHapley Additive exPlanations) is primarily an interpretation method rather than a complete fairness metric [29]. The CrisisMMD/MEDIC data do not provide reliable protected attributes such as gender, ethnicity, age group, disability status, nationality, or socioeconomic position, so demographic fairness measures like demographic parity, equal opportunity, predictive parity, and equalised odds cannot be responsibly calculated for this research.

SHAP values alone cannot prove that a model is socially fair; they can only help identify which input features influence a prediction and support bias investigation when meaningful group variables are available. A formal fairness evaluation would require clearly defined population groups and group-level error analysis, which the available annotations do not support.

In this research, fairness is therefore assessed only in a limited, class-level sense — comparing the recall of each target category to determine whether the classifier performs consistently across majority and minority disaster-information classes.

Informativeness was the most balanced (3.48-point gap). The humanitarian adapter was more balanced than LoRA, which despite perfect reported Affected-Individuals precision missed over half the true cases. Severity showed the largest disparity (up to 61 points between Severe and Mild recall). These gaps reflect performance differences between disaster-

Table 4.17. Class-Level Recall Disparity

Model	Highest Recall	Lowest Recall	Gap
Humanitarian adapter	93%	67%	26 pts
Humanitarian LoRA	91%	44%	47 pts
Severity adapter	94%	33%	61 pts
Severity LoRA	90%	37%	53 pts
Informativeness (partial FT)	90.96%	87.48%	3.48 pts

information categories, not evidence of demographic fairness.

SHAP Values for Main Features

Exact SHAP was applied only to a Random Forest informativeness baseline using two annotation-derived features: `label_image_feat` and `label_text_feat`. The model used 100 trees, entropy criterion, max depth 2, max leaf nodes 3, and `shap.TreeExplainer`.

However, the target label was created directly from `label_image`, while `label_image_feat` was also used as an input feature. This creates target leakage. Therefore, this SHAP result explains an annotation-based Random Forest rule, not the deployable CLIP-based image-text classifier. It should be presented only as a TreeSHAP demonstration.

Interpreting the SHAP Results

The SHAP summary plot showed `label_image_feat` as the dominant feature, as expected since the target was built from it. Because 1 = Not Informative, high `label_image_feat` values pushed predictions toward Not Informative; the text-annotation feature had a smaller but still visible effect. This confirms the Random Forest reproduced the annotation rule rather than revealing which raw content is more predictive.

For the actual CLIP models, more meaningful interpretation came from Grad-CAM-style saliency, token-gradient importance, signed Gradient×Input attribution, modality-gate weights, occlusion, word-deletion, and modality ablation.

Table 4.18. Average Humanitarian Modality Weights

Humanitarian Class	Text Weight	Image Weight	Dominant
Infrastructure	0.503	0.497	Nearly balanced
Other Relevant Information	0.564	0.436	Text
Not Humanitarian	0.529	0.471	Text
Rescue/Donation	0.575	0.425	Text
Affected Individuals	0.515	0.485	Nearly balanced

Text was slightly favoured across all categories, most strongly for Rescue/Donation (plausible, since relief and evacuation requests rely on written detail), while Infrastructure and Affected Individuals were more visually balanced given the relevance of visible structural

damage or human condition. These weights indicate relative modality reliance, not causal responsibility for correctness.

Visualising SHAP Impact

A SHAP summary plot places features on the vertical axis, SHAP value on the horizontal axis, and colours points by feature magnitude, ordered by mean absolute SHAP value. With only two features, `label_image_feat` showed the widest spread and greatest global influence over `label_text_feat`. Red/high points and blue/low points indicate direction of effect; points near zero have little influence.

SHAP should not be mixed with neural hyperparameters (LoRA rank, adapter dimension, learning rate, dropout) — those effects are better shown via accuracy/F1 comparison charts, trainable-parameter charts, GPU-memory charts, and learning curves. Consistent with this, LoRA used fewer parameters than the adapter while outperforming it on informativeness, whereas partial fine-tuning achieved the top accuracy at the highest memory cost.

The Streamlit app’s beeswarm-style text plot uses signed $\text{Gradient} \times \text{Input}$ values per embedding dimension, and should be labelled a $\text{Gradient} \times \text{Input}$ attribution plot rather than an exact SHAP summary plot, to avoid conflating it with the leakage-affected Random Forest SHAP figure.

4.3.2 State of Art

Direct comparison across studies is complicated by differing CrisisMMD versions, label-merging rules, agreed/non-agreed label subsets, train/test splits, modality inputs, F1-averaging conventions, and preprocessing choices, so numerical differences do not automatically prove superiority [6, 7, 10, 11, 13, 14].

Abavisani et al.’s cross-attention BERT-DenseNet model (Setting A) reported 89.33% informativeness accuracy (88.09% macro-F1), 91.14% humanitarian accuracy (68.41% macro-F1), and 72.65% severity accuracy (59.76% macro-F1) [36].

Table 4.19. Contextual Comparison with Abavisani et al.

Task	Previous Model	Current Best Model	Discussion
Informativeness	89.33% acc / 88.09% macro-F1	89.36% acc / $\sim$ 89.25% macro-F1 (partial FT)	Highly similar; check protocols before claiming improvement
Humanitarian	91.14% acc / 68.41% macro-F1	90.58% acc / 83.24% macro-F1 (adapter)	Slightly lower accuracy, much higher macro-F1
Severity	72.65% acc / 59.76% macro-F1	71.27% acc (adapter) / 58% macro-F1 (LoRA)	Previous model remained slightly stronger

The current humanitarian adapter matched the previous model’s accuracy closely while achieving substantially higher macro-F1, though dataset preparation must be confirmed identical before calling this an improvement. Severity remained slightly below the earlier benchmark, reinforcing that distinguishing Mild from Little/No Damage may need finer visual detail than CLIP ViT-B/32 captures. Khattar and Quadri’s CAMM model reported 84.08% average F1 for informative-vs-non-informative classification [37]; the present LoRA (87.15% F1) and partial fine-tuning ($\sim 89.36\%$) results are numerically higher, though this should be read as contextual rather than a strict ranking given protocol differences.

The earlier cross-attention study also reported that attention-based multimodal fusion improved macro-F1 more clearly than simple feature concatenation [36], which supports the use of gated humanitarian fusion and symmetric severity fusion adopted in the present framework.

Main advantages of the present framework:

- Shared CLIP backbone across three disaster-classification tasks with a direct zero-shot baseline;
- Adapter and LoRA parameter-efficient adaptation with task-specific multimodal fusion;
- Computational-resource measurement, CrisisMMD and MEDIC evaluation;
- Image/text explainability and deployment via an interactive Streamlit application.

Main limitations:

- Test loader used during parts of informativeness model development;
- ROC-AUC values and complete paired predictions (for McNemar testing) were not retained;
- Exact SHAP applied only to a target-leaking Random Forest baseline;
- Rare humanitarian categories had very few test samples; severity minority recall remained low;
- MEDIC protocols differed across tasks; no formal latency/usability study was conducted.

Task-specific fine-tuning improved CLIP in all three tasks. The adapter model achieved the best humanitarian result with 90.58% accuracy and 83.24% macro-F1. It also achieved the highest severity accuracy of 71.27%, while LoRA produced better severity macro-F1 and weighted F1.

Partial fine-tuning achieved the best informativeness accuracy of 89.36%. LoRA achieved 87.17% accuracy while updating only about 1.03% of the model parameters, making it a more efficient choice.

On MEDIC, the framework achieved 87% accuracy for informativeness, 88% for humanitarian classification, and 86% for severity. However, these results were obtained with different training settings. Affected Individuals, Mild Damage and None remained the most difficult classes due to limited samples, unclear visual evidence and class imbalance.

Chapter 5

Project Management and Finance

5.1 Project Planning

Project planning organised the thesis activities from August 2025 to July 2026, including topic selection, literature review, dataset preparation, model training, evaluation, XAI analysis, Streamlit deployment, thesis writing, formatting, and viva preparation.

Since the thesis is software-based, the planning focused on dataset management, model implementation, result analysis, web application development, and documentation. No hardware implementation or circuit simulation was included.

5.1.1 Work plan

The thesis work plan was prepared to complete the project systematically. It included topic selection, supervisor consultation, problem definition, literature review, dataset preparation, preprocessing, model training, evaluation, comparative analysis, XAI analysis, Streamlit integration, thesis writing, revision, formatting, final submission, and viva preparation.

The planning covered the period from August 2025 to July 2026. Topic selection and planning were completed in August–September 2025, followed by dataset preparation in October, model development and evaluation in November, and comparative analysis, XAI, and web integration in December. Thesis writing and revision were carried out from January to May 2026, while final formatting, submission preparation, and viva preparation were completed from June to July 2026.

5.1.2 Timeline Diagram

The overall timeline of the thesis work is shown in Figure 5.1. The timeline represents the complete structure of the research process, beginning with problem identification and literature review, followed by methodology, design and implementation, results and discussion, social and environmental analysis, complex engineering analysis, and conclusion with future plan.

5.2 Project Estimation

Project estimation was prepared to identify the time, resources, equipment, and financial support required to complete the thesis. As this thesis focused on a software-based multimodal disaster-classification system, the estimation mainly covered dataset preparation,

Table 5.1. Thesis Activity Planning Timeline

Period	Main Activities
August 2025	Initial planning, research area selection, and supervisor consultation
September 2025	Topic selection, problem definition, and literature review
October 2025	Dataset collection, inspection, label mapping, and preprocessing
November 2025	Zero-shot testing, adapter tuning, LoRA, partial fine-tuning, and model evaluation
December 2025	Comparative analysis, MEDIC evaluation, XAI analysis, and Streamlit integration
January 2026	Documentation, chapter preparation, revision, and thesis writing
February 2026	Improvement of methodology, result checking, and correction of analysis
March 2026	Thesis chapter refinement, figure preparation, and formatting
April 2026	Final result validation, discussion writing, and supervisor feedback
May 2026	Final thesis editing, plagiarism checking, and proofreading
June 2026	Final formatting, submission preparation, and presentation slide preparation
July 2026	Final submission, viva preparation, and thesis defense

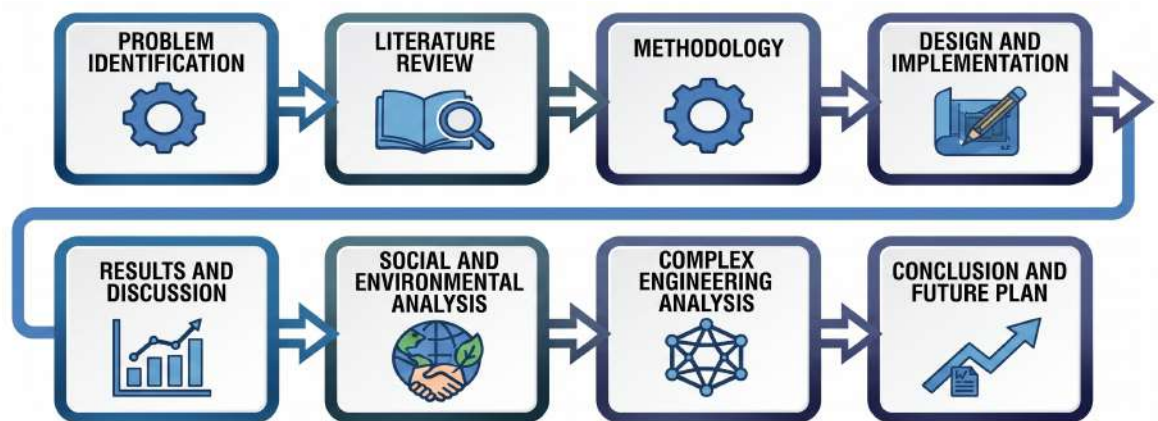


Fig. 5.1. Overview of the structural thesis outline.

model training, evaluation, explainable AI analysis, Streamlit deployment, documentation, and thesis preparation costs. No additional hardware implementation or physical circuit development was included.

5.2.1 Overall Estimation

The overall estimation covered the full process from research planning to final submission, including topic selection, literature review, dataset collection, preprocessing, model development, testing, result analysis, web application development, thesis writing, formatting, plagiarism checking, and presentation preparation.

The project required approximately one year, from August 2025 to July 2026. Most technical work was completed using existing computing resources, open-source machine-learning tools, and publicly available datasets. Since the project was purely software-based, no dedicated hardware design, embedded system, or circuit simulation was required.

Table 5.2. Overall Project Estimation

Estimation Area	Description
Project duration	August 2025 to July 2026
Project type	Software-based multimodal disaster-classification system
Main technical work	Dataset preparation, preprocessing, model training, evaluation, XAI analysis, and deployment
Implementation platform	Python, PyTorch, Hugging Face Transformers, PEFT, and Streamlit
Dataset source	Publicly available disaster-related datasets
Hardware requirement	Existing laptop/desktop computer with GPU support
Additional hardware	Not required
Circuit simulation	Not required
Final output	Trained models, comparative results, XAI outputs, Streamlit application, and thesis documentation

5.2.2 Equipment Estimation

The equipment required for this thesis was mainly related to software development and deep-learning experimentation. A Windows-based computer with GPU support was used for model training and evaluation, while CPU execution served as a fallback option.

The main tools included Anaconda Python, PyTorch, Hugging Face Transformers, PEFT, Streamlit, pandas, NumPy, scikit-learn, Matplotlib, and LaTeX/Overleaf. Since the project was software-based, no additional physical hardware components were required beyond a computer, internet connection, storage space, and documentation tools.

Table 5.3. Equipment Estimation of the Project

SL	Equipment / Tool	Purpose
1	Laptop/Desktop computer with GPU	Model development, training, testing, and documentation
2	Internet connection and storage	Dataset download, literature review, checkpoints, and thesis files
3	Anaconda Python environment	Package and environment management
4	PyTorch, Transformers, and PEFT	CLIP implementation and LoRA-based fine-tuning
5	Streamlit	Web application deployment
6	LaTeX/Overleaf	Thesis writing and formatting

5.2.3 Financial Estimation

The financial estimation of the project was comparatively low because most of the software tools and datasets used in this thesis were open-source or freely available for academic research. An existing computer system was used, so no major hardware purchase was required. The Streamlit application was developed locally; therefore, no paid deployment server was required for thesis demonstration. The only major direct cost considered for this thesis was final thesis printing and binding.

Table 5.4. Financial Estimation of the Project

SL	Cost Item	Estimated Cost (BDT)
1	Internet, electricity, and local computer usage	0
2	Software tools, libraries, and dataset usage	0
3	Cloud/GPU or backup computing support	0
4	Thesis printing and binding	3,000
5	Presentation and miscellaneous expenses	0
Total Estimated Cost		3,000

Therefore, the total estimated financial cost of the project is approximately BDT 3,000. Most of the project expenses were considered zero because the thesis was completed using existing resources, open-source software, and publicly available datasets.

Chapter 6

Social and Environmental Influence

This chapter discusses the socioeconomic, humanitarian, ethical, legal, cultural, and environmental implications of the proposed multimodal disaster-classification system [1, 2, 32]. The system analyses disaster-related images and social-media text to assess informativeness, humanitarian category, and damage severity, supporting rapid information screening and disaster monitoring [5, 13]. However, practical deployment must address risks such as misclassification, privacy concerns, dataset bias, cultural mismatch, misuse, and computational environmental cost through responsible and sustainable AI practices.

6.1 Societal Impact

During natural or human-induced disasters, a large volume of images, captions, and social-media posts are published within a very short timeframe [3, 4]. It is practically impossible for humanitarian agencies and emergency-response teams to review every post manually.

The proposed system identifies informative posts, classifies the type of humanitarian information, and estimates visible damage severity. It can help disaster-response organizations, government agencies, researchers, emergency volunteers, news analysts, and social-media monitoring teams by highlighting posts that may be useful.

However, the system is not intended to replace emergency managers, humanitarian experts, engineers, medical professionals, or field responders. Its predictions are meant to support decision-making rather than serve as confirmed evidence. Final decisions such as rescue operations, resource allocation, evacuation, or damage assessment must remain under human supervision.

6.1.1 Financial and Health Influences

Disasters cause billions of dollars in damage to infrastructure, homes, transport systems, utility lines, businesses, and crop resources. Quickly understanding where damage has occurred, who is affected, and where relief activities are needed is a key part of the response effort.

Automated multimodal classification can reduce the labour and cost of manually reviewing thousands of social-media posts. It can serve as a first-line filter, helping analysts determine which posts warrant closer human review.

The informativeness gate also prevents non-informative posts from being passed to the humanitarian and severity classifiers, unless the user enables an override. This reduces unnecessary computation and limits irrelevant downstream predictions.

From a humanitarian perspective, the system can help identify:

- victims, injured individuals, or displaced people;
- infrastructure and utility damage;
- rescue, volunteering, and donation activities;
- other relevant disaster information; and
- severe, moderate, or mild visual damage.

Nevertheless, misclassification can cause analysts to miss urgent information or focus on less critical content. Financial efficiency must therefore never come at the expense of human safety, accuracy, accountability, and expert review.

Humanitarian and Health Implications

Perhaps the system's most significant humanitarian contribution is its ability to facilitate rapid screening and prioritization of disaster-related information. Potential benefits include:

- identifying posts that are informative and need human review or monitoring, much faster than manual methods;
- aiding detection of victims, rescue and donation operations, and post-disaster research; and
- indirectly supporting health and safety by flagging posts referring to injuries, damaged hospitals, blocked roads, or damaged infrastructure.

However, a predicted humanitarian category does not confirm that the event is real or current. Social-media posts may contain misinformation, outdated images, edited content, or missing context. Therefore, predictions should be verified using reliable sources, field reports, geolocation evidence, and expert judgement. The system should provide probabilistic recommendations, not confirmed emergency information, and sensitive outputs should be accessible only to responsible users.

6.1.2 Safety, Legal and Cultural Issues

One safety concern is that classification errors can influence how disaster information is prioritized. Two important error types are:

- **False negatives:** informative or urgent posts are classified as non-informative, or assigned the wrong humanitarian class.

- **False positives:** irrelevant or low-priority posts are categorized as important or severely damaged.

A false negative can prevent an urgent post from ever being sent downstream for analysis, while a false positive can waste response resources or cause unnecessary alarm. Model probabilities and explanations should therefore be carefully scrutinized before any action with serious consequences is taken.

There are also legal and privacy concerns, since disaster-related social-media posts may contain:

- personal identities and faces;
- location and property details;
- medical, safety, or sensitive disaster information;
- copyrighted or graphic content.

Deployment must respect data-protection, copyright, consent, and ethical requirements, with particular care taken around personal and sensitive information. High-impact predictions must be reviewed by a human and verified in context, since language and cultural differences can reduce model accuracy.

6.2 Impact on Environment

AI systems require significant compute, memory, storage, and energy. CLIP ViT-B/32 is computationally demanding because it uses separate vision and text encoders, and repeated training of zero-shot, adapter, LoRA, and partial fine-tuning models increases energy use [30].

Adapter and LoRA reduce training cost by updating fewer parameters and creating smaller task-specific checkpoints [22]. However, the full CLIP backbone is still needed during inference. Sequential gating and Streamlit caching help reduce unnecessary computation, but multimodal transformers remain resource-intensive.

6.3 Sustainability Issues

Sustainability in this study spans social, ethical, environmental, and technical dimensions. The goal is to develop a long-lasting disaster-analysis system that is relevant, easy to use, resource-efficient, and maintainable, while remaining non-harmful.

6.3.1 Social Sustainability

Social sustainability depends on protecting disaster-affected communities and avoiding the aggravation of risk or inequality. Important principles include:

- maintaining human supervision and verifying predictions before intervention;
- protecting privacy, dignity, and avoiding public exposure of affected individuals;
- testing across different regions and disaster types while ensuring minority categories are not excluded.

At the class level, categories with fewer samples such as Affected Individuals and minority severity classes performed worse. Future work should therefore focus on balanced data collection and dedicated minority-class evaluation. The system is intended to assist analysts and response teams, not to replace field knowledge or professional judgement.

6.3.2 Ethical Sustainability

The main ethical concerns include:

- responsible use of social-media data;
- privacy and informed data handling;
- classification bias;
- misinformation and manipulated content;
- transparency of model decisions;
- human review of urgent predictions; and
- safe handling of distressing images.

The final application includes image saliency maps, token-importance graphs, and Gradient $\times$ Input attribution, which help clarify which parts of an image or text token influenced a given prediction. However, these explanations do not guarantee that a prediction is correct, fair, or causally justified. Model outputs should never be presented as validated humanitarian facts; uncertainty should always be communicated clearly.

6.3.3 Environmental Sustainability

Improving environmental sustainability requires reducing computation, storage, and energy use. Model selection should consider accuracy, F1-score, trainable parameters, GPU memory, inference latency, model size, request volume, and deployment hardware.

The highest-performing model is not always the most sustainable option. For example, partial fine-tuning achieved the best informativeness performance but required more trainable parameters, while LoRA used fewer parameters and offered better efficiency. Therefore, the deployed Streamlit application uses LoRA-based checkpoints to balance performance and computational cost.

Further improvements may include batching inference requests, using efficient GPU or cloud resources, avoiding unnecessary retraining, compressing checkpoints, applying lower-precision inference where feasible, and turning off inactive computational resources.

6.3.4 Technical Sustainability

Technical sustainability requires the system to remain maintainable, scalable, and adaptable. The implementation follows a modular structure, separating dataset preparation, model training, task-specific architectures, checkpoint loading, explainability, interface design, and Streamlit inference.

The web application is organised into separate components for the user interface, model definitions, explanation functions, and visual styling. This allows future updates without redesigning the entire system. Future improvements may include adding new datasets, supporting multilingual and geographic data, improving minority-class recognition, adding misinformation detection, improving calibration, and deploying the system through scalable cloud infrastructure.

Since social-media language, disaster terminology, and image characteristics change over time, the models should be updated regularly using recent, representative, and ethically collected data.

6.3.5 Positive and Negative Impacts

Positive Impacts

- Quick detection of informative disaster-related posts, reducing manual-review workload for researchers, analysts, and emergency-response teams.
- Humanitarian-category classification and visible damage assessment using both image and text inputs.
- Explainability methods that support more transparent predictions.
- A deployable and parameter-efficient system available through a web application.

Negative Impacts

- False negatives may miss urgent information, while false positives may over-flag less relevant posts.
- Privacy, copyright, and sensitive-image exposure may create ethical and legal risks.
- Cultural, linguistic, and class imbalance may reduce fairness across different groups and disaster contexts.
- Over-reliance on the system may lead to unsupported monitoring or automated decision-making.
- GPU-based training and inference increase energy use and environmental cost.

Therefore, the system should support human analysts rather than function as an autonomous emergency decision-maker.

This chapter discussed the social and environmental impacts of the proposed multimodal disaster-classification system. The system can support rapid information screening, humanitarian categorisation, and visible damage assessment, but risks such as misclassification, privacy violations, misinformation, bias, and misuse remain. Therefore, human oversight, privacy protection, accountability, and efficient deployment are essential to ensure responsible, transparent, and sustainable AI use.

Chapter 7

Complex Engineering Problems and Activities

This chapter relates the proposed multimodal disaster-classification system to complex engineering knowledge, problem-solving, and engineering activities [31,32]. The system integrates computer vision, NLP, multimodal learning, deep learning, statistics, explainable AI, ethical responsibility, web deployment, and sustainability [18–20]. Since disaster-related outputs may affect humanitarian monitoring and emergency response, the system requires careful data preparation, class-sensitive evaluation, cross-dataset testing, human oversight, and responsible engineering design.

7.1 Addressing Complex Engineering Problems

7.1.1 Overview

The proposed system analyses disaster-related images and social-media text for informativeness classification, humanitarian-category classification, and damage-severity assessment. Such posts are often noisy, incomplete, ambiguous, and context-dependent, making automated classification challenging [1, 2].

Captions may change the interpretation of an image, and humanitarian categories can overlap, such as infrastructure damage appearing together with affected people or rescue activity [13, 14]. Severity assessment is also difficult because damage levels are often gradual rather than clearly separated.

Therefore, a rule-based or single-modality approach is insufficient. This research combines computer vision, NLP, multimodal fusion, fine-tuning, statistical evaluation, explainable AI, and web deployment. The system is intended to support disaster information analysis, not replace human emergency decision-making.

7.1.2 Knowledge Profile

This thesis draws on the following engineering knowledge attributes:

The thesis therefore draws on mathematical, computational, technical, practical, ethical, and research-based knowledge – this is not simply a matter of building a single classifier.

7.1.3 Range of Complex Engineering Problem Solving

The study meets the key characteristics of complex engineering problem-solving:

Addressing this problem therefore requires technical analysis, experimentation, research-based judgement, and human-centred engineering decisions.

Table 7.1. Engineering Knowledge Profile

Attribute	Application in This Thesis
K1	Understanding disaster communication, humanitarian information, visual damage evidence, and social-media behaviour
K2	Probability, linear algebra, cosine similarity, softmax, cross-entropy, precision, recall, F1-score, confusion matrices, confidence intervals, and attribution analysis
K3	Structured engineering design covering data preparation, multimodal preprocessing, model architecture, training, testing, validation, and deployment
K4	Advanced AI methods including CLIP, Vision Transformers, text transformers, prompt engineering, adapters, LoRA, gated fusion, Grad-CAM, and Gradient $\times$ Input
K5	Complete system design involving three connected classification stages, checkpoint management, explainability, and a Streamlit interface
K6	Practical tools including Python, PyTorch, Hugging Face Transformers, PEFT, Scikit-learn, Pandas, NumPy, Matplotlib, Pillow, and Streamlit
K7	Public safety, privacy, ethical data use, bias, cultural and linguistic limitations, environmental cost, and sustainability
K8	Use of research literature on multimodal disaster analysis, humanitarian computing, transfer learning, parameter-efficient fine-tuning, and explainable AI

7.1.4 Engineering Complexity in the Proposed System

The proposed system involved several engineering challenges. CrisisMMD and MEDIC had different file structures, labels, class distributions, and image–text relationships, requiring careful preprocessing and label harmonisation. Missing images, invalid paths, unreadable files, noisy text, and weak image–text alignment also affected reliability.

The system used CLIP image and text encoders to extract, normalise, fuse, and classify multimodal features. Different methods, including zero-shot prompting, adapters, LoRA, and partial fine-tuning, required balancing accuracy, trainable parameters, GPU memory, and deployment feasibility. Class imbalance and dataset shift further affected generalisation.

Explainability and deployment added further complexity. Saliency, token attribution, occlusion, and ablation were used to analyse model decisions. Finally, the selected models were integrated into a sequential Streamlit application with an informativeness gate. Overall, the system balanced performance, fairness, interpretability, efficiency, usability, and responsible deployment.

Table 7.2. Complex Engineering Problem Characteristics

Attribute	Application in This Thesis
P1: Conflicting requirements	Balancing accuracy, minority-class recall, false predictions, trainable parameters, GPU memory, latency, explainability, and sustainability
P2: Depth of analysis	No direct rule-based solution exists; zero-shot, adapter, LoRA, partial fine-tuning, gated fusion, and multiple explanation methods were investigated
P3: Depth of knowledge	Required advanced knowledge of computer vision, NLP, transformers, multimodal embeddings, statistical evaluation, and XAI
P4: Unfamiliar issues	Disaster images and captions are noisy, incomplete, culturally variable, and sometimes contradictory
P5: Limited fixed standards	No universal engineering standard defines how social-media images and captions should be classified for humanitarian use
P6: Stakeholder involvement	Stakeholders include disaster-affected communities, humanitarian organisations, government agencies, emergency responders, researchers, volunteers, and platform operators
P7: Significant consequences	Incorrect predictions may cause urgent posts to be overlooked, resources to be misdirected, or sensitive information to be exposed
P8: Interdependence	Dataset preparation, image and text encoding, multimodal fusion, model training, evaluation, XAI, and web deployment are interconnected

7.1.5 Problem-Solving Approach

- Defined the three classification tasks: informativeness, humanitarian category, and damage severity.
- Selected CrisisMMD for model development and MEDIC for out-of-distribution evaluation.
- Analysed dataset structure, labels, image paths, class distribution, missing values, and invalid records.
- Preprocessed text and images, harmonised labels, and processed both modalities using the CLIP processor.
- Implemented zero-shot CLIP, adapter fine-tuning, LoRA, and partial fine-tuning methods.
- Designed task-specific fusion layers and trained models using suitable losses, optimisation, and checkpoint selection.

- Evaluated performance using accuracy, precision, recall, macro-F1, weighted F1, confusion matrices, confidence intervals, and computational cost.
- Applied explainability methods, compared CrisisMMD and MEDIC results, and deployed the final models through Streamlit.

7.1.6 Selected Attributes for Complex Engineering Problems

The attributes that best characterise this as a complex engineering problem are:

Table 7.3. Selected Complex Engineering Problem Attributes

Selected Attribute	Reason
P1	The system must balance accuracy, minority-class recall, false positives, false negatives, memory consumption, latency, explainability, and sustainability
P2	Multiple model families, adaptation strategies, fusion structures, and evaluation settings had to be investigated
P3	The work required advanced knowledge of CLIP, Vision Transformers, NLP transformers, multimodal fusion, parameter-efficient fine-tuning, statistics, and XAI
P6	Stakeholders include affected communities, humanitarian analysts, responders, researchers, volunteers, organisations, and platform operators
P7	Misclassification may influence the prioritisation of urgent disaster information and the allocation of human attention
P8	Dataset quality, encoders, fusion, losses, evaluation metrics, XAI, and deployment behaviour are strongly interconnected

These characteristics show both engineering and human-centred complexity in the research.

The main conflict was accuracy versus minority-class performance. A model could achieve high accuracy by predicting the majority class but still fail to detect Mild Damage or Affected Individuals.

The highest-accuracy model was also not always the best for deployment. Partial fine-tuning performed best for informativeness, while LoRA was more efficient in trainable parameters and computational cost. Therefore, model selection required engineering judgement rather than dependence on one metric.

7.2 Addressing Complex Engineering Activities

7.2.1 Complex Engineering Activities

The project also involves complex engineering activities at a significant scale:

Table 7.4. Complex Engineering Activity Characteristics

Attribute	Activity	Application in This Thesis
A1	Range of resources	CrisisMMD and MEDIC datasets, GPU computing, Python libraries, pretrained CLIP models, statistical tools, visualisation methods, and research literature.
A2	Level of interaction	Interaction among image quality, text quality, class imbalance, accuracy, recall, explainability, computational cost, privacy, and deployment.
A3	Innovation	Integration of prompt engineering, adapters, LoRA, task-specific multimodal fusion, cross-dataset evaluation, XAI, and a sequential Streamlit pipeline.
A4	Social and environmental consequences	Potential support for disaster monitoring, misclassification risks, privacy concerns, unequal class performance, and GPU-related environmental impact.
A5	Familiarity	Multimodal humanitarian AI is an evolving area without a single fixed engineering standard or universally accepted deployment procedure.

This work therefore extends beyond training a single model – it encompasses the design, comparative evaluation, interpretation, and deployment of a fully functioning multimodal AI system.

7.2.2 Complex Engineering Activity Attributes

Table 7.5. Selected Complex Engineering Activity Attributes

Selected Attribute	Reason
Range of resources	The project uses multimodal datasets, pre-trained models, GPUs, software libraries, visualisation tools, statistical methods, and research literature.
Level of interaction	It integrates preprocessing, image encoding, text encoding, fusion, training, evaluation, explainability, checkpoint management, and deployment.
Innovation	It combines a shared CLIP backbone with task-specific fusion, adapters, LoRA, sequential gating, cross-dataset evaluation, and live XAI.
Social and environmental consequences	It considers humanitarian benefits, prediction risks, privacy, bias, sensitive images, computational cost, and environmental impact.
Familiarity	It addresses an evolving multimodal disaster-analysis problem with limited fixed standards and changing social-media behaviour.

This thesis integrated multiple resources, including CrisisMMD, MEDIC, pretrained CLIP, PyTorch, Transformers, PEFT, and Streamlit for dataset use, model training, LoRA fine-tuning, and web deployment. The system components were strongly connected, as preprocessing, image quality, and informativeness classification could affect downstream humanitarian and severity predictions. Therefore, overall reliability depends on both the final classifiers and the informativeness gate.

Table 7.6. Complex engineering activities

Attribute	Description	Notation
A1	Use of diverse resources	✓
A2	Level of interaction	✓
A3	Innovation	×
A4	Consequences for society and the environment	×
A5	Familiarity	✓

This thesis represents an engineering “wicked problem” because it combines humanitarian, ethical, computational, and deployment challenges. Zero-shot CLIP was used as the baseline, while adapters, LoRA, and partial fine-tuning improved task-specific performance. Adapters achieved the best humanitarian result and highest severity accuracy, LoRA provided more balanced severity classification, and partial fine-tuning achieved the best informativeness accuracy.

Class-wise recall, confusion matrices, and MEDIC evaluation showed that minority classes and dataset shift remained challenging. Explainability was supported through saliency maps, token attribution, Gradient $\times$ Input, probability visualisation, gating logic, and Streamlit deployment. However, human supervision remains essential because disaster decisions involve safety, privacy, uncertainty, and ethical responsibility.

Overall, the proposed system is a complex engineering task involving computer vision, NLP, multimodal learning, deep learning, statistical evaluation, explainable AI, systems design, ethics, sustainability, and deployment. The final chapter presents the conclusions, limitations, and future directions of the study.

Chapter 8

Conclusion and Future Plan

This chapter presents the conclusion, limitations, and future directions of the CLIP-based multimodal disaster-classification framework. The study addressed informativeness classification, humanitarian-category classification, and damage-severity assessment using disaster-related images and social-media text. The framework was evaluated through zero-shot CLIP, adapter fine-tuning, LoRA, partial fine-tuning, multimodal fusion, explainability analysis, cross-dataset testing, and Streamlit-based deployment [7, 15, 18, 21, 22].

8.1 Conclusion

This thesis developed a multimodal pipeline that integrates preprocessing, CLIP-based feature extraction, model adaptation, fusion, evaluation, explainability, and web-based deployment [13, 14, 18]. The system first identifies whether a disaster-related post is informative and then classifies its humanitarian category and visible damage severity.

The system is intended as a decision-support tool for humanitarian analysts, disaster researchers, emergency-monitoring professionals, volunteers, and response organisations. It should not replace emergency responders, field verification, structural assessment, or human decision-making.

8.1.1 Development of Classification Models

CrisisMMD was used as the primary dataset because it includes disaster-related images, tweet text, and labels for informativeness, humanitarian classification, and damage severity. MEDIC was used as a supplementary dataset to assess out-of-distribution generalisation.

The final outputs were organised into three task groups:

- **Informativeness:** Informative and Not Informative.
- **Humanitarian type:** Infrastructure and Utility Damage; Other Relevant Information; Not Humanitarian; Rescue/Volunteering/Donation Effort; and Affected Individuals.
- **Damage severity:** Severe, Mild, and None.

Several methods were compared, including zero-shot CLIP, adapter fine-tuning, LoRA fine-tuning, partial CLIP fine-tuning, image-text concatenation, gated humanitarian fusion, and feature-level severity fusion. All approaches used `openai/clip-vit-base-patch32` as the shared backbone, with separate fusion and classification heads for each task.

The selected models were deployed in a Streamlit web application that supports image upload, caption input, sequential classification, probability display, checkpoint selection, token-level attribution, and CPU/GPU latency reporting. Since neural-network confidence scores may be poorly calibrated, the displayed probabilities are treated as decision-support indicators rather than confirmed certainty [41]. This demonstrates the practical usability of the proposed multimodal disaster-classification framework.

8.1.2 Evaluation of Performance Metrics

Model performance was evaluated using accuracy, precision, recall, F1-score, macro-F1, weighted F1, confusion matrices, confidence intervals, trainable parameters, and GPU-memory requirements. These metrics were important because disaster datasets are often imbalanced, making accuracy alone insufficient.

For humanitarian-category classification, the adapter model achieved the best performance with 90.58% accuracy and 83.24% macro-F1, using 5,302,279 trainable parameters, which represents 3.39% of the total parameters. The LoRA model achieved 87.02% accuracy and 81.17% macro-F1 with 8,919,815 trainable parameters, representing 5.57% of the total parameters, making it a competitive but slightly heavier alternative.

For damage-severity classification, the adapter model achieved the highest accuracy of 71.27%, while LoRA achieved stronger class balance with 58% macro-F1 and 69% weighted F1. This indicates that LoRA performed more consistently across the Severe, Mild, and None classes.

For informativeness classification, partial fine-tuning achieved the best result with 89.36% accuracy and F1-score. However, LoRA achieved 87.17% accuracy while training only about 1.03% of the parameters, compared with about 13.88% for partial fine-tuning.

The MEDIC out-of-distribution results showed 87% informativeness accuracy, 88% humanitarian accuracy, and 86% severity accuracy. These findings suggest reasonable generalisation, although MEDIC results should be interpreted separately because its training and evaluation procedures differed from CrisisMMD.

8.1.3 Baseline Model Comparison

Zero-shot CLIP prompt engineering was used as the main baseline because it uses pretrained visual–language knowledge without task-specific training. It provided a reference for comparing adapter fine-tuning, LoRA, partial fine-tuning, and multimodal fusion.

For humanitarian classification, zero-shot CLIP achieved 66.49% accuracy and 56.17% macro-F1, while adapter fine-tuning improved performance to 90.58% accuracy and 83.24% macro-F1. For severity classification, zero-shot CLIP achieved 60.11% accuracy, 38% macro-F1, and 54% weighted F1. Adapter fine-tuning achieved the highest severity accuracy of 71.27%, while LoRA provided better class balance with 58% macro-F1 and 69%

weighted F1. For informativeness classification, LoRA achieved 87.17% accuracy, while partial fine-tuning achieved the best result at 89.36%.

Overall, fine-tuned methods outperformed the zero-shot baseline, but the best method varied by task. Adapter fine-tuning was strongest for humanitarian classification, partial fine-tuning performed best for informativeness, and LoRA offered a strong balance between performance and parameter efficiency. Bootstrap evaluation supported these improvements, although a formal McNemar test could not be reconstructed because paired-prediction tables were not retained.

8.2 Limitations

Although the proposed system achieved promising results, several limitations remain. These are mainly related to dataset coverage, class imbalance, multimodal noise, interpretability, deployment scope, and external validation.

8.2.1 Limited and Unbalanced Disaster Data

CrisisMMD and MEDIC cover limited disaster types, locations, languages, platforms, and time periods. Therefore, the model may not generalise equally well to unseen disasters, non-English or mixed-language posts, and different social-media contexts.

Class imbalance was another limitation. Some humanitarian categories had very few samples, and severity classes were unevenly distributed. This may bias the model toward majority classes and reduce reliability for underrepresented but important categories.

8.2.2 Complexity of Multimodal Disaster Information

Disaster-related social-media data are noisy, sparse, and difficult to verify. Images may be blurred, edited, duplicated, outdated, or mismatched with captions. Text may contain abbreviations, spelling errors, hashtags, misinformation, and culture-specific expressions.

The current system assigns one main humanitarian label per post, although a single post may contain multiple meanings. Damage-severity assessment is also affected by image quality, angle, lighting, and occlusion. Therefore, automated predictions should not replace expert inspection or official field validation.

8.2.3 Model Interpretability

Although explanation methods were applied, interpretability remains challenging. Grad-CAM-style heatmaps can highlight important image regions, but they do not prove causal reasoning. Similarly, token attribution and Gradient $\times$ Input can show influential words, but they cannot fully explain transformer-based multimodal decisions.

Only the Random Forest annotation baseline received full SHAP analysis. However, because target leakage was present in that experiment, it should not be treated as an explanation of the deployed CLIP models. The Streamlit beeswarm-style visualisation is based on Gradient $\times$ Input, not exact SHAP values. Therefore, further testing of explanation faithfulness is needed.

8.2.4 Deployment and External Factors

The current system analyses disaster images and text but does not include external information such as location, time, source credibility, weather, satellite signals, network conditions, or field reports.

A single social-media post may provide only a partial or misleading view of a disaster situation. In addition, the system was mainly tested on desktop CPU/GPU hardware. Its scalability to embedded, FPGA, portable, or low-power devices remains a future research direction.

8.3 Future Plan

Future work should focus on making the framework more robust, generalisable, interpretable and deployable. System Structure : (i)The system can extend from the Streamlit CPU/GPU application to FPGA, embedded/edge/sensor/communication based applications for lightweight disaster analysis.

8.3.1 FPGA-Based Model Acceleration

One of the important research prospects is to accelerate intensive operations, such as preprocessing, matrix multiplication, attention computation, multimodal fusion and classification with FPGA. This, in turn, may lead to reduced latency, stable processing and energy efficiency compared to CPU-based execution.

CLIP ViT-B/32 is a large model that requires quantisation, pruning, knowledge distillation, fixed-point arithmetics and hardware-aware compression before FPGA deployment. The FPGA variant should be compared and contrasted against the CPU and GPU implementations with respect to accuracy, F1-score, latency, memory use, power consumption and energy-per-prediction.

8.3.2 Hardware–Software Co-Design

In future work, deploying hardware-software co-design to jointly optimise the model and hardware is a promising direction. It would handle input, preprocessing, communication, visualisation, storage and checkpoint control.

The FPGA or embedded accelerator would take over the heavy-lifting tasks like feature extraction, fusion, matrix multiplication and classification. The informativeness type model could serve as a preliminary gate, where information that carried little value may be discarded to limit processing and decrease energy expenditure.

8.3.3 Portable Edge-Device Implementation

These edge platforms could consist of FPGA boards, embedded GPUs, single-board computers, handheld emergency terminals and drone-mounted units. These devices may assist with local disaster analysis without constant reliance on the cloud.

At edge deployment, images and associated text were available from cameras, drones, smartphones or monitoring stations. This can help to decrease latency, preserve privacy, conserve bandwidth and promote operations in places with weak connectivity.

Given the size and limited resources of LAPIs, it may be necessary to quantise, prune or distil the model in order to build a smaller but efficient CLIP-based system .

8.3.4 Sensor and Communication Integration

The next iteration of hardware may incorporate cameras, GPS, and environmental sensors that bring together image + text + location + sensor data to provide more extensive disaster awareness.

For communication Wi-Fi, cellular networks like 3G/4G, low power and short range bluetooth transmissions (also build for IoT), LoRa communication, satellite transmission or Radio transmission might have been used. The final results could be displayed on the screen of a web dashboard, and/or recorded to local storage, e.g. when device-to-device network connectivity is necessary like in disaster zones or where reliable networks are non-existent.

8.3.5 Hardware Validation and Real-Time Monitoring

Hardware validation should test the system under realistic disaster conditions, including low-light or blurred images, corrupted inputs, limited connectivity, repeated inference, interrupted communication, and constrained power. Battery life, temperature, memory use, and power consumption should also be evaluated.

A real-time dashboard could show device status, location, inputs, predictions, confidence scores, latency, resource use, power use, and human-review alerts. Predictions related to Affected Individuals or Severe Damage should be checked by human analysts before any action.

This thesis developed a CLIP-based multimodal disaster-classification system. Adapter fine-tuning achieved the best humanitarian performance, LoRA provided efficient and balanced severity classification, and partial fine-tuning achieved the highest informativeness accuracy.

The evaluation used precision, recall, F1-score, confusion matrices, parameter efficiency, cross-dataset testing, and explainability. The selected models were deployed in a Streamlit application for image–text input, sequential prediction, probability display, attribution, and latency reporting.

Future work will focus on FPGA acceleration, embedded and edge deployment, sensor integration, improved multimodal models such as BLIP-style vision-language architectures [39] and wireless communication, while maintaining human supervision, privacy, source verification, transparency, and ethical responsibility [30–32].

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APPENDIX A

Dataset Details

A.1 CrisisMMD Dataset Details

The CrisisMMD dataset comprises Twitter data from seven natural disasters: Hurricanes Harvey, Irma, and Maria, the 2015 Nepal Earthquake, the 2016 Ecuador Earthquake, the 2017 Mexico Earthquake, and the 2018 California Wildfires. Each sample consists of an image and its associated tweet text, along with task-specific annotations. The dataset is split into training, development, and test sets for three tasks: informativeness, humanitarian categories, and damage severity. The dataset files used in this study are:

- `crisismmd_informativeness_train.tsv`
- `crisismmd_informativeness_dev.tsv`
- `crisismmd_informativeness_test.tsv`
- `crisismmd_humanitarian_train.tsv`
- `crisismmd_humanitarian_dev.tsv`
- `crisismmd_humanitarian_test.tsv`
- `crisismmd_severity_train.tsv`
- `crisismmd_severity_dev.tsv`
- `crisismmd_severity_test.tsv`

A.2 MEDIC Dataset Details

The MEDIC dataset, or Multimodal Dataset for Emergency and Disaster Image Classification, contains images from various disaster events annotated for informativeness, humanitarian category, damage severity, and disaster type. It is used for out-of-distribution evaluation and cross-dataset generalisation. The dataset includes a wider variety of disaster types and image sources than CrisisMMD, making it a useful benchmark for testing model robustness.

APPENDIX B

Label Mapping and Model Configuration

All labels were converted into integer values before training. The same mapping was used during validation, testing, and deployment. CLIP ViT-B/32 was used as the backbone model. LoRA and adapter tuning were applied for parameter-efficient task adaptation.

Table B.1. Label Mapping and Model Configuration Summary

Component	Details
Humanitarian labels	Infrastructure and Utility Damage = 0; Other Relevant Information = 1; Not Humanitarian = 2; Rescue/Volunteering/Donation Effort = 3; Affected Individuals = 4.
Severity labels	Severe = 0; Mild = 1; None = 2.
Informativeness labels	Informative = 0; Not Informative = 1.
LoRA settings	Informativeness: $r = 8$, $\alpha = 16$, dropout = 0.1; Humanitarian: $r = 128$, $\alpha = 256$, dropout = 0.05; Severity: $r = 32$, $\alpha = 64$, dropout = 0.1; MEDIC models: $r = 8$, $\alpha = 8$, dropout = 0.1.
LoRA target modules	q_proj, k_proj, v_proj, and out_proj were used where applicable. Humanitarian and severity CrisisMMD LoRA mainly used q_proj and v_proj.
Adapter setting	Bottleneck adapters were inserted into selected CLIP transformer layers. The bottleneck dimension was 64, with GELU activation.
Fusion methods	Image and text features were combined using concatenation, gated fusion, or symmetric fusion depending on task requirements.

APPENDIX C

Training, Evaluation, and Deployment

The models were trained using task-specific settings based on class imbalance, sample size, and computational cost. AdamW was used as the optimiser. Dropout, weight decay, early stopping, and validation-based checkpoint selection were applied to reduce overfitting. The trained models were integrated into a Streamlit application for demonstration and decision-support use.

Table C.1. Training, Evaluation, XAI, and Deployment Summary

Component	Details
Training setup	CLIP ViT-B/32 backbone, AdamW optimiser, dropout, weight decay, early stopping, and best validation checkpoint selection.
Loss functions	Cross-entropy, weighted loss, and focal loss were used depending on task and class imbalance.
Imbalance handling	Class weighting, weighted sampling, macro-F1, class-wise recall, and confusion-matrix analysis.
Evaluation metrics	Accuracy, precision, recall, macro-F1, weighted F1, confusion matrix, latency, trainable parameters, and GPU memory.
XAI methods	Grad-CAM-style heatmap, Integrated Gradients, Gradient $\times$ Input, occlusion analysis, modality ablation, and probability display.
Streamlit features	Image upload, text input, informativeness gate, humanitarian prediction, severity prediction, probability display, XAI output, and latency report.
Deployment caution	The system supports human decision-making only. Sensitive outputs related to affected people or severe damage require expert review.